DETAILED PROJECT REPORT

Construction of a 200-Bed District Hospital Block at Nalbari, Assam

including Diagnostic Wing and Staff Quarters

Submitted by: Assam Public Works Department (Buildings)

Implementing agency / SPV: Assam Health Infrastructure Development Society

DPR prepared by: Brahmaputra Architects and Engineers LLP

Prepared in accordance with the KIIFB Buildings DPR Template (chapters 1-18).

TABLE OF CONTENTS

1. SALIENT FEATURES

Sl.	Particular	Detail
1	Title of the project	Construction of a 200-bed district hospital block at Nalbari with diagnostic wing and staff quarte
2	Department	Health and Family Welfare
3	District / Taluk / Local body / Constituency	Nalbari / Nalbari / Nalbari Municipal Board / Nalbari LAC
4	Implementing agency / SPV	Assam Health Infrastructure Development Society
5	DPR prepared by	Brahmaputra Architects and Engineers LLP
6	Project outlay	Rs. 186.40 crore
7	Budget provision	Rs. 62.00 crore in the current financial year
8	Budget speech reference	Para 118, State Budget Speech 2026-27
9	Administrative sanction	AS accorded vide G.O. No. HLB/294/2026 dated 22 March 2026
10	Nature of the project	New building
11	Present status of existing building	Existing 60-bed block constructed in 1974; structurally distressed, assessed as beyond econor
12	Need for the project	Bed occupancy of the existing block exceeds 140 per cent; the district refers 3,400 cases a yea
13	Plinth area	18,640 sqm across four blocks
14	Number of floors	Ground plus four
15	Land available / required	3.20 ha available in departmental possession; no acquisition required
16	Basis of estimate	Assam PWD Schedule of Rates 2025-26; detailed estimate at Annexure IV
17	Details of revenue streams	User charges as per state schedule; estimated Rs. 4.20 crore per annum
18	Cost Benefit Analysis	BCR 1.38
19	Details of project risks	Chapter 11 — 11 risks identified
20	Project management organisation	Chapter 12
21	Contract management strategy	Item rate — Chapter 13
22	Implementation schedule and WBS	28 months from zero date — Chapter 14
23	Details of statutory clearances	Chapter 15
24	Quality control mechanism	Chapter 16
25	O&M arrangements after completion	Chapter 17
26	Details of attached drawings	Annexures I to III
27	Other attachments	Annexures IV to VII

1.1 Salient Features (continued) — Sheet 1

Building services comprise electrical distribution, water supply and sanitation, HVAC for the identified conditioned areas, medical gas pipeline and vertical transportation. Connected load and demand factors are given in the load schedule.

Quantities have been computed from the approved architectural and structural drawings and cross-checked against the detailed measurement sheets enclosed at the Annexure. Rates are as per the schedule of rates in force with lead and lift adjustments.

Consultations were held with the head of each clinical department on adjacency, patient flow and equipment layout. Minutes of the consultation meetings are placed at the Annexure to this chapter.

The space programme has been developed from the functional brief agreed with the user department. Departmental areas, circulation and the gross-to-carpet ratio are tabulated in the area statement enclosed at the Annexure.

Reference: drawing no. ST-716, revision A.

1.1 Salient Features (continued) — Sheet 2

Bore Log Summary

Depth (m)	Strata	N value	Sample
20.9	Medium sand	25	SPT
2.6	Medium sand	14	UDS
13.6	Clayey silt	27	UDS
14.4	Coarse sand	19	DS
23.8	Weathered rock	17	UDS
13.2	Coarse sand	17	UDS
19.7	Clayey silt	47	SPT
20.1	Coarse sand	29	SPT
7.3	Silty clay	29	DS
20.2	Silty clay	59	SPT
6.9	Silty clay	26	UDS
11.4	Coarse sand	4	SPT
17.2	Medium sand	6	SPT
5.8	Silty clay	52	UDS
12.4	Gravelly sand	56	SPT
26.9	Clayey silt	34	UDS
15.2	Clayey silt	23	UDS
20.9	Gravelly sand	59	UDS
28.6	Silty clay	6	DS
10.7	Gravelly sand	60	SPT
25.2	Silty clay	47	DS
18.5	Clayey silt	55	SPT
23.9	Medium sand	47	UDS

Building services comprise electrical distribution, water supply and sanitation, HVAC for the identified conditioned areas, medical gas pipeline and vertical transportation. Connected load and demand factors are given in the load schedule.

2. EXECUTIVE SUMMARY

The project provides for including diagnostic wing and staff quarters. 3.1 Introduction — Nalbari district has a population of 7.7 lakh served by a single district hospital constructed in 1974 with a sanctioned strength of 60 beds. The block is a load-bearing masonry structure showing distr

The total project cost is estimated at Rs. 186.40 crore inclusive of all taxes and duties, at the price level of the current financial year. The implementation period is 28 months from the zero date.

2.1 Executive Summary (continued) — Sheet 1

The structural design conforms to IS 456 for reinforced concrete and IS 1893 for earthquake resistant design. The site falls in seismic zone V and the importance factor applicable to a hospital building has been adopted.

Quantities have been computed from the approved architectural and structural drawings and cross-checked against the detailed measurement sheets enclosed at the Annexure. Rates are as per the schedule of rates in force with lead and lift adjustments.

The structural design conforms to IS 456 for reinforced concrete and IS 1893 for earthquake resistant design. The site falls in seismic zone V and the importance factor applicable to a hospital building has been adopted.

The structural design conforms to IS 456 for reinforced concrete and IS 1893 for earthquake resistant design. The site falls in seismic zone V and the importance factor applicable to a hospital building has been adopted.

Reference: drawing no. PH-399, revision C.

2.1 Executive Summary (continued) — Sheet 2

Electrical Load Schedule

Panel	Description	Connected load (kW)	Demand factor	Demand (kVA)
DB-35	Medical equipment	105.47	0.80	93.56
DB-12	Pumps	134.85	0.89	132.86
DB-33	Power outlets	142.77	0.64	101.98
DB-12	Medical equipment	17.12	0.43	8.16
DB-15	Pumps	104.41	0.89	103.73
DB-40	Lifts	96.41	0.48	51.41
DB-16	Medical equipment	114.17	0.77	97.51
DB-13	Medical equipment	18.38	0.58	11.91
DB-26	Power outlets	78.08	0.73	63.47
DB-14	Lighting	163.55	0.49	89.38
DB-28	Power outlets	106.81	0.75	88.68
DB-18	Sterilisation	101.33	0.64	72.32
DB-35	Lifts	58.25	0.81	52.31
DB-33	Pumps	157.38	0.94	165.17
DB-14	Medical equipment	38.91	0.53	23.10
DB-11	Medical equipment	106.48	0.70	82.65
DB-04	Lifts	151.00	0.60	101.22
DB-12	HVAC	23.23	0.66	17.05
DB-25	Pumps	14.89	0.87	14.43
DB-12	Power outlets	47.34	0.78	40.84
DB-11	Power outlets	58.03	0.51	32.58
DB-39	Lifts	93.78	0.48	49.55
DB-16	Lifts	96.65	0.41	43.56
DB-10	Lighting	51.04	0.46	26.29
DB-29	Lifts	76.35	0.57	48.76

Building services comprise electrical distribution, water supply and sanitation, HVAC for the identified conditioned areas, medical gas pipeline and vertical transportation. Connected load and demand factors are given in the load schedule.

3 PROJECT BACKGROUND

3.1 Introduction — Nalbari district has a population of 7.7 lakh served by a single district hospital constructed in 1974 with a sanctioned strength of 60 beds. The block is a load-bearing masonry structure showing distress in the form of differential settlement and corrosion of embedded steel.

3.2 Project Objective — to provide a 200-bed secondary care facility meeting Indian Public Health Standards for a district hospital, with diagnostic services presently unavailable within the district.

3.3 Methodology — the requirement was assessed from five years of hospital records, referral registers and the district health profile. Space programming follows the Indian Public Health Standards 2022 for district hospitals.

3.4 Overview of the Project Area — the site is 3.20 hectares of departmental land adjoining the existing hospital campus, level and with existing access from the district road.

3.x Project Background — Working Papers — Sheet 1

Building services comprise electrical distribution, water supply and sanitation, HVAC for the identified conditioned areas, medical gas pipeline and vertical transportation. Connected load and demand factors are given in the load schedule.

Quantities have been computed from the approved architectural and structural drawings and cross-checked against the detailed measurement sheets enclosed at the Annexure. Rates are as per the schedule of rates in force with lead and lift adjustments.

Consultations were held with the head of each clinical department on adjacency, patient flow and equipment layout. Minutes of the consultation meetings are placed at the Annexure to this chapter.

Building services comprise electrical distribution, water supply and sanitation, HVAC for the identified conditioned areas, medical gas pipeline and vertical transportation. Connected load and demand factors are given in the load schedule.

Reference: drawing no. PH-526, revision C.

3.x Project Background — Working Papers — Sheet 2

Departmental Area Statement

Department	Floor	Units	Area per unit (sqm)	Total (sqm)
ICU	First	7	101.33	709.33
Administration	Second	15	65.28	979.16
Pharmacy	Third	13	40.26	523.39
ICU	Third	2	52.08	104.16
Administration	Ground	14	20.20	282.78
Staff Quarters	First	10	24.71	247.10
OPD	Ground	11	42.45	466.94
Diagnostics	Third	14	92.25	1291.52
IPD Ward	First	16	122.19	1955.11
Staff Quarters	Ground	1	33.57	33.57
Staff Quarters	First	12	100.65	1207.82
Administration	Third	5	86.35	431.76
ICU	First	1	133.66	133.66
Diagnostics	Second	5	80.48	402.38
Pharmacy	Ground	2	69.91	139.83
Pharmacy	First	4	123.76	495.04
Staff Quarters	Ground	7	87.97	615.78
ICU	Ground	17	63.09	1072.54
IPD Ward	Third	10	44.90	448.98
IPD Ward	Third	5	36.23	181.16
OPD	First	16	47.21	755.28
IPD Ward	Ground	8	64.99	519.94
Diagnostics	Second	2	80.14	160.29
Diagnostics	Second	7	126.85	887.98
Diagnostics	First	6	64.60	387.60
Pharmacy	Third	13	77.71	1010.20

Soil investigation was carried out by rotary drilling at the locations shown on the layout plan. The safe bearing capacity adopted for design is based on the lower of the shear and settlement criteria as reported in the geotechnical report.

3.x Project Background — Working Papers — Sheet 3

Design Computation

Deflection = 74.821 × 6.572 = 491.697 mm   (permissible 900.594) — safe

Shear = 59.767 × 9.786 = 584.860 kN   (permissible 874.871) — safe

Deflection = 10.902 × 10.942 = 119.288 mm   (permissible 159.728) — safe

Moment = 54.403 × 3.118 = 169.601 kNm   (permissible 286.267) — safe

Axial load = 74.862 × 3.949 = 295.606 kN   (permissible 464.585) — safe

Bearing pressure = 52.128 × 5.993 = 312.389 kPa   (permissible 419.375) — safe

Moment = 26.762 × 1.259 = 33.701 kNm   (permissible 63.572) — safe

Deflection = 2.140 × 7.779 = 16.647 mm   (permissible 21.573) — safe

Building services comprise electrical distribution, water supply and sanitation, HVAC for the identified conditioned areas, medical gas pipeline and vertical transportation. Connected load and demand factors are given in the load schedule.

3.x Project Background — Working Papers — Sheet 4

Fire safety provisions follow Part 4 of the National Building Code. Means of egress, travel distances, staircase widths and the fire fighting installations have been designed for the occupancy classification of the block.

Consultations were held with the head of each clinical department on adjacency, patient flow and equipment layout. Minutes of the consultation meetings are placed at the Annexure to this chapter.

The space programme has been developed from the functional brief agreed with the user department. Departmental areas, circulation and the gross-to-carpet ratio are tabulated in the area statement enclosed at the Annexure.

The structural design conforms to IS 456 for reinforced concrete and IS 1893 for earthquake resistant design. The site falls in seismic zone V and the importance factor applicable to a hospital building has been adopted.

Reference: drawing no. AR-509, revision B.

3.x Project Background — Working Papers — Sheet 5

Departmental Area Statement

Department	Floor	Units	Area per unit (sqm)	Total (sqm)
Pharmacy	Third	9	43.88	394.91
Administration	Third	2	98.30	196.60
Administration	Third	5	118.07	590.35
Administration	First	15	68.59	1028.87
Staff Quarters	First	8	105.30	842.42
ICU	Second	15	44.62	669.29
Diagnostics	Third	10	49.44	494.39
ICU	First	16	122.47	1959.56
ICU	Third	9	31.22	281.02
Pharmacy	Second	8	92.00	736.02
OPD	Third	17	101.51	1725.71
OPD	Ground	9	39.90	359.06
OPD	First	4	38.70	154.80
Staff Quarters	Ground	15	117.25	1758.68
Staff Quarters	Third	5	74.47	372.34
Operation Theatre	Second	13	16.37	212.86
Administration	Ground	8	111.55	892.44
ICU	Ground	17	18.21	309.60
Diagnostics	Ground	14	42.17	590.38

Building services comprise electrical distribution, water supply and sanitation, HVAC for the identified conditioned areas, medical gas pipeline and vertical transportation. Connected load and demand factors are given in the load schedule.

3.x Project Background — Working Papers — Sheet 6

Design Computation

Axial load = 62.168 × 1.124 = 69.878 kN   (permissible 85.935) — revise section

Shear = 2.695 × 6.030 = 16.251 kN   (permissible 29.633) — safe

Bearing pressure = 46.493 × 11.901 = 553.310 kPa   (permissible 663.810) — safe

Deflection = 3.212 × 7.632 = 24.511 mm   (permissible 46.515) — safe

Shear = 56.030 × 6.316 = 353.878 kN   (permissible 446.817) — safe

Axial load = 42.809 × 10.149 = 434.456 kN   (permissible 462.622) — safe

Soil investigation was carried out by rotary drilling at the locations shown on the layout plan. The safe bearing capacity adopted for design is based on the lower of the shear and settlement criteria as reported in the geotechnical report.

3.x Project Background — Working Papers — Sheet 7

Barrier-free access has been provided in accordance with the Harmonised Guidelines and Standards for Universal Accessibility. Ramps, lifts, signage and accessible toilets are provided on every floor of the block.

Building services comprise electrical distribution, water supply and sanitation, HVAC for the identified conditioned areas, medical gas pipeline and vertical transportation. Connected load and demand factors are given in the load schedule.

Fire safety provisions follow Part 4 of the National Building Code. Means of egress, travel distances, staircase widths and the fire fighting installations have been designed for the occupancy classification of the block.

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Reference: drawing no. PH-148, revision C.

3.x Project Background — Working Papers — Sheet 8

Bore Log Summary

Depth (m)	Strata	N value	Sample
13.9	Silty clay	7	SPT
18.8	Weathered rock	53	DS
2.4	Weathered rock	42	UDS
3.3	Clayey silt	16	SPT
5.2	Gravelly sand	41	UDS
6.2	Coarse sand	17	SPT
14.9	Silty clay	6	DS
24.3	Clayey silt	60	DS
5.6	Medium sand	11	UDS
24.1	Coarse sand	60	DS
12.4	Medium sand	29	DS
3.5	Weathered rock	33	DS
8.6	Coarse sand	34	UDS
18.0	Gravelly sand	49	SPT
18.8	Medium sand	33	DS
9.9	Silty clay	24	SPT
29.9	Clayey silt	22	DS
0.8	Coarse sand	17	DS
21.3	Silty clay	45	DS
25.3	Medium sand	57	SPT
22.9	Gravelly sand	10	DS
23.6	Gravelly sand	48	UDS
21.2	Gravelly sand	50	UDS
2.7	Coarse sand	38	UDS
7.3	Medium sand	23	UDS

Soil investigation was carried out by rotary drilling at the locations shown on the layout plan. The safe bearing capacity adopted for design is based on the lower of the shear and settlement criteria as reported in the geotechnical report.

4 PROJECT FEASIBILITY STUDIES

4.1 Requirement / Demand Analysis — average bed occupancy over the last three years is 141 per cent against the sanctioned 60 beds. Outpatient attendance has grown from 218 to 396 per day over five years. The district referred 3,412 cases to tertiary facilities in Guwahati in the last year, of which an assessed 61 per cent could have been managed locally with the proposed diagnostic and surgical capacity.

4.2 Existing Situation Assessment — the structural assessment carried out by the Assam Engineering College concluded that retrofitting the existing block would cost 68 per cent of replacement value with a residual life of 15 years, and recommended replacement.

4.3 Stakeholders Consultation — consultations were held with the District Health Society, the Indian Medical Association district branch and three resident welfare associations. Minutes are at Annexure VII.

4.4 Environmental and Sustainability Aspects — the project is a building project below the threshold requiring prior environmental clearance under the EIA Notification 2006 as amended. Consent to establish has been obtained from the Assam Pollution Control Board. Biomedical waste will be handled through the existing common facility. The design provides for rainwater harvesting across 14,200 sqm of roof area and a 180 kWp rooftop solar installation.

4.x Project Feasibility Studies — Working Papers — Sheet 1

Consultations were held with the head of each clinical department on adjacency, patient flow and equipment layout. Minutes of the consultation meetings are placed at the Annexure to this chapter.

Fire safety provisions follow Part 4 of the National Building Code. Means of egress, travel distances, staircase widths and the fire fighting installations have been designed for the occupancy classification of the block.

Consultations were held with the head of each clinical department on adjacency, patient flow and equipment layout. Minutes of the consultation meetings are placed at the Annexure to this chapter.

Barrier-free access has been provided in accordance with the Harmonised Guidelines and Standards for Universal Accessibility. Ramps, lifts, signage and accessible toilets are provided on every floor of the block.

Reference: drawing no. ST-471, revision A.

4.x Project Feasibility Studies — Working Papers — Sheet 2

Equipment Schedule

Item	Department	Nos	Unit rate (Rs)	Amount (Rs)
X-ray unit	Casualty	2	1,654,313	3,308,626
Autoclave	Ward	30	1,982,525	59,475,750
Ventilator	Casualty	13	971,669	12,631,697
Ultrasound	Radiology	29	2,200,896	63,825,984
Autoclave	ICU	40	1,691,543	67,661,720
X-ray unit	Ward	5	305,141	1,525,705
Hospital bed	ICU	26	383,771	9,978,046
Patient monitor	ICU	16	1,088,263	17,412,208
Ventilator	Ward	17	1,841,105	31,298,785
Ventilator	CSSD	36	1,846,235	66,464,460
X-ray unit	CSSD	40	435,643	17,425,720
Ventilator	Radiology	17	1,003,516	17,059,772
Ventilator	Casualty	26	993,377	25,827,802
OT table	CSSD	5	1,338,370	6,691,850
Ultrasound	ICU	7	2,306,843	16,147,901
Patient monitor	CSSD	31	441,234	13,678,254
Autoclave	ICU	33	702,181	23,171,973
Hospital bed	Casualty	1	2,182,198	2,182,198
Autoclave	Radiology	22	1,575,588	34,662,936
X-ray unit	Casualty	31	1,404,047	43,525,457
OT table	CSSD	20	368,500	7,370,000
Defibrillator	ICU	9	1,747,179	15,724,611
Defibrillator	CSSD	9	1,370,748	12,336,732

Building services comprise electrical distribution, water supply and sanitation, HVAC for the identified conditioned areas, medical gas pipeline and vertical transportation. Connected load and demand factors are given in the load schedule.

4.x Project Feasibility Studies — Working Papers — Sheet 3

Design Computation

Moment = 29.943 × 4.662 = 139.579 kNm   (permissible 197.876) — safe

Deflection = 70.370 × 0.876 = 61.619 mm   (permissible 105.650) — safe

Deflection = 84.750 × 9.975 = 845.422 mm   (permissible 1491.830) — safe

Axial load = 84.693 × 3.654 = 309.465 kN   (permissible 546.348) — revise section

Axial load = 27.627 × 2.416 = 66.754 kN   (permissible 120.926) — safe

Bearing pressure = 54.978 × 9.074 = 498.892 kPa   (permissible 898.083) — safe

Shear = 21.370 × 7.135 = 152.487 kN   (permissible 229.362) — safe

Shear = 76.065 × 3.716 = 282.637 kN   (permissible 345.727) — safe

Building services comprise electrical distribution, water supply and sanitation, HVAC for the identified conditioned areas, medical gas pipeline and vertical transportation. Connected load and demand factors are given in the load schedule.

4.x Project Feasibility Studies — Working Papers — Sheet 4

Quantities have been computed from the approved architectural and structural drawings and cross-checked against the detailed measurement sheets enclosed at the Annexure. Rates are as per the schedule of rates in force with lead and lift adjustments.

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Soil investigation was carried out by rotary drilling at the locations shown on the layout plan. The safe bearing capacity adopted for design is based on the lower of the shear and settlement criteria as reported in the geotechnical report.

Consultations were held with the head of each clinical department on adjacency, patient flow and equipment layout. Minutes of the consultation meetings are placed at the Annexure to this chapter.

Reference: drawing no. ST-523, revision B.

4.x Project Feasibility Studies — Working Papers — Sheet 5

Equipment Schedule

Item	Department	Nos	Unit rate (Rs)	Amount (Rs)
OT table	OT	18	20,216	363,888
X-ray unit	Casualty	4	2,326,588	9,306,352
X-ray unit	Radiology	21	656,172	13,779,612
Defibrillator	ICU	27	1,782,439	48,125,853
Patient monitor	Casualty	30	806,168	24,185,040
OT table	Ward	9	2,367,315	21,305,835
Defibrillator	Casualty	4	983,007	3,932,028
X-ray unit	ICU	7	282,445	1,977,115
Defibrillator	OT	24	236,508	5,676,192
OT table	Casualty	33	1,369,428	45,191,124
X-ray unit	Ward	21	1,971,286	41,397,006
X-ray unit	OT	8	299,676	2,397,408
OT table	Radiology	21	2,327,809	48,883,989
OT table	Casualty	15	195,818	2,937,270
Defibrillator	OT	12	832,228	9,986,736
X-ray unit	CSSD	12	334,075	4,008,900

Fire safety provisions follow Part 4 of the National Building Code. Means of egress, travel distances, staircase widths and the fire fighting installations have been designed for the occupancy classification of the block.

4.x Project Feasibility Studies — Working Papers — Sheet 6

Design Computation

Deflection = 78.553 × 0.540 = 42.413 mm   (permissible 65.034) — safe

Moment = 18.542 × 4.954 = 91.866 kNm   (permissible 104.330) — safe

Axial load = 26.630 × 8.623 = 229.628 kN   (permissible 317.398) — safe

Deflection = 29.395 × 8.076 = 237.393 mm   (permissible 397.322) — safe

Shear = 88.939 × 9.510 = 845.829 kN   (permissible 1454.157) — safe

Deflection = 29.565 × 5.858 = 173.196 mm   (permissible 295.436) — safe

Deflection = 38.433 × 7.034 = 270.331 mm   (permissible 435.865) — safe

Shear = 55.679 × 10.623 = 591.471 kN   (permissible 772.419) — safe

The structural design conforms to IS 456 for reinforced concrete and IS 1893 for earthquake resistant design. The site falls in seismic zone V and the importance factor applicable to a hospital building has been adopted.

4.x Project Feasibility Studies — Working Papers — Sheet 7

Quantities have been computed from the approved architectural and structural drawings and cross-checked against the detailed measurement sheets enclosed at the Annexure. Rates are as per the schedule of rates in force with lead and lift adjustments.

Building services comprise electrical distribution, water supply and sanitation, HVAC for the identified conditioned areas, medical gas pipeline and vertical transportation. Connected load and demand factors are given in the load schedule.

Consultations were held with the head of each clinical department on adjacency, patient flow and equipment layout. Minutes of the consultation meetings are placed at the Annexure to this chapter.

Fire safety provisions follow Part 4 of the National Building Code. Means of egress, travel distances, staircase widths and the fire fighting installations have been designed for the occupancy classification of the block.

Reference: drawing no. PH-960, revision B.

4.x Project Feasibility Studies — Working Papers — Sheet 8

Bore Log Summary

Depth (m)	Strata	N value	Sample
7.5	Gravelly sand	50	SPT
20.6	Silty clay	32	DS
29.3	Gravelly sand	38	UDS
13.7	Coarse sand	43	UDS
12.3	Weathered rock	33	SPT
21.0	Silty clay	28	UDS
27.7	Silty clay	43	SPT
26.6	Coarse sand	29	DS
17.2	Clayey silt	53	DS
14.2	Coarse sand	13	SPT
19.7	Coarse sand	46	SPT
1.5	Silty clay	4	DS
15.4	Medium sand	52	SPT
20.9	Clayey silt	47	DS
16.7	Silty clay	34	SPT
9.6	Silty clay	39	DS
28.7	Gravelly sand	46	SPT
10.2	Gravelly sand	34	UDS
21.7	Weathered rock	29	DS

Barrier-free access has been provided in accordance with the Harmonised Guidelines and Standards for Universal Accessibility. Ramps, lifts, signage and accessible toilets are provided on every floor of the block.

4.x Project Feasibility Studies — Working Papers — Sheet 9

Design Computation

Shear = 61.884 × 2.910 = 180.058 kN   (permissible 299.235) — safe

Bearing pressure = 64.300 × 1.084 = 69.677 kPa   (permissible 91.813) — safe

Axial load = 77.882 × 11.307 = 880.644 kN   (permissible 1168.197) — safe

Axial load = 70.511 × 6.822 = 481.003 kN   (permissible 685.096) — safe

Deflection = 79.832 × 9.820 = 783.953 mm   (permissible 1130.963) — safe

Fire safety provisions follow Part 4 of the National Building Code. Means of egress, travel distances, staircase widths and the fire fighting installations have been designed for the occupancy classification of the block.

4.x Project Feasibility Studies — Working Papers — Sheet 10

The space programme has been developed from the functional brief agreed with the user department. Departmental areas, circulation and the gross-to-carpet ratio are tabulated in the area statement enclosed at the Annexure.

Barrier-free access has been provided in accordance with the Harmonised Guidelines and Standards for Universal Accessibility. Ramps, lifts, signage and accessible toilets are provided on every floor of the block.

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Reference: drawing no. ST-771, revision A.

4.x Project Feasibility Studies — Working Papers — Sheet 11

Electrical Load Schedule

Panel	Description	Connected load (kW)	Demand factor	Demand (kVA)
DB-32	Sterilisation	173.72	0.88	170.55
DB-05	HVAC	154.36	0.66	112.86
DB-01	Lifts	115.41	0.81	103.38
DB-05	Sterilisation	62.96	0.41	28.56
DB-22	Lighting	141.23	0.48	74.83
DB-28	HVAC	166.40	0.53	97.18
DB-14	HVAC	163.26	0.71	128.96
DB-33	Lighting	74.56	0.57	47.51
DB-12	HVAC	118.17	0.94	124.06
DB-31	Sterilisation	79.06	0.49	43.37
DB-11	Pumps	92.18	0.88	90.31
DB-02	Sterilisation	35.48	0.54	21.14
DB-07	HVAC	122.36	0.65	88.00
DB-16	HVAC	49.97	0.79	43.97
DB-34	Lighting	27.60	0.77	23.66
DB-08	Lifts	34.02	0.61	23.07
DB-21	Power outlets	132.70	0.70	103.41
DB-15	Medical equipment	154.72	0.78	133.47
DB-01	Power outlets	99.57	0.74	82.18
DB-33	Pumps	117.19	0.60	78.76
DB-19	Medical equipment	88.46	0.62	60.73

Consultations were held with the head of each clinical department on adjacency, patient flow and equipment layout. Minutes of the consultation meetings are placed at the Annexure to this chapter.

4.x Project Feasibility Studies — Working Papers — Sheet 12

Design Computation

Shear = 75.121 × 1.300 = 97.637 kN   (permissible 120.139) — safe

Moment = 33.108 × 10.465 = 346.467 kNm   (permissible 536.567) — safe

Moment = 78.973 × 5.026 = 396.913 kNm   (permissible 534.661) — safe

Bearing pressure = 53.446 × 10.112 = 540.452 kPa   (permissible 768.414) — safe

Deflection = 9.397 × 9.464 = 88.931 mm   (permissible 106.065) — safe

Deflection = 65.121 × 10.416 = 678.289 mm   (permissible 1016.482) — safe

Building services comprise electrical distribution, water supply and sanitation, HVAC for the identified conditioned areas, medical gas pipeline and vertical transportation. Connected load and demand factors are given in the load schedule.

5 SITE SURVEYS AND INVESTIGATIONS

5.1 Ocular / Reconnaissance Survey — carried out in January 2026. The site is level with a fall of 0.8 m across its length and no waterlogging observed.

5.2 Topographical Survey — total station survey on a 10 m grid, connected to the GTS benchmark at Nalbari town, reduced level 47.216 m.

5.3 Soil Investigation — nine boreholes to 20 m depth. Safe bearing capacity assessed at 180 kN/sqm at 2.5 m founding depth. Ground water encountered at 4.2 m below existing ground level.

5.4 Hydro-Geological Study — the aquifer is unconfined at 12 to 18 m. A borewell yield of 42,000 litres per day has been assessed against a requirement of 168,000 litres per day, the balance to be drawn from the municipal supply.

5.5 Primary Surveys — utility mapping identified an 11 kV overhead line crossing the northern boundary, to be shifted under a deposit work.

5.x Site Surveys And Investigations — Working Papers — Sheet 1

The space programme has been developed from the functional brief agreed with the user department. Departmental areas, circulation and the gross-to-carpet ratio are tabulated in the area statement enclosed at the Annexure.

Fire safety provisions follow Part 4 of the National Building Code. Means of egress, travel distances, staircase widths and the fire fighting installations have been designed for the occupancy classification of the block.

The space programme has been developed from the functional brief agreed with the user department. Departmental areas, circulation and the gross-to-carpet ratio are tabulated in the area statement enclosed at the Annexure.

Consultations were held with the head of each clinical department on adjacency, patient flow and equipment layout. Minutes of the consultation meetings are placed at the Annexure to this chapter.

Reference: drawing no. EL-344, revision B.

5.x Site Surveys And Investigations — Working Papers — Sheet 2

Room Data Sheet

Room	Department	Carpet area (sqm)	Occupancy	Floor finish
Room 408	Surgery	58.68	16	Kota stone
Room 409	Radiology	37.12	7	Epoxy screed
Room 493	Paediatrics	69.65	2	Ceramic tile
Room 460	Surgery	89.34	25	Antistatic vinyl
Room 720	Paediatrics	62.66	22	Epoxy screed
Room 268	Paediatrics	42.75	9	Antistatic vinyl
Room 297	Obstetrics	10.33	15	Antistatic vinyl
Room 135	Obstetrics	56.31	24	Kota stone
Room 642	Casualty	60.13	5	Antistatic vinyl
Room 129	Casualty	22.92	16	Vitrified tile
Room 628	General Medicine	51.94	20	Vitrified tile
Room 319	Paediatrics	54.05	28	Ceramic tile
Room 415	Surgery	13.22	15	Vitrified tile
Room 806	Obstetrics	58.24	8	Ceramic tile
Room 341	Radiology	25.66	21	Antistatic vinyl
Room 384	Pathology	34.70	8	Vitrified tile
Room 751	Surgery	81.89	25	Antistatic vinyl

Consultations were held with the head of each clinical department on adjacency, patient flow and equipment layout. Minutes of the consultation meetings are placed at the Annexure to this chapter.

5.x Site Surveys And Investigations — Working Papers — Sheet 3

Design Computation

Axial load = 58.578 × 2.759 = 161.635 kN   (permissible 219.888) — safe

Deflection = 2.373 × 5.153 = 12.225 mm   (permissible 17.196) — safe

Bearing pressure = 32.276 × 4.745 = 153.160 kPa   (permissible 219.686) — safe

Bearing pressure = 80.947 × 6.994 = 566.145 kPa   (permissible 675.200) — safe

Deflection = 74.275 × 3.154 = 234.281 mm   (permissible 286.418) — safe

Bearing pressure = 18.963 × 2.482 = 47.072 kPa   (permissible 88.276) — safe

Fire safety provisions follow Part 4 of the National Building Code. Means of egress, travel distances, staircase widths and the fire fighting installations have been designed for the occupancy classification of the block.

5.x Site Surveys And Investigations — Working Papers — Sheet 4

Fire safety provisions follow Part 4 of the National Building Code. Means of egress, travel distances, staircase widths and the fire fighting installations have been designed for the occupancy classification of the block.

Consultations were held with the head of each clinical department on adjacency, patient flow and equipment layout. Minutes of the consultation meetings are placed at the Annexure to this chapter.

Barrier-free access has been provided in accordance with the Harmonised Guidelines and Standards for Universal Accessibility. Ramps, lifts, signage and accessible toilets are provided on every floor of the block.

The space programme has been developed from the functional brief agreed with the user department. Departmental areas, circulation and the gross-to-carpet ratio are tabulated in the area statement enclosed at the Annexure.

Reference: drawing no. EL-976, revision C.

5.x Site Surveys And Investigations — Working Papers — Sheet 5

Equipment Schedule

Item	Department	Nos	Unit rate (Rs)	Amount (Rs)
X-ray unit	Ward	38	2,064,083	78,435,154
Hospital bed	Radiology	32	1,140,431	36,493,792
Autoclave	CSSD	24	446,106	10,706,544
Hospital bed	CSSD	10	901,929	9,019,290
Patient monitor	CSSD	17	622,170	10,576,890
Ultrasound	CSSD	19	198,302	3,767,738
Ventilator	Casualty	23	1,192,291	27,422,693
Autoclave	OT	17	1,407,604	23,929,268
X-ray unit	OT	1	430,645	430,645
Autoclave	Casualty	27	1,769,606	47,779,362
OT table	OT	36	1,509,331	54,335,916
Ventilator	Radiology	11	1,743,501	19,178,511
Autoclave	Ward	27	1,964,248	53,034,696
Ventilator	OT	18	78,390	1,411,020
Hospital bed	Casualty	4	382,331	1,529,324
Defibrillator	Radiology	7	1,589,520	11,126,640
Ultrasound	CSSD	16	56,347	901,552
Ventilator	Ward	28	2,157,092	60,398,576
Patient monitor	Ward	33	756,978	24,980,274
Defibrillator	Radiology	33	1,037,803	34,247,499
OT table	Ward	36	1,577,353	56,784,708
Hospital bed	CSSD	40	2,245,350	89,814,000
Ventilator	Ward	29	373,739	10,838,431
Hospital bed	Casualty	9	205,560	1,850,040
Patient monitor	Casualty	33	1,280,250	42,248,250

Building services comprise electrical distribution, water supply and sanitation, HVAC for the identified conditioned areas, medical gas pipeline and vertical transportation. Connected load and demand factors are given in the load schedule.

5.x Site Surveys And Investigations — Working Papers — Sheet 6

Design Computation

Bearing pressure = 70.038 × 2.597 = 181.872 kPa   (permissible 220.927) — safe

Moment = 52.527 × 1.368 = 71.877 kNm   (permissible 101.020) — safe

Bearing pressure = 39.651 × 5.163 = 204.737 kPa   (permissible 256.043) — safe

Deflection = 69.404 × 3.467 = 240.648 mm   (permissible 410.639) — safe

Moment = 81.280 × 7.161 = 582.068 kNm   (permissible 941.435) — safe

Consultations were held with the head of each clinical department on adjacency, patient flow and equipment layout. Minutes of the consultation meetings are placed at the Annexure to this chapter.

5.x Site Surveys And Investigations — Working Papers — Sheet 7

Quantities have been computed from the approved architectural and structural drawings and cross-checked against the detailed measurement sheets enclosed at the Annexure. Rates are as per the schedule of rates in force with lead and lift adjustments.

Soil investigation was carried out by rotary drilling at the locations shown on the layout plan. The safe bearing capacity adopted for design is based on the lower of the shear and settlement criteria as reported in the geotechnical report.

Soil investigation was carried out by rotary drilling at the locations shown on the layout plan. The safe bearing capacity adopted for design is based on the lower of the shear and settlement criteria as reported in the geotechnical report.

Barrier-free access has been provided in accordance with the Harmonised Guidelines and Standards for Universal Accessibility. Ramps, lifts, signage and accessible toilets are provided on every floor of the block.

Reference: drawing no. PH-193, revision C.

5.x Site Surveys And Investigations — Working Papers — Sheet 8

Detailed Measurement Sheet

Item	L (m)	B (m)	D (m)	Nos	Quantity
Item 214	20.95	4.04	1.84	18	11.777
Item 233	16.83	3.43	0.20	20	90.332
Item 850	27.26	0.70	1.73	10	289.360
Item 610	34.10	4.45	2.54	19	90.045
Item 376	39.51	5.28	0.81	4	310.483
Item 601	23.01	1.60	2.98	17	45.186
Item 141	6.89	0.39	1.64	10	186.010
Item 692	6.50	5.61	2.99	24	327.737
Item 877	27.16	1.80	0.23	22	26.875
Item 163	15.86	2.38	0.44	3	34.105
Item 593	17.28	4.41	2.68	1	96.732
Item 850	10.23	1.45	1.61	24	159.386
Item 211	4.79	4.10	0.97	24	12.056
Item 623	12.80	2.79	2.08	6	288.001
Item 752	13.90	4.94	2.38	24	262.099
Item 361	16.53	3.52	0.51	21	342.204
Item 643	36.15	2.98	2.11	23	316.991
Item 180	35.50	4.69	2.17	14	16.368
Item 645	26.21	2.54	1.55	9	133.667
Item 158	5.91	1.80	1.21	20	218.867
Item 354	11.35	0.84	2.16	6	87.845
Item 162	5.93	4.41	2.60	11	178.512
Item 106	9.48	3.15	0.95	7	329.804
Item 564	3.97	0.67	2.07	3	201.616

The space programme has been developed from the functional brief agreed with the user department. Departmental areas, circulation and the gross-to-carpet ratio are tabulated in the area statement enclosed at the Annexure.

5.x Site Surveys And Investigations — Working Papers — Sheet 9

Design Computation

Deflection = 20.831 × 11.043 = 230.030 mm   (permissible 280.170) — safe

Bearing pressure = 85.856 × 6.226 = 534.566 kPa   (permissible 773.915) — safe

Axial load = 24.203 × 8.078 = 195.514 kN   (permissible 285.235) — safe

Moment = 86.413 × 9.380 = 810.567 kNm   (permissible 1053.125) — safe

Deflection = 31.377 × 9.530 = 299.039 mm   (permissible 542.177) — safe

Axial load = 58.198 × 6.117 = 356.000 kN   (permissible 655.043) — safe

Bearing pressure = 78.340 × 5.962 = 467.058 kPa   (permissible 848.108) — safe

Bearing pressure = 34.992 × 10.552 = 369.248 kPa   (permissible 637.947) — safe

The space programme has been developed from the functional brief agreed with the user department. Departmental areas, circulation and the gross-to-carpet ratio are tabulated in the area statement enclosed at the Annexure.

5.x Site Surveys And Investigations — Working Papers — Sheet 10

Fire safety provisions follow Part 4 of the National Building Code. Means of egress, travel distances, staircase widths and the fire fighting installations have been designed for the occupancy classification of the block.

Fire safety provisions follow Part 4 of the National Building Code. Means of egress, travel distances, staircase widths and the fire fighting installations have been designed for the occupancy classification of the block.

The space programme has been developed from the functional brief agreed with the user department. Departmental areas, circulation and the gross-to-carpet ratio are tabulated in the area statement enclosed at the Annexure.

The structural design conforms to IS 456 for reinforced concrete and IS 1893 for earthquake resistant design. The site falls in seismic zone V and the importance factor applicable to a hospital building has been adopted.

Reference: drawing no. EL-495, revision A.

5.x Site Surveys And Investigations — Working Papers — Sheet 11

Electrical Load Schedule

Panel	Description	Connected load (kW)	Demand factor	Demand (kVA)
DB-04	Power outlets	33.63	0.52	19.43
DB-17	HVAC	150.27	0.66	109.52
DB-26	Medical equipment	151.11	0.87	146.38
DB-40	Pumps	143.38	0.47	75.04
DB-15	Pumps	29.17	0.64	20.63
DB-02	Lifts	84.57	0.82	77.12
DB-32	Lifts	27.78	0.54	16.66
DB-07	Medical equipment	7.99	0.88	7.83
DB-37	HVAC	141.36	0.75	117.75
DB-09	Lifts	126.85	0.67	95.07
DB-15	HVAC	111.87	0.78	97.22
DB-01	Medical equipment	9.70	0.83	8.99
DB-16	Medical equipment	5.89	0.93	6.09
DB-04	Pumps	140.30	0.80	124.08
DB-16	Sterilisation	59.83	0.73	48.34
DB-40	Sterilisation	92.60	0.48	48.90
DB-20	Sterilisation	167.43	0.76	141.70
DB-12	HVAC	156.99	0.80	140.30
DB-11	Lifts	48.09	0.85	45.50
DB-40	Lighting	100.66	0.46	50.91
DB-33	Sterilisation	6.33	0.80	5.66

The structural design conforms to IS 456 for reinforced concrete and IS 1893 for earthquake resistant design. The site falls in seismic zone V and the importance factor applicable to a hospital building has been adopted.

5.x Site Surveys And Investigations — Working Papers — Sheet 12

Design Computation

Shear = 2.809 × 1.998 = 5.612 kN   (permissible 10.312) — safe

Deflection = 39.529 × 2.156 = 85.223 mm   (permissible 134.217) — safe

Axial load = 40.494 × 3.173 = 128.492 kN   (permissible 185.345) — revise section

Bearing pressure = 76.401 × 4.959 = 378.894 kPa   (permissible 419.451) — safe

Moment = 37.457 × 11.618 = 435.160 kNm   (permissible 658.701) — safe

Deflection = 38.598 × 0.612 = 23.623 mm   (permissible 39.760) — safe

Bearing pressure = 45.599 × 0.683 = 31.137 kPa   (permissible 41.841) — safe

Consultations were held with the head of each clinical department on adjacency, patient flow and equipment layout. Minutes of the consultation meetings are placed at the Annexure to this chapter.

5.x Site Surveys And Investigations — Working Papers — Sheet 13

Fire safety provisions follow Part 4 of the National Building Code. Means of egress, travel distances, staircase widths and the fire fighting installations have been designed for the occupancy classification of the block.

Soil investigation was carried out by rotary drilling at the locations shown on the layout plan. The safe bearing capacity adopted for design is based on the lower of the shear and settlement criteria as reported in the geotechnical report.

Quantities have been computed from the approved architectural and structural drawings and cross-checked against the detailed measurement sheets enclosed at the Annexure. Rates are as per the schedule of rates in force with lead and lift adjustments.

Consultations were held with the head of each clinical department on adjacency, patient flow and equipment layout. Minutes of the consultation meetings are placed at the Annexure to this chapter.

Reference: drawing no. AR-920, revision B.

5.x Site Surveys And Investigations — Working Papers — Sheet 14

Detailed Measurement Sheet

Item	L (m)	B (m)	D (m)	Nos	Quantity
Item 377	27.26	1.52	0.29	22	155.386
Item 522	28.39	3.60	0.97	17	59.175
Item 365	35.56	4.25	0.43	9	112.067
Item 588	26.40	0.69	1.54	21	301.775
Item 776	18.93	2.28	1.91	8	183.360
Item 525	22.91	2.92	1.73	1	114.910
Item 545	2.79	1.74	2.78	16	229.467
Item 550	18.32	2.83	1.50	7	23.616
Item 130	4.25	5.39	2.12	18	181.426
Item 211	39.44	3.87	0.23	6	114.575
Item 275	28.69	0.64	1.88	3	33.758
Item 647	20.19	0.81	0.36	20	16.526
Item 479	17.00	1.84	2.94	5	308.214
Item 796	5.88	3.12	1.84	2	171.481
Item 447	7.38	0.91	0.68	16	59.008
Item 626	27.24	3.67	1.12	1	266.123
Item 299	33.22	0.93	1.09	2	83.110
Item 121	20.08	4.32	2.76	10	199.806
Item 573	28.16	3.16	1.13	14	140.646
Item 674	11.23	0.83	2.99	18	213.918
Item 310	19.94	4.60	2.68	2	286.049
Item 712	25.30	4.07	1.78	15	296.287
Item 633	33.82	5.51	1.25	12	317.345
Item 120	37.37	4.36	0.82	17	64.077

Building services comprise electrical distribution, water supply and sanitation, HVAC for the identified conditioned areas, medical gas pipeline and vertical transportation. Connected load and demand factors are given in the load schedule.

5.x Site Surveys And Investigations — Working Papers — Sheet 15

Design Computation

Shear = 9.318 × 6.046 = 56.337 kN (permissible 93.874) — safe

Axial load = 80.534 × 1.298 = 104.557 kN (permissible 123.975) — safe

Deflection = 21.600 × 10.654 = 230.131 mm (permissible 353.338) — safe

Axial load = 29.552 × 3.901 = 115.293 kN (permissible 122.985) — safe

Axial load = 58.091 × 5.021 = 291.681 kN (permissible 352.934) — safe

Shear = 18.774 × 10.456 = 196.298 kN (permissible 239.033) — safe

Soil investigation was carried out by rotary drilling at the locations shown on the layout plan. The safe bearing capacity adopted for design is based on the lower of the shear and settlement criteria as reported in the geotechnical report.

5.x Site Surveys And Investigations — Working Papers — Sheet 16

Barrier-free access has been provided in accordance with the Harmonised Guidelines and Standards for Universal Accessibility. Ramps, lifts, signage and accessible toilets are provided on every floor of the block.

Quantities have been computed from the approved architectural and structural drawings and cross-checked against the detailed measurement sheets enclosed at the Annexure. Rates are as per the schedule of rates in force with lead and lift adjustments.

The space programme has been developed from the functional brief agreed with the user department. Departmental areas, circulation and the gross-to-carpet ratio are tabulated in the area statement enclosed at the Annexure.

Consultations were held with the head of each clinical department on adjacency, patient flow and equipment layout. Minutes of the consultation meetings are placed at the Annexure to this chapter.

Reference: drawing no. EL-997, revision B.

5.x Site Surveys And Investigations — Working Papers — Sheet 17

Departmental Area Statement

Department	Floor	Units	Area per unit (sqm)	Total (sqm)
Staff Quarters	Third	12	45.98	551.75
OPD	First	14	61.08	855.18
Operation Theatre	Ground	8	51.88	415.02
OPD	First	18	138.69	2496.50
Pharmacy	Second	4	98.20	392.80
Pharmacy	First	9	60.25	542.29
IPD Ward	First	10	25.26	252.62
ICU	First	4	111.08	444.33
IPD Ward	Third	15	46.05	690.72
Administration	Second	4	61.23	244.91
IPD Ward	Second	10	69.82	698.16
Operation Theatre	Ground	18	26.72	480.98
Staff Quarters	Second	13	90.39	1175.11
Pharmacy	First	17	112.98	1920.69
Operation Theatre	First	18	118.93	2140.77
Operation Theatre	Second	15	101.69	1525.28
Diagnostics	Ground	11	139.96	1539.52
Operation Theatre	Third	5	90.95	454.73
Staff Quarters	Second	8	128.76	1030.09
OPD	Third	15	137.27	2059.09
Operation Theatre	Third	6	90.76	544.58
Administration	Third	7	122.34	856.36
OPD	First	9	129.88	1168.93
OPD	Second	2	73.02	146.05
Diagnostics	Ground	5	116.54	582.72

Consultations were held with the head of each clinical department on adjacency, patient flow and equipment layout. Minutes of the consultation meetings are placed at the Annexure to this chapter.

5.x Site Surveys And Investigations — Working Papers — Sheet 18

Design Computation

Moment = 85.359 × 9.659 = 824.473 kNm   (permissible 1543.980) — safe

Bearing pressure = 16.073 × 9.980 = 160.406 kPa   (permissible 219.990) — safe

Bearing pressure = 61.452 × 5.291 = 325.166 kPa   (permissible 534.280) — safe

Shear = 82.526 × 8.664 = 714.967 kN   (permissible 1307.097) — safe

Bearing pressure = 32.861 × 3.119 = 102.507 kPa   (permissible 120.451) — safe

Axial load = 79.719 × 10.434 = 831.807 kN   (permissible 878.504) — safe

Soil investigation was carried out by rotary drilling at the locations shown on the layout plan. The safe bearing capacity adopted for design is based on the lower of the shear and settlement criteria as reported in the geotechnical report.

6 FUNCTIONAL DESIGN

Space programming follows Indian Public Health Standards 2022. Three massing options were examined. The proposed option separates outpatient, inpatient and diagnostic flows with distinct entries, places the operation theatre complex on the second floor directly above the emergency and diagnostic block, and keeps the service core on the north to limit heat gain.

6.x Functional Design — Working Papers — Sheet 1

Building services comprise electrical distribution, water supply and sanitation, HVAC for the identified conditioned areas, medical gas pipeline and vertical transportation. Connected load and demand factors are given in the load schedule.

Soil investigation was carried out by rotary drilling at the locations shown on the layout plan. The safe bearing capacity adopted for design is based on the lower of the shear and settlement criteria as reported in the geotechnical report.

Quantities have been computed from the approved architectural and structural drawings and cross-checked against the detailed measurement sheets enclosed at the Annexure. Rates are as per the schedule of rates in force with lead and lift adjustments.

Building services comprise electrical distribution, water supply and sanitation, HVAC for the identified conditioned areas, medical gas pipeline and vertical transportation. Connected load and demand factors are given in the load schedule.

Reference: drawing no. AR-257, revision C.

6.x Functional Design — Working Papers — Sheet 2

Equipment Schedule

Item	Department	Nos	Unit rate (Rs)	Amount (Rs)
OT table	CSSD	34	1,251,823	42,561,982
OT table	ICU	24	1,434,691	34,432,584
OT table	CSSD	30	1,675,631	50,268,930
Ultrasound	CSSD	34	1,282,318	43,598,812
X-ray unit	Radiology	28	1,947,338	54,525,464
Autoclave	Casualty	15	1,421,212	21,318,180
Ventilator	CSSD	7	350,634	2,454,438
Ventilator	CSSD	17	1,466,423	24,929,191
Ultrasound	ICU	13	2,029,076	26,377,988
OT table	Ward	34	603,464	20,517,776
X-ray unit	Radiology	22	275,329	6,057,238
Ultrasound	ICU	6	1,158,939	6,953,634
X-ray unit	Radiology	14	1,006,832	14,095,648
Patient monitor	Radiology	6	631,318	3,787,908
OT table	Ward	23	1,532,914	35,257,022
Ventilator	ICU	28	333,150	9,328,200
Hospital bed	CSSD	17	2,127,298	36,164,066
Hospital bed	Ward	32	174,530	5,584,960
Hospital bed	Radiology	23	1,145,209	26,339,807
Autoclave	CSSD	9	1,124,927	10,124,343
Patient monitor	Ward	35	1,896,074	66,362,590
OT table	OT	22	163,060	3,587,320
Defibrillator	ICU	33	1,160,207	38,286,831
OT table	Radiology	29	2,170,540	62,945,660
Defibrillator	OT	28	2,270,401	63,571,228
Ventilator	OT	30	2,159,649	64,789,470

Barrier-free access has been provided in accordance with the Harmonised Guidelines and Standards for Universal Accessibility. Ramps, lifts, signage and accessible toilets are provided on every floor of the block.

6.x Functional Design — Working Papers — Sheet 3

Design Computation

Moment = 66.267 × 10.419 = 690.443 kNm   (permissible 1206.167) — safe

Bearing pressure = 54.823 × 7.422 = 406.913 kPa   (permissible 648.661) — safe

Moment = 32.054 × 1.641 = 52.587 kNm   (permissible 63.065) — safe

Shear = 46.996 × 2.588 = 121.633 kN   (permissible 132.653) — revise section

Bearing pressure = 59.427 × 10.779 = 640.546 kPa   (permissible 1021.147) — safe

Fire safety provisions follow Part 4 of the National Building Code. Means of egress, travel distances, staircase widths and the fire fighting installations have been designed for the occupancy classification of the block.

6.x Functional Design — Working Papers — Sheet 4

Soil investigation was carried out by rotary drilling at the locations shown on the layout plan. The safe bearing capacity adopted for design is based on the lower of the shear and settlement criteria as reported in the geotechnical report.

The structural design conforms to IS 456 for reinforced concrete and IS 1893 for earthquake resistant design. The site falls in seismic zone V and the importance factor applicable to a hospital building has been adopted.

Building services comprise electrical distribution, water supply and sanitation, HVAC for the identified conditioned areas, medical gas pipeline and vertical transportation. Connected load and demand factors are given in the load schedule.

Barrier-free access has been provided in accordance with the Harmonised Guidelines and Standards for Universal Accessibility. Ramps, lifts, signage and accessible toilets are provided on every floor of the block.

Reference: drawing no. ST-548, revision B.

6.x Functional Design — Working Papers — Sheet 5

Equipment Schedule

Item	Department	Nos	Unit rate (Rs)	Amount (Rs)
Autoclave	Casualty	20	2,228,013	44,560,260
Defibrillator	Casualty	29	1,267,034	36,743,986
Ultrasound	Casualty	38	188,546	7,164,748
X-ray unit	ICU	13	1,850,589	24,057,657
Ventilator	Ward	6	469,339	2,816,034
Autoclave	Casualty	14	2,126,496	29,770,944
OT table	CSSD	35	1,103,339	38,616,865
X-ray unit	Radiology	31	393,816	12,208,296
Defibrillator	ICU	31	1,926,240	59,713,440
Ultrasound	ICU	13	1,910,892	24,841,596
OT table	CSSD	27	1,221,389	32,977,503
Patient monitor	Radiology	1	1,776,402	1,776,402
Ventilator	Casualty	32	414,545	13,265,440
OT table	Ward	35	35,784	1,252,440
OT table	Radiology	16	2,339,424	37,430,784
Hospital bed	OT	16	1,605,904	25,694,464
Defibrillator	OT	8	1,716,122	13,728,976

Quantities have been computed from the approved architectural and structural drawings and cross-checked against the detailed measurement sheets enclosed at the Annexure. Rates are as per the schedule of rates in force with lead and lift adjustments.

6.x Functional Design — Working Papers — Sheet 6

Design Computation

Moment = 14.274 × 10.611 = 151.456 kNm   (permissible 261.077) — safe

Axial load = 14.265 × 10.506 = 149.858 kN   (permissible 166.283) — safe

Moment = 42.320 × 1.023 = 43.281 kNm   (permissible 52.604) — safe

Axial load = 11.027 × 1.212 = 13.363 kN   (permissible 23.833) — safe

Deflection = 1.653 × 4.555 = 7.532 mm   (permissible 10.862) — safe

Axial load = 30.892 × 9.341 = 288.571 kN   (permissible 408.335) — safe

Deflection = 88.746 × 1.594 = 141.492 mm   (permissible 199.673) — revise section

Building services comprise electrical distribution, water supply and sanitation, HVAC for the identified conditioned areas, medical gas pipeline and vertical transportation. Connected load and demand factors are given in the load schedule.

6.x Functional Design — Working Papers — Sheet 7

Fire safety provisions follow Part 4 of the National Building Code. Means of egress, travel distances, staircase widths and the fire fighting installations have been designed for the occupancy classification of the block.

Consultations were held with the head of each clinical department on adjacency, patient flow and equipment layout. Minutes of the consultation meetings are placed at the Annexure to this chapter.

Consultations were held with the head of each clinical department on adjacency, patient flow and equipment layout. Minutes of the consultation meetings are placed at the Annexure to this chapter.

Building services comprise electrical distribution, water supply and sanitation, HVAC for the identified conditioned areas, medical gas pipeline and vertical transportation. Connected load and demand factors are given in the load schedule.

Reference: drawing no. PH-536, revision C.

6.x Functional Design — Working Papers — Sheet 8

Equipment Schedule

Item	Department	Nos	Unit rate (Rs)	Amount (Rs)
Defibrillator	ICU	32	136,875	4,380,000
Patient monitor	Ward	12	850,965	10,211,580
Ultrasound	ICU	30	719,171	21,575,130
Autoclave	Casualty	5	1,427,167	7,135,835
OT table	CSSD	17	1,653,917	28,116,589
Patient monitor	Radiology	30	1,338,078	40,142,340
OT table	Radiology	21	1,379,878	28,977,438
X-ray unit	Casualty	30	191,464	5,743,920
Ventilator	Casualty	7	1,456,909	10,198,363
Hospital bed	Casualty	10	1,495,258	14,952,580
X-ray unit	Ward	34	178,803	6,079,302
X-ray unit	Casualty	38	913,559	34,715,242
Ventilator	Casualty	25	149,522	3,738,050
Autoclave	Radiology	24	2,094,567	50,269,608
Defibrillator	Radiology	29	775,723	22,495,967
Defibrillator	ICU	23	284,945	6,553,735
Ventilator	CSSD	22	1,895,618	41,703,596
Defibrillator	Ward	20	424,485	8,489,700
Ultrasound	Radiology	25	2,034,329	50,858,225
Defibrillator	CSSD	14	1,466,518	20,531,252
Ultrasound	Casualty	25	1,768,598	44,214,950
Ventilator	ICU	34	1,496,025	50,864,850
Patient monitor	ICU	20	2,083,164	41,663,280

Soil investigation was carried out by rotary drilling at the locations shown on the layout plan. The safe bearing capacity adopted for design is based on the lower of the shear and settlement criteria as reported in the geotechnical report.

6.x Functional Design — Working Papers — Sheet 9

Design Computation

Shear = 77.830 × 2.532 = 197.096 kN   (permissible 331.453) — safe

Axial load = 81.432 × 9.356 = 761.919 kN   (permissible 858.934) — safe

Shear = 9.675 × 6.780 = 65.593 kN   (permissible 70.025) — safe

Moment = 80.457 × 6.855 = 551.508 kNm   (permissible 744.680) — safe

Axial load = 17.964 × 11.090 = 199.222 kN   (permissible 277.018) — safe

Fire safety provisions follow Part 4 of the National Building Code. Means of egress, travel distances, staircase widths and the fire fighting installations have been designed for the occupancy classification of the block.

6.x Functional Design — Working Papers — Sheet 10

The space programme has been developed from the functional brief agreed with the user department. Departmental areas, circulation and the gross-to-carpet ratio are tabulated in the area statement enclosed at the Annexure.

Barrier-free access has been provided in accordance with the Harmonised Guidelines and Standards for Universal Accessibility. Ramps, lifts, signage and accessible toilets are provided on every floor of the block.

The space programme has been developed from the functional brief agreed with the user department. Departmental areas, circulation and the gross-to-carpet ratio are tabulated in the area statement enclosed at the Annexure.

Fire safety provisions follow Part 4 of the National Building Code. Means of egress, travel distances, staircase widths and the fire fighting installations have been designed for the occupancy classification of the block.

Reference: drawing no. EL-881, revision B.

6.x Functional Design — Working Papers — Sheet 11

Detailed Measurement Sheet

Item	L (m)	B (m)	D (m)	Nos	Quantity
Item 602	20.83	5.69	0.28	19	96.190
Item 147	11.23	3.46	0.28	8	50.192
Item 546	5.11	4.35	0.92	4	124.999
Item 395	2.33	4.04	1.53	16	250.528
Item 808	27.13	1.04	0.21	14	12.923
Item 296	9.96	2.80	1.42	20	205.534
Item 371	2.95	4.69	1.22	15	51.869
Item 493	22.23	4.71	0.55	4	37.233
Item 644	23.62	2.63	2.82	16	389.761
Item 687	21.57	1.20	2.93	6	73.159
Item 439	19.72	1.18	1.98	24	187.797
Item 767	11.51	0.42	1.14	18	240.112
Item 162	12.82	3.58	0.66	4	258.351
Item 396	15.29	5.11	1.97	16	290.329
Item 442	6.67	4.81	0.26	3	357.907
Item 488	13.40	5.13	0.87	5	369.845
Item 808	4.35	2.87	1.65	6	255.048
Item 241	20.90	1.79	2.37	8	9.672

Barrier-free access has been provided in accordance with the Harmonised Guidelines and Standards for Universal Accessibility. Ramps, lifts, signage and accessible toilets are provided on every floor of the block.

6.x Functional Design — Working Papers — Sheet 12

Design Computation

Deflection = 50.163 × 10.355 = 519.424 mm   (permissible 547.137) — safe

Bearing pressure = 73.612 × 9.111 = 670.660 kPa   (permissible 775.145) — safe

Axial load = 34.787 × 4.949 = 172.159 kN   (permissible 305.725) — safe

Shear = 60.481 × 7.458 = 451.090 kN   (permissible 583.275) — safe

Shear = 25.678 × 3.972 = 101.997 kN   (permissible 128.375) — safe

Axial load = 23.852 × 1.415 = 33.742 kN   (permissible 48.935) — safe

Deflection = 29.418 × 2.220 = 65.294 mm   (permissible 74.749) — safe

Deflection = 6.703 × 11.976 = 80.279 mm   (permissible 95.659) — safe

Soil investigation was carried out by rotary drilling at the locations shown on the layout plan. The safe bearing capacity adopted for design is based on the lower of the shear and settlement criteria as reported in the geotechnical report.

6.x Functional Design — Working Papers — Sheet 13

Barrier-free access has been provided in accordance with the Harmonised Guidelines and Standards for Universal Accessibility. Ramps, lifts, signage and accessible toilets are provided on every floor of the block.

Fire safety provisions follow Part 4 of the National Building Code. Means of egress, travel distances, staircase widths and the fire fighting installations have been designed for the occupancy classification of the block.

Soil investigation was carried out by rotary drilling at the locations shown on the layout plan. The safe bearing capacity adopted for design is based on the lower of the shear and settlement criteria as reported in the geotechnical report.

Quantities have been computed from the approved architectural and structural drawings and cross-checked against the detailed measurement sheets enclosed at the Annexure. Rates are as per the schedule of rates in force with lead and lift adjustments.

Reference: drawing no. ST-146, revision B.

6.x Functional Design — Working Papers — Sheet 14

Departmental Area Statement

Department	Floor	Units	Area per unit (sqm)	Total (sqm)
Staff Quarters	Ground	9	39.94	359.44
Operation Theatre	Second	17	50.55	859.38
Staff Quarters	Second	3	95.19	285.56
OPD	Second	18	33.32	599.80
Administration	Ground	9	16.17	145.54
Staff Quarters	Ground	1	125.17	125.17
IPD Ward	Second	7	87.11	609.76
Pharmacy	Ground	14	49.21	689.00
Diagnostics	Third	3	76.87	230.62
Pharmacy	Third	15	48.59	728.78
Operation Theatre	First	15	55.00	825.03
ICU	First	7	53.36	373.50
ICU	Third	14	132.94	1861.13
Operation Theatre	First	18	76.08	1369.47
IPD Ward	First	18	73.28	1319.03
Pharmacy	Second	5	31.87	159.33
Pharmacy	Ground	3	121.55	364.65
Staff Quarters	Third	15	33.77	506.49
Administration	Third	11	37.22	409.41
Diagnostics	Second	14	126.92	1776.91
Pharmacy	Ground	18	79.31	1427.59

Quantities have been computed from the approved architectural and structural drawings and cross-checked against the detailed measurement sheets enclosed at the Annexure. Rates are as per the schedule of rates in force with lead and lift adjustments.

7 ENGINEERING DESIGN

The structure is a reinforced concrete moment resisting frame designed to IS:456-2000, IS:1893 (Part 1)-2016 for seismic zone V with importance factor 1.5, and IS:875 for loads. Foundations are isolated footings on the assessed safe bearing capacity, with combined footings where column spacing requires. Fire safety follows the National Building Code 2016 Part 4.

7.x Engineering Design — Working Papers — Sheet 1

Fire safety provisions follow Part 4 of the National Building Code. Means of egress, travel distances, staircase widths and the fire fighting installations have been designed for the occupancy classification of the block.

Building services comprise electrical distribution, water supply and sanitation, HVAC for the identified conditioned areas, medical gas pipeline and vertical transportation. Connected load and demand factors are given in the load schedule.

Soil investigation was carried out by rotary drilling at the locations shown on the layout plan. The safe bearing capacity adopted for design is based on the lower of the shear and settlement criteria as reported in the geotechnical report.

Soil investigation was carried out by rotary drilling at the locations shown on the layout plan. The safe bearing capacity adopted for design is based on the lower of the shear and settlement criteria as reported in the geotechnical report.

Reference: drawing no. EL-872, revision C.

7.x Engineering Design — Working Papers — Sheet 2

Bore Log Summary

Depth (m)	Strata	N value	Sample
27.6	Coarse sand	47	SPT
14.7	Clayey silt	39	UDS
5.0	Coarse sand	59	UDS
29.0	Gravelly sand	41	UDS
5.4	Medium sand	37	UDS
26.1	Medium sand	54	DS
25.0	Gravelly sand	31	UDS
21.6	Clayey silt	25	DS
7.8	Gravelly sand	49	SPT
18.2	Medium sand	20	SPT
25.8	Clayey silt	59	UDS
14.6	Gravelly sand	25	UDS
26.3	Clayey silt	43	SPT
25.5	Clayey silt	47	DS
22.6	Coarse sand	16	DS
19.7	Clayey silt	25	DS
8.1	Clayey silt	44	DS
5.6	Gravelly sand	9	SPT
18.3	Silty clay	6	UDS
29.9	Silty clay	36	SPT
12.8	Gravelly sand	28	SPT
24.4	Coarse sand	27	SPT
5.5	Silty clay	36	UDS
4.0	Silty clay	24	DS
26.9	Silty clay	55	SPT
25.2	Gravelly sand	25	DS

The structural design conforms to IS 456 for reinforced concrete and IS 1893 for earthquake resistant design. The site falls in seismic zone V and the importance factor applicable to a hospital building has been adopted.

7.x Engineering Design — Working Papers — Sheet 3

Design Computation

Deflection = 39.769 × 11.587 = 460.789 mm   (permissible 679.073) — safe

Deflection = 71.012 × 3.961 = 281.314 mm   (permissible 375.194) — safe

Moment = 10.316 × 8.291 = 85.533 kNm   (permissible 154.204) — safe

Shear = 10.122 × 11.230 = 113.675 kN   (permissible 196.502) — safe

Deflection = 4.077 × 2.045 = 8.338 mm   (permissible 14.713) — safe

Shear = 23.177 × 9.028 = 209.250 kN   (permissible 224.592) — safe

Deflection = 60.469 × 11.332 = 685.215 mm   (permissible 771.232) — safe

Building services comprise electrical distribution, water supply and sanitation, HVAC for the identified conditioned areas, medical gas pipeline and vertical transportation. Connected load and demand factors are given in the load schedule.

7.x Engineering Design — Working Papers — Sheet 4

Soil investigation was carried out by rotary drilling at the locations shown on the layout plan. The safe bearing capacity adopted for design is based on the lower of the shear and settlement criteria as reported in the geotechnical report.

Fire safety provisions follow Part 4 of the National Building Code. Means of egress, travel distances, staircase widths and the fire fighting installations have been designed for the occupancy classification of the block.

The structural design conforms to IS 456 for reinforced concrete and IS 1893 for earthquake resistant design. The site falls in seismic zone V and the importance factor applicable to a hospital building has been adopted.

Soil investigation was carried out by rotary drilling at the locations shown on the layout plan. The safe bearing capacity adopted for design is based on the lower of the shear and settlement criteria as reported in the geotechnical report.

Reference: drawing no. AR-691, revision B.

7.x Engineering Design — Working Papers — Sheet 5

Room Data Sheet

Room	Department	Carpet area (sqm)	Occupancy	Floor finish
Room 858	General Medicine	87.67	8	Vitrified tile
Room 340	Casualty	37.66	11	Antistatic vinyl
Room 360	Casualty	24.12	20	Vitrified tile
Room 428	Pathology	39.68	16	Kota stone
Room 541	General Medicine	85.35	6	Epoxy screed
Room 295	Casualty	63.83	17	Epoxy screed
Room 841	Obstetrics	27.72	13	Kota stone
Room 297	Radiology	34.27	18	Ceramic tile
Room 106	Paediatrics	41.47	27	Vitrified tile
Room 393	Casualty	27.35	27	Antistatic vinyl
Room 488	Casualty	60.17	27	Epoxy screed
Room 516	Surgery	36.88	20	Antistatic vinyl
Room 416	Paediatrics	74.09	8	Epoxy screed
Room 502	Pathology	79.10	15	Vitrified tile
Room 243	Obstetrics	14.87	9	Vitrified tile
Room 892	Paediatrics	91.22	28	Antistatic vinyl
Room 891	Casualty	58.78	18	Kota stone
Room 115	Pathology	30.61	10	Ceramic tile
Room 412	Pathology	18.80	22	Vitrified tile
Room 567	General Medicine	46.72	26	Kota stone
Room 760	Pathology	13.21	19	Ceramic tile
Room 388	Paediatrics	49.06	9	Vitrified tile

Building services comprise electrical distribution, water supply and sanitation, HVAC for the identified conditioned areas, medical gas pipeline and vertical transportation. Connected load and demand factors are given in the load schedule.

7.x Engineering Design — Working Papers — Sheet 6

Design Computation

Deflection = 49.883 × 9.226 = 460.210 mm (permissible 615.280) — safe

Deflection = 80.721 × 7.710 = 622.335 mm (permissible 890.160) — safe

Bearing pressure = 4.272 × 5.855 = 25.013 kPa (permissible 42.086) — safe

Deflection = 89.910 × 6.801 = 611.485 mm (permissible 715.764) — safe

Moment = 13.132 × 6.895 = 90.547 kNm (permissible 140.510) — safe

Deflection = 30.844 × 0.760 = 23.427 mm (permissible 25.278) — safe

Moment = 10.903 × 10.557 = 115.103 kNm (permissible 126.267) — safe

The structural design conforms to IS 456 for reinforced concrete and IS 1893 for earthquake resistant design. The site falls in seismic zone V and the importance factor applicable to a hospital building has been adopted.

7.x Engineering Design — Working Papers — Sheet 7

Barrier-free access has been provided in accordance with the Harmonised Guidelines and Standards for Universal Accessibility. Ramps, lifts, signage and accessible toilets are provided on every floor of the block.

Quantities have been computed from the approved architectural and structural drawings and cross-checked against the detailed measurement sheets enclosed at the Annexure. Rates are as per the schedule of rates in force with lead and lift adjustments.

The space programme has been developed from the functional brief agreed with the user department. Departmental areas, circulation and the gross-to-carpet ratio are tabulated in the area statement enclosed at the Annexure.

Building services comprise electrical distribution, water supply and sanitation, HVAC for the identified conditioned areas, medical gas pipeline and vertical transportation. Connected load and demand factors are given in the load schedule.

Reference: drawing no. PH-148, revision A.

7.x Engineering Design — Working Papers — Sheet 8

Equipment Schedule

Item	Department	Nos	Unit rate (Rs)	Amount (Rs)
Hospital bed	Radiology	2	1,142,075	2,284,150
Autoclave	CSSD	19	1,280,550	24,330,450
Autoclave	OT	5	1,293,947	6,469,735
Hospital bed	ICU	9	1,512,118	13,609,062
Hospital bed	ICU	11	620,944	6,830,384
Ventilator	OT	11	1,863,308	20,496,388
Hospital bed	Ward	10	117,583	1,175,830
Ultrasound	ICU	6	1,220,479	7,322,874
Ventilator	Casualty	39	918,872	35,836,008
OT table	Casualty	37	497,300	18,400,100
OT table	Casualty	24	1,847,647	44,343,528
Hospital bed	Radiology	30	239,369	7,181,070
Autoclave	ICU	19	2,184,680	41,508,920
Ventilator	Radiology	39	245,072	9,557,808
Patient monitor	OT	4	483,865	1,935,460
Autoclave	Radiology	11	1,474,197	16,216,167
Patient monitor	Casualty	22	387,768	8,530,896
Hospital bed	ICU	38	2,243,142	85,239,396
Ultrasound	Ward	11	2,126,481	23,391,291
Ultrasound	Radiology	27	1,402,834	37,876,518
Autoclave	OT	33	106,060	3,499,980
Autoclave	Casualty	12	135,424	1,625,088

Building services comprise electrical distribution, water supply and sanitation, HVAC for the identified conditioned areas, medical gas pipeline and vertical transportation. Connected load and demand factors are given in the load schedule.

7.x Engineering Design — Working Papers — Sheet 9

Design Computation

Bearing pressure = 2.738 × 3.856 = 10.556 kPa   (permissible 12.173) — safe

Axial load = 10.163 × 7.648 = 77.728 kN   (permissible 125.545) — safe

Moment = 46.139 × 3.146 = 145.140 kNm   (permissible 239.324) — safe

Axial load = 83.724 × 5.470 = 457.983 kN   (permissible 594.866) — safe

Shear = 30.739 × 10.773 = 331.145 kN   (permissible 556.370) — safe

Deflection = 3.735 × 6.243 = 23.319 mm   (permissible 34.475) — safe

Shear = 47.437 × 10.066 = 477.493 kN   (permissible 906.317) — safe

Bearing pressure = 86.692 × 5.419 = 469.768 kPa   (permissible 623.036) — safe

Building services comprise electrical distribution, water supply and sanitation, HVAC for the identified conditioned areas, medical gas pipeline and vertical transportation. Connected load and demand factors are given in the load schedule.

7.x Engineering Design — Working Papers — Sheet 10

The structural design conforms to IS 456 for reinforced concrete and IS 1893 for earthquake resistant design. The site falls in seismic zone V and the importance factor applicable to a hospital building has been adopted.

Consultations were held with the head of each clinical department on adjacency, patient flow and equipment layout. Minutes of the consultation meetings are placed at the Annexure to this chapter.

Fire safety provisions follow Part 4 of the National Building Code. Means of egress, travel distances, staircase widths and the fire fighting installations have been designed for the occupancy classification of the block.

Consultations were held with the head of each clinical department on adjacency, patient flow and equipment layout. Minutes of the consultation meetings are placed at the Annexure to this chapter.

Reference: drawing no. AR-275, revision A.

7.x Engineering Design — Working Papers — Sheet 11

Equipment Schedule

Item	Department	Nos	Unit rate (Rs)	Amount (Rs)
OT table	Ward	20	551,065	11,021,300
Ventilator	CSSD	35	838,640	29,352,400
Hospital bed	OT	38	171,739	6,526,082
X-ray unit	OT	19	329,900	6,268,100
OT table	CSSD	23	2,324,854	53,471,642
Autoclave	Radiology	8	1,304,521	10,436,168
X-ray unit	Casualty	17	1,356,035	23,052,595
OT table	OT	32	1,813,215	58,022,880
Hospital bed	Ward	8	2,002,781	16,022,248
Autoclave	Radiology	40	1,818,407	72,736,280
OT table	Casualty	24	749,716	17,993,184
Hospital bed	Casualty	12	19,463	233,556
Ventilator	Radiology	21	1,199,976	25,199,496
Ultrasound	OT	8	2,077,769	16,622,152
Hospital bed	Ward	32	699,822	22,394,304
X-ray unit	Ward	26	236,342	6,144,892
Autoclave	Casualty	9	1,069,707	9,627,363

Building services comprise electrical distribution, water supply and sanitation, HVAC for the identified conditioned areas, medical gas pipeline and vertical transportation. Connected load and demand factors are given in the load schedule.

7.x Engineering Design — Working Papers — Sheet 12

Design Computation

Bearing pressure = 27.340 × 7.612 = 208.110 kPa   (permissible 304.717) — safe

Axial load = 78.985 × 4.513 = 356.466 kN   (permissible 405.590) — safe

Moment = 7.600 × 9.087 = 69.060 kNm   (permissible 91.138) — safe

Shear = 88.518 × 9.668 = 855.807 kN   (permissible 1572.383) — safe

Moment = 40.135 × 6.717 = 269.604 kNm   (permissible 325.900) — safe

Axial load = 69.726 × 3.211 = 223.868 kN   (permissible 294.943) — safe

Deflection = 16.399 × 6.959 = 114.121 mm   (permissible 199.152) — revise section

Axial load = 78.487 × 4.530 = 355.576 kN   (permissible 487.470) — revise section

The space programme has been developed from the functional brief agreed with the user department. Departmental areas, circulation and the gross-to-carpet ratio are tabulated in the area statement enclosed at the Annexure.

7.x Engineering Design — Working Papers — Sheet 13

Quantities have been computed from the approved architectural and structural drawings and cross-checked against the detailed measurement sheets enclosed at the Annexure. Rates are as per the schedule of rates in force with lead and lift adjustments.

Quantities have been computed from the approved architectural and structural drawings and cross-checked against the detailed measurement sheets enclosed at the Annexure. Rates are as per the schedule of rates in force with lead and lift adjustments.

Consultations were held with the head of each clinical department on adjacency, patient flow and equipment layout. Minutes of the consultation meetings are placed at the Annexure to this chapter.

Quantities have been computed from the approved architectural and structural drawings and cross-checked against the detailed measurement sheets enclosed at the Annexure. Rates are as per the schedule of rates in force with lead and lift adjustments.

Reference: drawing no. AR-616, revision C.

7.x Engineering Design — Working Papers — Sheet 14

Electrical Load Schedule

Panel	Description	Connected load (kW)	Demand factor	Demand (kVA)
DB-37	Power outlets	120.49	0.74	98.76
DB-31	Lighting	158.37	0.70	122.36
DB-10	Power outlets	133.02	0.73	108.38
DB-08	HVAC	151.34	0.57	95.03
DB-33	Pumps	26.61	0.58	17.03
DB-07	Pumps	102.90	0.64	73.30
DB-12	Power outlets	138.71	0.45	69.55
DB-31	Medical equipment	58.31	0.47	30.71
DB-39	Power outlets	89.42	0.91	90.78
DB-33	Lighting	116.75	0.82	106.98
DB-19	Medical equipment	23.12	0.65	16.70
DB-29	Medical equipment	52.92	0.61	36.00
DB-07	Lifts	82.66	0.91	83.91
DB-13	Medical equipment	8.43	0.59	5.49
DB-40	Sterilisation	141.39	0.50	78.92
DB-39	Lighting	10.73	0.72	8.60
DB-11	Power outlets	56.78	0.78	49.21
DB-02	Power outlets	99.71	0.81	90.02
DB-30	Lifts	115.37	0.60	76.50
DB-06	Sterilisation	179.51	0.55	110.44
DB-02	Lighting	9.10	0.55	5.59
DB-16	Sterilisation	174.77	0.76	146.83
DB-15	Power outlets	73.42	0.79	64.72
DB-13	Power outlets	64.76	0.92	66.05
DB-36	Power outlets	101.59	0.52	58.90
DB-05	HVAC	41.67	0.83	38.47

The space programme has been developed from the functional brief agreed with the user department. Departmental areas, circulation and the gross-to-carpet ratio are tabulated in the area statement enclosed at the Annexure.

7.x Engineering Design — Working Papers — Sheet 15

Design Computation

Axial load = 74.163 × 5.634 = 417.826 kN   (permissible 651.459) — safe

Deflection = 10.545 × 5.041 = 53.159 mm   (permissible 65.452) — safe

Moment = 45.334 × 11.673 = 529.173 kNm   (permissible 680.528) — safe

Axial load = 14.274 × 8.440 = 120.478 kN   (permissible 157.201) — safe

Deflection = 66.861 × 5.259 = 351.588 mm   (permissible 396.221) — safe

Moment = 11.332 × 8.849 = 100.278 kNm   (permissible 168.255) — safe

Axial load = 26.184 × 11.170 = 292.470 kN   (permissible 310.180) — revise section

Fire safety provisions follow Part 4 of the National Building Code. Means of egress, travel distances, staircase widths and the fire fighting installations have been designed for the occupancy classification of the block.

7.x Engineering Design — Working Papers — Sheet 16

Soil investigation was carried out by rotary drilling at the locations shown on the layout plan. The safe bearing capacity adopted for design is based on the lower of the shear and settlement criteria as reported in the geotechnical report.

Barrier-free access has been provided in accordance with the Harmonised Guidelines and Standards for Universal Accessibility. Ramps, lifts, signage and accessible toilets are provided on every floor of the block.

Quantities have been computed from the approved architectural and structural drawings and cross-checked against the detailed measurement sheets enclosed at the Annexure. Rates are as per the schedule of rates in force with lead and lift adjustments.

Barrier-free access has been provided in accordance with the Harmonised Guidelines and Standards for Universal Accessibility. Ramps, lifts, signage and accessible toilets are provided on every floor of the block.

Reference: drawing no. ST-891, revision A.

7.x Engineering Design — Working Papers — Sheet 17

Detailed Measurement Sheet

Item	L (m)	B (m)	D (m)	Nos	Quantity
Item 212	6.40	2.84	1.81	10	16.745
Item 865	5.88	5.15	0.64	7	253.088
Item 807	32.78	4.24	1.67	22	241.167
Item 777	31.90	3.43	1.32	5	40.864
Item 470	10.74	3.55	2.12	18	269.315
Item 704	22.09	2.26	0.70	1	180.773
Item 404	7.64	0.91	2.29	4	363.660
Item 293	31.61	0.91	1.58	6	171.964
Item 665	10.22	5.23	0.23	11	121.803
Item 324	32.32	5.13	1.06	24	334.834
Item 224	27.64	5.30	1.46	9	171.398
Item 596	3.39	3.56	2.03	20	372.272
Item 655	11.01	4.92	0.87	9	353.211
Item 880	24.11	3.15	0.68	14	297.681
Item 119	27.32	0.99	2.70	1	84.693
Item 212	9.41	2.19	2.34	23	141.522
Item 336	12.77	3.18	1.25	10	390.604
Item 626	35.09	2.99	1.99	9	319.754
Item 425	30.52	0.63	0.36	22	146.790
Item 717	19.32	2.86	2.98	23	227.011
Item 477	22.76	5.19	2.86	1	265.363
Item 373	6.05	4.59	1.67	8	327.838

The space programme has been developed from the functional brief agreed with the user department. Departmental areas, circulation and the gross-to-carpet ratio are tabulated in the area statement enclosed at the Annexure.

7.x Engineering Design — Working Papers — Sheet 18

Design Computation

Moment = 22.813 × 4.905 = 111.893 kNm   (permissible 118.098) — safe

Moment = 79.844 × 0.777 = 62.011 kNm   (permissible 108.681) — safe

Deflection = 15.611 × 4.239 = 66.170 mm   (permissible 100.723) — safe

Bearing pressure = 23.274 × 3.062 = 71.275 kPa   (permissible 128.344) — safe

Moment = 32.168 × 5.014 = 161.299 kNm   (permissible 305.253) — safe

Axial load = 22.543 × 1.808 = 40.754 kN   (permissible 62.039) — safe

Bearing pressure = 35.769 × 5.635 = 201.549 kPa   (permissible 373.438) — safe

Quantities have been computed from the approved architectural and structural drawings and cross-checked against the detailed measurement sheets enclosed at the Annexure. Rates are as per the schedule of rates in force with lead and lift adjustments.

7.x Engineering Design — Working Papers — Sheet 19

Soil investigation was carried out by rotary drilling at the locations shown on the layout plan. The safe bearing capacity adopted for design is based on the lower of the shear and settlement criteria as reported in the geotechnical report.

The space programme has been developed from the functional brief agreed with the user department. Departmental areas, circulation and the gross-to-carpet ratio are tabulated in the area statement enclosed at the Annexure.

The space programme has been developed from the functional brief agreed with the user department. Departmental areas, circulation and the gross-to-carpet ratio are tabulated in the area statement enclosed at the Annexure.

Fire safety provisions follow Part 4 of the National Building Code. Means of egress, travel distances, staircase widths and the fire fighting installations have been designed for the occupancy classification of the block.

Reference: drawing no. PH-805, revision B.

7.x Engineering Design — Working Papers — Sheet 20

Electrical Load Schedule

Panel	Description	Connected load (kW)	Demand factor	Demand (kVA)
DB-07	Medical equipment	85.08	0.57	53.83
DB-26	Lifts	94.13	0.64	67.42
DB-39	Pumps	173.07	0.46	89.21
DB-17	Lighting	165.86	0.71	131.69
DB-08	Lifts	15.01	0.61	10.15
DB-19	Lighting	66.10	0.90	65.90
DB-33	Power outlets	65.02	0.81	58.23
DB-33	Medical equipment	64.31	0.58	41.14
DB-09	Pumps	44.60	0.62	30.65
DB-35	Medical equipment	149.17	0.56	92.98
DB-16	Lifts	68.87	0.49	37.25
DB-24	Pumps	143.92	0.71	113.74
DB-37	Medical equipment	100.58	0.60	67.56
DB-23	Medical equipment	139.11	0.87	135.00
DB-37	Lighting	154.45	0.76	129.57
DB-20	Pumps	73.22	0.50	40.40
DB-12	Medical equipment	66.39	0.63	46.76
DB-38	Medical equipment	77.97	0.94	81.68
DB-20	Lighting	91.97	0.68	69.09
DB-27	Power outlets	39.21	0.78	34.14
DB-12	HVAC	62.78	0.79	54.85
DB-38	Sterilisation	102.24	0.43	48.29
DB-16	Medical equipment	130.97	0.49	71.46

Quantities have been computed from the approved architectural and structural drawings and cross-checked against the detailed measurement sheets enclosed at the Annexure. Rates are as per the schedule of rates in force with lead and lift adjustments.

7.x Engineering Design — Working Papers — Sheet 21

Design Computation

Deflection = 14.510 × 3.096 = 44.924 mm   (permissible 55.196) — safe

Moment = 11.666 × 8.021 = 93.579 kNm   (permissible 116.445) — safe

Deflection = 45.494 × 9.514 = 432.815 mm   (permissible 751.148) — safe

Bearing pressure = 40.094 × 2.170 = 86.986 kPa   (permissible 119.847) — safe

Moment = 20.115 × 5.567 = 111.981 kNm   (permissible 193.875) — safe

Moment = 35.122 × 1.016 = 35.679 kNm   (permissible 54.603) — revise section

Moment = 80.372 × 9.188 = 738.450 kNm   (permissible 1058.615) — safe

Moment = 5.147 × 3.794 = 19.526 kNm   (permissible 23.635) — safe

The space programme has been developed from the functional brief agreed with the user department. Departmental areas, circulation and the gross-to-carpet ratio are tabulated in the area statement enclosed at the Annexure.

7.x Engineering Design — Working Papers — Sheet 22

Consultations were held with the head of each clinical department on adjacency, patient flow and equipment layout. Minutes of the consultation meetings are placed at the Annexure to this chapter.

Barrier-free access has been provided in accordance with the Harmonised Guidelines and Standards for Universal Accessibility. Ramps, lifts, signage and accessible toilets are provided on every floor of the block.

Soil investigation was carried out by rotary drilling at the locations shown on the layout plan. The safe bearing capacity adopted for design is based on the lower of the shear and settlement criteria as reported in the geotechnical report.

Barrier-free access has been provided in accordance with the Harmonised Guidelines and Standards for Universal Accessibility. Ramps, lifts, signage and accessible toilets are provided on every floor of the block.

Reference: drawing no. PH-942, revision C.

7.x Engineering Design — Working Papers — Sheet 23

Room Data Sheet

Room	Department	Carpet area (sqm)	Occupancy	Floor finish
Room 375	Radiology	21.40	12	Vitrified tile
Room 423	Obstetrics	32.83	12	Ceramic tile
Room 511	General Medicine	72.56	26	Vitrified tile
Room 788	Obstetrics	61.17	6	Epoxy screed
Room 683	Paediatrics	23.79	19	Vitrified tile
Room 738	Casualty	42.67	4	Epoxy screed
Room 273	Obstetrics	17.78	26	Antistatic vinyl
Room 672	Casualty	55.21	18	Kota stone
Room 314	Obstetrics	79.25	22	Epoxy screed
Room 762	Paediatrics	36.63	18	Ceramic tile
Room 619	Casualty	65.13	19	Antistatic vinyl
Room 465	Obstetrics	85.06	26	Antistatic vinyl
Room 730	Surgery	35.33	16	Antistatic vinyl
Room 616	Pathology	87.84	12	Ceramic tile
Room 624	Radiology	69.98	16	Ceramic tile
Room 600	Surgery	52.87	24	Vitrified tile

The space programme has been developed from the functional brief agreed with the user department. Departmental areas, circulation and the gross-to-carpet ratio are tabulated in the area statement enclosed at the Annexure.

7.x Engineering Design — Working Papers — Sheet 24

Design Computation

Moment = 83.484 × 1.054 = 87.988 kNm   (permissible 129.445) — safe

Axial load = 10.802 × 11.782 = 127.271 kN   (permissible 235.937) — safe

Deflection = 16.099 × 6.498 = 104.622 mm   (permissible 162.691) — safe

Deflection = 45.216 × 4.338 = 196.128 mm   (permissible 300.339) — safe

Deflection = 7.489 × 5.844 = 43.767 mm   (permissible 78.392) — revise section

Axial load = 71.920 × 5.189 = 373.168 kN   (permissible 471.692) — safe

Barrier-free access has been provided in accordance with the Harmonised Guidelines and Standards for Universal Accessibility. Ramps, lifts, signage and accessible toilets are provided on every floor of the block.

7.x Engineering Design — Working Papers — Sheet 25

The space programme has been developed from the functional brief agreed with the user department. Departmental areas, circulation and the gross-to-carpet ratio are tabulated in the area statement enclosed at the Annexure.

The space programme has been developed from the functional brief agreed with the user department. Departmental areas, circulation and the gross-to-carpet ratio are tabulated in the area statement enclosed at the Annexure.

The structural design conforms to IS 456 for reinforced concrete and IS 1893 for earthquake resistant design. The site falls in seismic zone V and the importance factor applicable to a hospital building has been adopted.

The space programme has been developed from the functional brief agreed with the user department. Departmental areas, circulation and the gross-to-carpet ratio are tabulated in the area statement enclosed at the Annexure.

Reference: drawing no. EL-568, revision C.

7.x Engineering Design — Working Papers — Sheet 26

Detailed Measurement Sheet

Item	L (m)	B (m)	D (m)	Nos	Quantity
Item 416	20.55	0.68	1.89	21	335.066
Item 665	12.82	4.99	2.69	9	211.031
Item 675	16.43	5.72	0.83	19	380.661
Item 337	13.62	5.63	2.05	17	217.426
Item 627	33.10	4.72	0.51	21	300.984
Item 392	28.67	3.22	0.38	12	91.918
Item 111	28.72	4.65	2.65	10	17.341
Item 775	19.81	3.85	2.66	2	339.773
Item 628	39.50	4.14	2.28	2	235.580
Item 898	27.18	4.57	0.53	23	265.773
Item 560	23.11	0.88	1.38	8	318.531
Item 534	10.24	3.30	2.25	7	19.248
Item 780	33.25	0.32	1.74	23	226.689
Item 648	18.45	1.72	1.25	7	344.941
Item 149	5.33	1.01	1.21	10	176.596
Item 159	26.51	4.26	0.98	9	212.425
Item 646	4.91	4.90	1.31	12	375.411
Item 258	8.96	4.83	0.45	16	99.317
Item 480	9.46	3.27	2.82	20	94.005
Item 353	24.00	5.02	2.14	15	28.343
Item 215	33.61	0.31	1.63	1	258.816
Item 636	22.87	3.61	1.11	9	175.462
Item 559	7.79	4.49	2.11	1	100.393

The space programme has been developed from the functional brief agreed with the user department. Departmental areas, circulation and the gross-to-carpet ratio are tabulated in the area statement enclosed at the Annexure.

7.x Engineering Design — Working Papers — Sheet 27

Design Computation

Deflection = 73.127 × 7.460 = 545.505 mm   (permissible 770.126) — safe

Axial load = 5.957 × 5.156 = 30.713 kN   (permissible 39.445) — safe

Moment = 78.149 × 6.838 = 534.346 kNm   (permissible 570.791) — safe

Bearing pressure = 4.554 × 4.685 = 21.331 kPa   (permissible 25.636) — safe

Axial load = 51.804 × 6.874 = 356.120 kN   (permissible 662.641) — safe

Moment = 19.685 × 1.392 = 27.408 kNm   (permissible 39.081) — safe

Moment = 10.678 × 1.865 = 19.915 kNm   (permissible 26.334) — safe

Quantities have been computed from the approved architectural and structural drawings and cross-checked against the detailed measurement sheets enclosed at the Annexure. Rates are as per the schedule of rates in force with lead and lift adjustments.

7.x Engineering Design — Working Papers — Sheet 28

Building services comprise electrical distribution, water supply and sanitation, HVAC for the identified conditioned areas, medical gas pipeline and vertical transportation. Connected load and demand factors are given in the load schedule.

Consultations were held with the head of each clinical department on adjacency, patient flow and equipment layout. Minutes of the consultation meetings are placed at the Annexure to this chapter.

Fire safety provisions follow Part 4 of the National Building Code. Means of egress, travel distances, staircase widths and the fire fighting installations have been designed for the occupancy classification of the block.

Soil investigation was carried out by rotary drilling at the locations shown on the layout plan. The safe bearing capacity adopted for design is based on the lower of the shear and settlement criteria as reported in the geotechnical report.

Reference: drawing no. PH-878, revision C.

7.x Engineering Design — Working Papers — Sheet 29

Equipment Schedule

Item	Department	Nos	Unit rate (Rs)	Amount (Rs)
Patient monitor	ICU	15	912,653	13,689,795
OT table	ICU	40	894,672	35,786,880
X-ray unit	CSSD	25	164,109	4,102,725
X-ray unit	OT	17	687,906	11,694,402
Hospital bed	Radiology	29	74,975	2,174,275
OT table	CSSD	21	682,317	14,328,657
X-ray unit	Ward	3	702,224	2,106,672
X-ray unit	Casualty	17	1,052,508	17,892,636
Ventilator	ICU	16	704,297	11,268,752
Hospital bed	OT	39	483,932	18,873,348
Defibrillator	Ward	4	504,642	2,018,568
Ultrasound	Radiology	27	1,196,528	32,306,256
OT table	Radiology	2	2,064,216	4,128,432
OT table	Ward	36	413,967	14,902,812
Hospital bed	OT	31	1,560,370	48,371,470
Ventilator	CSSD	7	1,327,942	9,295,594
Defibrillator	OT	24	2,315,694	55,576,656
Patient monitor	ICU	1	733,564	733,564
Autoclave	Radiology	24	1,916,365	45,992,760
OT table	OT	38	553,192	21,021,296
Patient monitor	Casualty	13	364,934	4,744,142
Patient monitor	OT	19	1,165,237	22,139,503

Consultations were held with the head of each clinical department on adjacency, patient flow and equipment layout. Minutes of the consultation meetings are placed at the Annexure to this chapter.

7.x Engineering Design — Working Papers — Sheet 30

Design Computation

Moment = 31.606 × 4.886 = 154.435 kNm   (permissible 280.523) — safe

Axial load = 65.088 × 7.129 = 463.992 kN   (permissible 835.960) — safe

Deflection = 75.697 × 2.079 = 157.339 mm   (permissible 168.206) — safe

Deflection = 77.803 × 10.561 = 821.649 mm   (permissible 1191.109) — safe

Moment = 40.280 × 5.062 = 203.904 kNm   (permissible 327.269) — safe

Axial load = 59.686 × 8.901 = 531.240 kN   (permissible 584.869) — safe

Deflection = 63.373 × 5.142 = 325.890 mm   (permissible 453.316) — safe

Quantities have been computed from the approved architectural and structural drawings and cross-checked against the detailed measurement sheets enclosed at the Annexure. Rates are as per the schedule of rates in force with lead and lift adjustments.

7.x Engineering Design — Working Papers — Sheet 31

Soil investigation was carried out by rotary drilling at the locations shown on the layout plan. The safe bearing capacity adopted for design is based on the lower of the shear and settlement criteria as reported in the geotechnical report.

The space programme has been developed from the functional brief agreed with the user department. Departmental areas, circulation and the gross-to-carpet ratio are tabulated in the area statement enclosed at the Annexure.

Fire safety provisions follow Part 4 of the National Building Code. Means of egress, travel distances, staircase widths and the fire fighting installations have been designed for the occupancy classification of the block.

Fire safety provisions follow Part 4 of the National Building Code. Means of egress, travel distances, staircase widths and the fire fighting installations have been designed for the occupancy classification of the block.

Reference: drawing no. AR-497, revision A.

7.x Engineering Design — Working Papers — Sheet 32

Equipment Schedule

Item	Department	Nos	Unit rate (Rs)	Amount (Rs)
Defibrillator	Ward	21	1,948,831	40,925,451
Defibrillator	Casualty	16	718,498	11,495,968
Hospital bed	ICU	18	1,545,240	27,814,320
X-ray unit	Casualty	7	1,068,842	7,481,894
X-ray unit	CSSD	5	218,351	1,091,755
Ultrasound	Ward	26	795,345	20,678,970
Defibrillator	CSSD	38	1,719,344	65,335,072
OT table	Casualty	8	401,372	3,210,976
Ventilator	CSSD	20	1,891,583	37,831,660
OT table	Ward	37	1,755,235	64,943,695
Autoclave	ICU	18	1,812,271	32,620,878
Autoclave	Casualty	2	1,119,704	2,239,408
Hospital bed	Ward	35	2,359,350	82,577,250
OT table	CSSD	34	1,779,174	60,491,916
X-ray unit	ICU	2	900,139	1,800,278
Ventilator	Casualty	6	297,226	1,783,356
Patient monitor	CSSD	27	63,305	1,709,235
Defibrillator	CSSD	17	921,580	15,666,860
Ultrasound	Radiology	8	1,891,738	15,133,904
OT table	ICU	23	1,803,971	41,491,333
Patient monitor	Radiology	29	85,426	2,477,354
Patient monitor	Ward	5	1,277,285	6,386,425
Ultrasound	Ward	32	2,241,850	71,739,200
Defibrillator	OT	4	33,816	135,264

The structural design conforms to IS 456 for reinforced concrete and IS 1893 for earthquake resistant design. The site falls in seismic zone V and the importance factor applicable to a hospital building has been adopted.

7.x Engineering Design — Working Papers — Sheet 33

Design Computation

Axial load = 63.741 × 5.952 = 379.413 kN   (permissible 591.581) — safe

Shear = 15.392 × 3.894 = 59.931 kN   (permissible 72.756) — safe

Deflection = 11.034 × 10.458 = 115.390 mm   (permissible 218.329) — safe

Axial load = 2.165 × 6.758 = 14.631 kN   (permissible 19.264) — safe

Bearing pressure = 87.307 × 9.885 = 863.009 kPa   (permissible 1364.146) — safe

Deflection = 67.154 × 2.010 = 135.010 mm   (permissible 189.234) — safe

Deflection = 24.336 × 4.538 = 110.434 mm   (permissible 146.926) — safe

Deflection = 12.838 × 4.361 = 55.993 mm   (permissible 103.401) — safe

Fire safety provisions follow Part 4 of the National Building Code. Means of egress, travel distances, staircase widths and the fire fighting installations have been designed for the occupancy classification of the block.

7.x Engineering Design — Working Papers — Sheet 34

Fire safety provisions follow Part 4 of the National Building Code. Means of egress, travel distances, staircase widths and the fire fighting installations have been designed for the occupancy classification of the block.

Fire safety provisions follow Part 4 of the National Building Code. Means of egress, travel distances, staircase widths and the fire fighting installations have been designed for the occupancy classification of the block.

Soil investigation was carried out by rotary drilling at the locations shown on the layout plan. The safe bearing capacity adopted for design is based on the lower of the shear and settlement criteria as reported in the geotechnical report.

The structural design conforms to IS 456 for reinforced concrete and IS 1893 for earthquake resistant design. The site falls in seismic zone V and the importance factor applicable to a hospital building has been adopted.

Reference: drawing no. PH-970, revision B.

7.x Engineering Design — Working Papers — Sheet 35

Equipment Schedule

Item	Department	Nos	Unit rate (Rs)	Amount (Rs)
Defibrillator	Casualty	20	1,797,567	35,951,340
OT table	OT	39	475,409	18,540,951
Ultrasound	OT	20	1,206,768	24,135,360
OT table	Casualty	30	1,898,790	56,963,700
X-ray unit	Ward	31	467,410	14,489,710
Defibrillator	OT	37	1,071,093	39,630,441
Ventilator	CSSD	36	1,116,762	40,203,432
Ultrasound	CSSD	13	256,286	3,331,718
OT table	Radiology	10	1,633,629	16,336,290
Autoclave	CSSD	26	1,504,400	39,114,400
Patient monitor	ICU	2	106,079	212,158
Autoclave	Ward	19	421,632	8,011,008
OT table	Casualty	32	1,362,912	43,613,184
Defibrillator	CSSD	39	1,192,907	46,523,373
Autoclave	OT	26	2,338,696	60,806,096
OT table	ICU	19	507,768	9,647,592
Ultrasound	Radiology	12	639,347	7,672,164
Autoclave	CSSD	21	1,351,372	28,378,812
Ultrasound	Casualty	10	1,409,942	14,099,420
OT table	Ward	1	1,239,431	1,239,431
Hospital bed	Ward	24	1,241,417	29,794,008
Hospital bed	Radiology	18	2,237,891	40,282,038
Defibrillator	Ward	22	930,410	20,469,020
OT table	ICU	10	649,548	6,495,480
X-ray unit	Casualty	23	2,115,040	48,645,920

The space programme has been developed from the functional brief agreed with the user department. Departmental areas, circulation and the gross-to-carpet ratio are tabulated in the area statement enclosed at the Annexure.

7.x Engineering Design — Working Papers — Sheet 36

Design Computation

Deflection = 71.043 × 5.145 = 365.549 mm   (permissible 395.688) — safe

Moment = 17.719 × 10.402 = 184.315 kNm   (permissible 264.924) — safe

Moment = 59.897 × 8.562 = 512.843 kNm   (permissible 555.466) — safe

Moment = 75.498 × 6.486 = 489.650 kNm   (permissible 525.285) — safe

Bearing pressure = 60.801 × 8.875 = 539.602 kPa   (permissible 736.967) — safe

Deflection = 53.948 × 9.857 = 531.777 mm   (permissible 828.004) — revise section

Deflection = 44.050 × 5.433 = 239.304 mm   (permissible 397.473) — safe

Bearing pressure = 21.333 × 0.444 = 9.480 kPa   (permissible 17.077) — safe

The space programme has been developed from the functional brief agreed with the user department. Departmental areas, circulation and the gross-to-carpet ratio are tabulated in the area statement enclosed at the Annexure.

7.x Engineering Design — Working Papers — Sheet 37

Fire safety provisions follow Part 4 of the National Building Code. Means of egress, travel distances, staircase widths and the fire fighting installations have been designed for the occupancy classification of the block.

Consultations were held with the head of each clinical department on adjacency, patient flow and equipment layout. Minutes of the consultation meetings are placed at the Annexure to this chapter.

Quantities have been computed from the approved architectural and structural drawings and cross-checked against the detailed measurement sheets enclosed at the Annexure. Rates are as per the schedule of rates in force with lead and lift adjustments.

Building services comprise electrical distribution, water supply and sanitation, HVAC for the identified conditioned areas, medical gas pipeline and vertical transportation. Connected load and demand factors are given in the load schedule.

Reference: drawing no. ST-585, revision B.

7.x Engineering Design — Working Papers — Sheet 38

Room Data Sheet

Room	Department	Carpet area (sqm)	Occupancy	Floor finish
Room 779	General Medicine	89.65	10	Kota stone
Room 170	Obstetrics	78.24	26	Ceramic tile
Room 494	Surgery	20.28	15	Kota stone
Room 189	Surgery	11.33	15	Kota stone
Room 203	Obstetrics	45.66	9	Epoxy screed
Room 426	General Medicine	48.52	18	Antistatic vinyl
Room 679	Surgery	89.27	5	Kota stone
Room 823	Surgery	87.22	27	Kota stone
Room 121	Surgery	13.79	17	Vitrified tile
Room 682	Casualty	68.22	7	Kota stone
Room 239	Radiology	15.93	10	Ceramic tile
Room 379	Surgery	42.30	14	Epoxy screed
Room 458	General Medicine	45.55	1	Kota stone
Room 290	Radiology	83.83	21	Antistatic vinyl
Room 232	General Medicine	20.88	27	Antistatic vinyl
Room 358	Pathology	32.17	12	Antistatic vinyl
Room 304	Radiology	60.58	15	Kota stone
Room 374	Radiology	29.10	6	Epoxy screed
Room 372	Paediatrics	23.45	6	Ceramic tile

The space programme has been developed from the functional brief agreed with the user department. Departmental areas, circulation and the gross-to-carpet ratio are tabulated in the area statement enclosed at the Annexure.

7.x Engineering Design — Working Papers — Sheet 39

Design Computation

Deflection = 51.042 × 5.461 = 278.764 mm   (permissible 345.848) — safe

Bearing pressure = 7.063 × 8.371 = 59.118 kPa   (permissible 102.497) — safe

Shear = 73.989 × 2.727 = 201.762 kN   (permissible 276.436) — revise section

Shear = 56.411 × 8.341 = 470.521 kN   (permissible 636.182) — safe

Bearing pressure = 84.077 × 9.683 = 814.086 kPa   (permissible 1120.069) — safe

Axial load = 70.625 × 3.944 = 278.538 kN   (permissible 327.280) — safe

Shear = 60.156 × 7.669 = 461.308 kN   (permissible 782.337) — safe

Axial load = 82.334 × 11.781 = 969.957 kN   (permissible 1181.353) — safe

The structural design conforms to IS 456 for reinforced concrete and IS 1893 for earthquake resistant design. The site falls in seismic zone V and the importance factor applicable to a hospital building has been adopted.

7.x Engineering Design — Working Papers — Sheet 40

Soil investigation was carried out by rotary drilling at the locations shown on the layout plan. The safe bearing capacity adopted for design is based on the lower of the shear and settlement criteria as reported in the geotechnical report.

Barrier-free access has been provided in accordance with the Harmonised Guidelines and Standards for Universal Accessibility. Ramps, lifts, signage and accessible toilets are provided on every floor of the block.

Consultations were held with the head of each clinical department on adjacency, patient flow and equipment layout. Minutes of the consultation meetings are placed at the Annexure to this chapter.

Fire safety provisions follow Part 4 of the National Building Code. Means of egress, travel distances, staircase widths and the fire fighting installations have been designed for the occupancy classification of the block.

Reference: drawing no. ST-936, revision A.

7.x Engineering Design — Working Papers — Sheet 41

Room Data Sheet

Room	Department	Carpet area (sqm)	Occupancy	Floor finish
Room 165	Paediatrics	74.90	3	Vitrified tile
Room 764	Obstetrics	68.34	30	Vitrified tile
Room 234	Obstetrics	50.38	2	Antistatic vinyl
Room 742	Pathology	28.58	14	Ceramic tile
Room 134	Surgery	67.47	26	Kota stone
Room 837	Obstetrics	16.90	26	Antistatic vinyl
Room 862	General Medicine	23.21	4	Kota stone
Room 349	Surgery	60.43	13	Ceramic tile
Room 325	Pathology	86.69	22	Ceramic tile
Room 526	Radiology	27.55	28	Vitrified tile
Room 543	Obstetrics	9.08	30	Epoxy screed
Room 583	Surgery	14.16	12	Epoxy screed
Room 120	Surgery	23.10	27	Epoxy screed
Room 830	Paediatrics	41.63	2	Vitrified tile
Room 697	Surgery	14.22	5	Epoxy screed
Room 842	Casualty	77.67	23	Ceramic tile
Room 165	Obstetrics	51.82	23	Antistatic vinyl
Room 420	Pathology	10.69	6	Vitrified tile
Room 492	Pathology	54.74	5	Ceramic tile
Room 854	General Medicine	15.98	30	Ceramic tile
Room 439	Paediatrics	11.29	16	Vitrified tile
Room 723	General Medicine	54.99	25	Vitrified tile
Room 841	Obstetrics	51.97	14	Ceramic tile

Barrier-free access has been provided in accordance with the Harmonised Guidelines and Standards for Universal Accessibility. Ramps, lifts, signage and accessible toilets are provided on every floor of the block.

7.x Engineering Design — Working Papers — Sheet 42

Design Computation

Deflection = 48.120 × 1.979 = 95.219 mm   (permissible 102.279) — safe

Bearing pressure = 79.512 × 3.199 = 254.330 kPa   (permissible 295.025) — safe

Deflection = 40.963 × 1.765 = 72.295 mm   (permissible 92.211) — safe

Bearing pressure = 70.466 × 5.618 = 395.898 kPa   (permissible 434.051) — safe

Moment = 43.980 × 9.498 = 417.725 kNm   (permissible 490.864) — safe

Axial load = 89.575 × 9.757 = 873.951 kN   (permissible 1444.862) — safe

Moment = 48.872 × 8.257 = 403.546 kNm   (permissible 484.199) — safe

Axial load = 14.911 × 7.580 = 113.026 kN   (permissible 159.525) — safe

The structural design conforms to IS 456 for reinforced concrete and IS 1893 for earthquake resistant design. The site falls in seismic zone V and the importance factor applicable to a hospital building has been adopted.

7.x Engineering Design — Working Papers — Sheet 43

Soil investigation was carried out by rotary drilling at the locations shown on the layout plan. The safe bearing capacity adopted for design is based on the lower of the shear and settlement criteria as reported in the geotechnical report.

Fire safety provisions follow Part 4 of the National Building Code. Means of egress, travel distances, staircase widths and the fire fighting installations have been designed for the occupancy classification of the block.

Consultations were held with the head of each clinical department on adjacency, patient flow and equipment layout. Minutes of the consultation meetings are placed at the Annexure to this chapter.

Quantities have been computed from the approved architectural and structural drawings and cross-checked against the detailed measurement sheets enclosed at the Annexure. Rates are as per the schedule of rates in force with lead and lift adjustments.

Reference: drawing no. ST-192, revision B.

7.x Engineering Design — Working Papers — Sheet 44

Room Data Sheet

Room	Department	Carpet area (sqm)	Occupancy	Floor finish
Room 735	Radiology	86.12	12	Antistatic vinyl
Room 114	General Medicine	28.96	7	Antistatic vinyl
Room 621	Radiology	61.20	18	Vitrified tile
Room 899	Paediatrics	59.58	25	Kota stone
Room 645	Paediatrics	78.49	6	Ceramic tile
Room 536	Pathology	81.55	16	Antistatic vinyl
Room 891	Casualty	62.85	17	Antistatic vinyl
Room 229	Pathology	26.58	3	Epoxy screed
Room 630	Surgery	59.97	3	Ceramic tile
Room 108	Radiology	20.57	2	Vitrified tile
Room 127	Obstetrics	15.95	2	Ceramic tile
Room 236	Obstetrics	33.14	1	Ceramic tile
Room 816	Surgery	20.07	15	Vitrified tile
Room 348	Obstetrics	79.84	15	Ceramic tile
Room 215	Obstetrics	86.58	20	Antistatic vinyl
Room 305	Radiology	21.10	14	Ceramic tile
Room 646	Casualty	38.35	4	Ceramic tile
Room 853	Casualty	63.74	26	Vitrified tile
Room 473	Obstetrics	69.04	5	Antistatic vinyl
Room 561	Pathology	84.43	7	Vitrified tile
Room 691	General Medicine	91.54	28	Antistatic vinyl
Room 470	Casualty	34.57	5	Epoxy screed
Room 497	Radiology	81.19	13	Ceramic tile
Room 294	Surgery	13.93	26	Ceramic tile

The space programme has been developed from the functional brief agreed with the user department. Departmental areas, circulation and the gross-to-carpet ratio are tabulated in the area statement enclosed at the Annexure.

8 FINANCIAL ESTIMATES AND COST PROJECTIONS

Rates are derived from the Schedule of Rates in force with lead and lift adjustments. Items outside the Schedule have been rate-analysed and are placed at the detailed estimate.

Sl.	Description of item	Amount (Rs. crore)
1	Civil works — main hospital block	88.20
2	Civil works — diagnostic wing	24.60
3	Civil works — staff quarters	18.40
4	Electrical, HVAC and medical gas	31.80
5	Lifts, fire fighting and BMS	9.40
6	External development and site works	7.20
7	Contingencies at 3 per cent	5.40
8	Quality control and supervision	1.40
	TOTAL PROJECT COST	192.70

The total project cost of the scheme is Rs. 192.70 crore.

8.x Financial Estimates And Cost Projections — Working Papers — Sheet 1

Fire safety provisions follow Part 4 of the National Building Code. Means of egress, travel distances, staircase widths and the fire fighting installations have been designed for the occupancy classification of the block.

The space programme has been developed from the functional brief agreed with the user department. Departmental areas, circulation and the gross-to-carpet ratio are tabulated in the area statement enclosed at the Annexure.

Soil investigation was carried out by rotary drilling at the locations shown on the layout plan. The safe bearing capacity adopted for design is based on the lower of the shear and settlement criteria as reported in the geotechnical report.

The structural design conforms to IS 456 for reinforced concrete and IS 1893 for earthquake resistant design. The site falls in seismic zone V and the importance factor applicable to a hospital building has been adopted.

Reference: drawing no. PH-595, revision A.

8.x Financial Estimates And Cost Projections — Working Papers — Sheet 2

Bore Log Summary

Depth (m)	Strata	N value	Sample
26.5	Weathered rock	59	DS
2.7	Silty clay	24	DS
10.0	Silty clay	39	DS
3.6	Weathered rock	32	UDS
20.7	Coarse sand	13	SPT
16.9	Clayey silt	33	DS
6.6	Weathered rock	38	SPT
29.9	Coarse sand	59	SPT
1.7	Gravelly sand	4	SPT
20.7	Gravelly sand	48	UDS
10.5	Clayey silt	44	UDS
1.1	Weathered rock	24	SPT
7.3	Weathered rock	30	UDS
28.8	Gravelly sand	56	SPT
22.4	Gravelly sand	51	SPT
11.8	Silty clay	60	SPT
2.1	Coarse sand	58	SPT
1.8	Silty clay	13	SPT
15.4	Silty clay	43	DS
28.2	Medium sand	34	UDS
16.7	Weathered rock	30	DS
17.0	Weathered rock	46	SPT

Fire safety provisions follow Part 4 of the National Building Code. Means of egress, travel distances, staircase widths and the fire fighting installations have been designed for the occupancy classification of the block.

8.x Financial Estimates And Cost Projections — Working Papers — Sheet 3

Design Computation

Moment = 35.451 × 0.857 = 30.386 kNm   (permissible 37.314) — safe

Bearing pressure = 48.391 × 1.717 = 83.080 kPa   (permissible 106.084) — safe

Bearing pressure = 25.270 × 2.925 = 73.924 kPa   (permissible 101.061) — safe

Moment = 72.109 × 9.515 = 686.116 kNm   (permissible 754.486) — safe

Shear = 70.303 × 10.393 = 730.660 kN   (permissible 928.925) — safe

Bearing pressure = 15.390 × 4.299 = 66.156 kPa   (permissible 85.269) — safe

Deflection = 1.660 × 9.872 = 16.385 mm   (permissible 17.763) — safe

Quantities have been computed from the approved architectural and structural drawings and cross-checked against the detailed measurement sheets enclosed at the Annexure. Rates are as per the schedule of rates in force with lead and lift adjustments.

8.x Financial Estimates And Cost Projections — Working Papers — Sheet 4

Consultations were held with the head of each clinical department on adjacency, patient flow and equipment layout. Minutes of the consultation meetings are placed at the Annexure to this chapter.

Barrier-free access has been provided in accordance with the Harmonised Guidelines and Standards for Universal Accessibility. Ramps, lifts, signage and accessible toilets are provided on every floor of the block.

The space programme has been developed from the functional brief agreed with the user department. Departmental areas, circulation and the gross-to-carpet ratio are tabulated in the area statement enclosed at the Annexure.

Fire safety provisions follow Part 4 of the National Building Code. Means of egress, travel distances, staircase widths and the fire fighting installations have been designed for the occupancy classification of the block.

Reference: drawing no. AR-960, revision A.

8.x Financial Estimates And Cost Projections — Working Papers — Sheet 5

Detailed Measurement Sheet

Item	L (m)	B (m)	D (m)	Nos	Quantity
Item 695	6.17	4.94	1.23	20	280.283
Item 547	18.08	0.45	2.44	7	169.435
Item 761	6.72	5.04	2.79	8	74.741
Item 620	2.72	1.45	1.71	10	368.960
Item 204	25.28	5.74	1.03	16	325.652
Item 881	27.00	3.29	2.94	1	29.997
Item 567	32.61	0.35	2.64	12	290.421
Item 203	34.85	1.92	1.15	20	138.679
Item 574	20.26	1.30	2.64	23	20.120
Item 734	12.04	4.94	2.40	17	340.725
Item 884	29.05	5.98	1.96	21	186.376
Item 377	32.25	1.12	2.32	20	119.270
Item 320	6.60	3.45	1.57	8	381.633
Item 307	32.43	0.84	0.73	7	24.295
Item 107	36.94	3.57	0.94	19	226.241
Item 513	26.04	5.83	2.04	21	34.243
Item 759	22.21	5.21	2.32	5	188.236
Item 166	3.17	4.73	2.49	2	22.941
Item 701	8.59	4.82	2.09	21	98.986
Item 446	14.71	0.69	0.30	12	356.578
Item 366	8.30	1.97	2.49	13	77.871
Item 251	12.38	4.93	2.00	4	253.291
Item 151	16.49	0.48	1.15	16	9.591

Quantities have been computed from the approved architectural and structural drawings and cross-checked against the detailed measurement sheets enclosed at the Annexure. Rates are as per the schedule of rates in force with lead and lift adjustments.

8.x Financial Estimates And Cost Projections — Working Papers — Sheet 6

Design Computation

Shear = 24.455 × 2.757 = 67.423 kN   (permissible 103.195) — safe

Bearing pressure = 37.595 × 1.341 = 50.418 kPa   (permissible 66.364) — safe

Deflection = 19.202 × 4.906 = 94.195 mm   (permissible 149.670) — safe

Shear = 30.140 × 2.735 = 82.436 kN   (permissible 144.412) — safe

Moment = 66.830 × 0.840 = 56.171 kNm   (permissible 71.671) — safe

The space programme has been developed from the functional brief agreed with the user department. Departmental areas, circulation and the gross-to-carpet ratio are tabulated in the area statement enclosed at the Annexure.

8.x Financial Estimates And Cost Projections — Working Papers — Sheet 7

Barrier-free access has been provided in accordance with the Harmonised Guidelines and Standards for Universal Accessibility. Ramps, lifts, signage and accessible toilets are provided on every floor of the block.

Soil investigation was carried out by rotary drilling at the locations shown on the layout plan. The safe bearing capacity adopted for design is based on the lower of the shear and settlement criteria as reported in the geotechnical report.

Fire safety provisions follow Part 4 of the National Building Code. Means of egress, travel distances, staircase widths and the fire fighting installations have been designed for the occupancy classification of the block.

Soil investigation was carried out by rotary drilling at the locations shown on the layout plan. The safe bearing capacity adopted for design is based on the lower of the shear and settlement criteria as reported in the geotechnical report.

Reference: drawing no. ST-631, revision B.

8.x Financial Estimates And Cost Projections — Working Papers — Sheet 8

Departmental Area Statement

Department	Floor	Units	Area per unit (sqm)	Total (sqm)
Administration	Ground	12	71.92	863.07
OPD	Second	1	113.02	113.02
OPD	Ground	6	21.46	128.74
Staff Quarters	Second	6	135.87	815.20
OPD	Third	14	49.53	693.35
Staff Quarters	Third	7	72.41	506.86
Operation Theatre	First	18	24.57	442.32
ICU	Ground	14	112.46	1574.37
Diagnostics	Second	15	56.93	853.99
Staff Quarters	Ground	8	46.74	373.92
Diagnostics	Third	9	127.87	1150.84
Staff Quarters	Ground	14	69.63	974.77
Pharmacy	First	3	133.32	399.96
IPD Ward	Second	5	72.16	360.82
OPD	Second	18	62.29	1121.25
OPD	Ground	14	120.40	1685.61
Staff Quarters	First	17	50.01	850.19
Pharmacy	Third	18	108.71	1956.79
OPD	Second	3	75.02	225.06
ICU	Third	17	90.63	1540.77
OPD	Second	16	21.12	337.96
Operation Theatre	Third	6	13.16	78.98
Administration	Third	16	24.50	392.00
OPD	Ground	17	112.07	1905.26
Operation Theatre	Ground	7	97.51	682.57
Operation Theatre	Third	8	52.85	422.79

The space programme has been developed from the functional brief agreed with the user department. Departmental areas, circulation and the gross-to-carpet ratio are tabulated in the area statement enclosed at the Annexure.

8.x Financial Estimates And Cost Projections — Working Papers — Sheet 9

Design Computation

Shear = 22.394 × 10.761 = 240.973 kN   (permissible 396.185) — safe

Bearing pressure = 25.225 × 3.285 = 82.874 kPa   (permissible 136.775) — safe

Axial load = 21.282 × 10.689 = 227.480 kN   (permissible 320.012) — safe

Shear = 80.655 × 3.982 = 321.194 kN   (permissible 415.056) — safe

Axial load = 31.892 × 0.631 = 20.118 kN   (permissible 32.055) — safe

Bearing pressure = 55.377 × 5.289 = 292.901 kPa   (permissible 431.229) — safe

Axial load = 24.122 × 3.932 = 94.849 kN   (permissible 105.563) — safe

Barrier-free access has been provided in accordance with the Harmonised Guidelines and Standards for Universal Accessibility. Ramps, lifts, signage and accessible toilets are provided on every floor of the block.

8.x Financial Estimates And Cost Projections — Working Papers — Sheet 10

Fire safety provisions follow Part 4 of the National Building Code. Means of egress, travel distances, staircase widths and the fire fighting installations have been designed for the occupancy classification of the block.

Building services comprise electrical distribution, water supply and sanitation, HVAC for the identified conditioned areas, medical gas pipeline and vertical transportation. Connected load and demand factors are given in the load schedule.

Barrier-free access has been provided in accordance with the Harmonised Guidelines and Standards for Universal Accessibility. Ramps, lifts, signage and accessible toilets are provided on every floor of the block.

The structural design conforms to IS 456 for reinforced concrete and IS 1893 for earthquake resistant design. The site falls in seismic zone V and the importance factor applicable to a hospital building has been adopted.

Reference: drawing no. ST-908, revision A.

8.x Financial Estimates And Cost Projections — Working Papers — Sheet 11

Bore Log Summary

Depth (m)	Strata	N value	Sample
29.4	Weathered rock	5	DS
4.7	Gravelly sand	51	SPT
25.4	Silty clay	44	SPT
7.4	Weathered rock	56	DS
12.9	Medium sand	60	UDS
26.7	Weathered rock	32	UDS
5.5	Coarse sand	14	DS
17.5	Clayey silt	39	DS
9.4	Weathered rock	12	DS
0.5	Clayey silt	9	DS
17.4	Gravelly sand	20	DS
10.8	Clayey silt	23	UDS
6.5	Silty clay	57	UDS
14.5	Silty clay	9	UDS
13.2	Coarse sand	25	SPT
29.5	Medium sand	41	SPT
18.6	Gravelly sand	29	SPT
18.7	Gravelly sand	15	UDS
18.5	Silty clay	51	SPT
26.1	Gravelly sand	58	DS
24.1	Clayey silt	7	SPT
27.1	Clayey silt	59	SPT
0.9	Weathered rock	10	UDS

Quantities have been computed from the approved architectural and structural drawings and cross-checked against the detailed measurement sheets enclosed at the Annexure. Rates are as per the schedule of rates in force with lead and lift adjustments.

8.x Financial Estimates And Cost Projections — Working Papers — Sheet 12

Design Computation

Shear = 72.225 × 10.733 = 775.207 kN   (permissible 1328.106) — safe

Moment = 24.303 × 8.176 = 198.697 kNm   (permissible 240.734) — safe

Bearing pressure = 65.823 × 2.737 = 180.153 kPa   (permissible 289.983) — safe

Deflection = 20.194 × 7.471 = 150.864 mm   (permissible 198.003) — safe

Bearing pressure = 84.055 × 2.941 = 247.188 kPa   (permissible 372.565) — safe

Quantities have been computed from the approved architectural and structural drawings and cross-checked against the detailed measurement sheets enclosed at the Annexure. Rates are as per the schedule of rates in force with lead and lift adjustments.

8.x Financial Estimates And Cost Projections — Working Papers — Sheet 13

The structural design conforms to IS 456 for reinforced concrete and IS 1893 for earthquake resistant design. The site falls in seismic zone V and the importance factor applicable to a hospital building has been adopted.

Soil investigation was carried out by rotary drilling at the locations shown on the layout plan. The safe bearing capacity adopted for design is based on the lower of the shear and settlement criteria as reported in the geotechnical report.

Consultations were held with the head of each clinical department on adjacency, patient flow and equipment layout. Minutes of the consultation meetings are placed at the Annexure to this chapter.

Fire safety provisions follow Part 4 of the National Building Code. Means of egress, travel distances, staircase widths and the fire fighting installations have been designed for the occupancy classification of the block.

Reference: drawing no. EL-110, revision B.

8.x Financial Estimates And Cost Projections — Working Papers — Sheet 14

Equipment Schedule

Item	Department	Nos	Unit rate (Rs)	Amount (Rs)
Hospital bed	OT	1	2,390,451	2,390,451
Autoclave	CSSD	29	1,199,240	34,777,960
Defibrillator	Casualty	12	1,259,304	15,111,648
Hospital bed	Casualty	13	2,041,653	26,541,489
Hospital bed	Casualty	26	538,398	13,998,348
Defibrillator	Casualty	18	1,323,927	23,830,686
Ultrasound	Ward	30	1,261,398	37,841,940
Autoclave	Ward	5	1,091,110	5,455,550
Patient monitor	Radiology	8	1,126,323	9,010,584
OT table	Ward	12	1,205,529	14,466,348
Hospital bed	Casualty	26	1,643,414	42,728,764
Patient monitor	Casualty	10	1,460,824	14,608,240
Ultrasound	Casualty	19	2,215,393	42,092,467
Hospital bed	ICU	31	1,419,361	44,000,191
Ultrasound	ICU	33	1,952,678	64,438,374
OT table	ICU	35	1,366,225	47,817,875
X-ray unit	Casualty	28	257,315	7,204,820
Defibrillator	ICU	30	1,477,494	44,324,820
Autoclave	Casualty	33	800,802	26,426,466

The structural design conforms to IS 456 for reinforced concrete and IS 1893 for earthquake resistant design. The site falls in seismic zone V and the importance factor applicable to a hospital building has been adopted.

8.x Financial Estimates And Cost Projections — Working Papers — Sheet 15

Design Computation

Shear = 53.625 × 7.843 = 420.565 kN   (permissible 767.311) — safe

Deflection = 28.510 × 7.665 = 218.530 mm   (permissible 288.392) — safe

Bearing pressure = 88.704 × 6.386 = 566.430 kPa   (permissible 1033.685) — safe

Bearing pressure = 19.786 × 10.539 = 208.521 kPa   (permissible 289.379) — safe

Axial load = 41.339 × 5.906 = 244.160 kN   (permissible 286.954) — safe

Quantities have been computed from the approved architectural and structural drawings and cross-checked against the detailed measurement sheets enclosed at the Annexure. Rates are as per the schedule of rates in force with lead and lift adjustments.

8.x Financial Estimates And Cost Projections — Working Papers — Sheet 16

Building services comprise electrical distribution, water supply and sanitation, HVAC for the identified conditioned areas, medical gas pipeline and vertical transportation. Connected load and demand factors are given in the load schedule.

Barrier-free access has been provided in accordance with the Harmonised Guidelines and Standards for Universal Accessibility. Ramps, lifts, signage and accessible toilets are provided on every floor of the block.

Soil investigation was carried out by rotary drilling at the locations shown on the layout plan. The safe bearing capacity adopted for design is based on the lower of the shear and settlement criteria as reported in the geotechnical report.

The structural design conforms to IS 456 for reinforced concrete and IS 1893 for earthquake resistant design. The site falls in seismic zone V and the importance factor applicable to a hospital building has been adopted.

Reference: drawing no. AR-754, revision B.

8.x Financial Estimates And Cost Projections — Working Papers — Sheet 17

Detailed Measurement Sheet

Item	L (m)	B (m)	D (m)	Nos	Quantity
Item 714	25.58	1.18	1.27	24	100.221
Item 214	23.68	2.02	2.74	8	374.676
Item 202	10.48	1.53	1.86	13	36.993
Item 206	28.67	5.31	0.69	9	193.370
Item 276	9.28	5.02	2.60	23	4.210
Item 332	12.02	3.68	0.66	21	124.071
Item 105	7.20	2.19	1.00	22	65.568
Item 349	39.89	1.78	1.92	17	207.934
Item 174	7.46	1.69	2.66	4	46.216
Item 776	20.32	0.63	0.92	15	384.863
Item 174	7.97	4.09	2.42	4	195.035
Item 703	19.20	4.58	1.19	12	79.948
Item 143	33.52	1.36	0.42	20	93.696
Item 798	25.83	3.77	1.73	5	99.583
Item 371	15.00	1.77	2.55	13	130.732
Item 143	36.12	0.66	0.98	4	329.712
Item 451	7.33	4.81	1.21	15	275.108
Item 850	32.85	4.57	3.00	5	192.060
Item 674	13.10	4.09	1.72	4	168.107

The structural design conforms to IS 456 for reinforced concrete and IS 1893 for earthquake resistant design. The site falls in seismic zone V and the importance factor applicable to a hospital building has been adopted.

8.x Financial Estimates And Cost Projections — Working Papers — Sheet 18

Design Computation

Deflection = 9.746 × 8.097 = 78.915 mm   (permissible 122.494) — safe

Bearing pressure = 41.898 × 10.109 = 423.543 kPa   (permissible 458.496) — safe

Shear = 57.702 × 0.668 = 38.527 kN   (permissible 42.866) — safe

Axial load = 58.853 × 5.732 = 337.367 kN   (permissible 383.100) — revise section

Axial load = 86.438 × 4.110 = 355.222 kN   (permissible 488.117) — safe

Moment = 30.655 × 6.048 = 185.391 kNm   (permissible 264.152) — safe

Barrier-free access has been provided in accordance with the Harmonised Guidelines and Standards for Universal Accessibility. Ramps, lifts, signage and accessible toilets are provided on every floor of the block.

9 REVENUE STREAMS

User charges are levied as per the state schedule of rates for government hospitals, with exemptions for beneficiaries of the state health assurance scheme. Estimated annual collection is Rs. 4.20 crore, retained by the Hospital Management Society under the Rogi Kalyan Samiti framework.

9.x Revenue Streams — Working Papers — Sheet 1

Building services comprise electrical distribution, water supply and sanitation, HVAC for the identified conditioned areas, medical gas pipeline and vertical transportation. Connected load and demand factors are given in the load schedule.

The structural design conforms to IS 456 for reinforced concrete and IS 1893 for earthquake resistant design. The site falls in seismic zone V and the importance factor applicable to a hospital building has been adopted.

The structural design conforms to IS 456 for reinforced concrete and IS 1893 for earthquake resistant design. The site falls in seismic zone V and the importance factor applicable to a hospital building has been adopted.

Building services comprise electrical distribution, water supply and sanitation, HVAC for the identified conditioned areas, medical gas pipeline and vertical transportation. Connected load and demand factors are given in the load schedule.

Reference: drawing no. PH-963, revision A.

9.x Revenue Streams — Working Papers — Sheet 2

Equipment Schedule

Item	Department	Nos	Unit rate (Rs)	Amount (Rs)
Ultrasound	Ward	8	2,229,413	17,835,304
Ventilator	Ward	30	151,650	4,549,500
Ventilator	Radiology	28	299,133	8,375,724
Defibrillator	ICU	25	1,791,328	44,783,200
Ventilator	CSSD	11	1,799,861	19,798,471
Patient monitor	Radiology	10	2,045,722	20,457,220
Ultrasound	ICU	15	742,343	11,135,145
Hospital bed	ICU	38	351,252	13,347,576
Ultrasound	Radiology	14	1,414,471	19,802,594
Autoclave	Radiology	17	1,391,763	23,659,971
X-ray unit	Casualty	36	1,408,366	50,701,176
Hospital bed	Ward	10	1,564,920	15,649,200
Hospital bed	CSSD	19	2,309,150	43,873,850
Hospital bed	Ward	30	1,391,116	41,733,480
Defibrillator	Casualty	29	186,762	5,416,098
Patient monitor	CSSD	12	224,616	2,695,392
Defibrillator	ICU	29	1,209,088	35,063,552
X-ray unit	CSSD	8	1,251,026	10,008,208
OT table	Radiology	25	1,215,001	30,375,025
Hospital bed	Casualty	18	1,439,195	25,905,510
Patient monitor	OT	13	1,109,058	14,417,754
Ultrasound	Radiology	30	285,402	8,562,060

Quantities have been computed from the approved architectural and structural drawings and cross-checked against the detailed measurement sheets enclosed at the Annexure. Rates are as per the schedule of rates in force with lead and lift adjustments.

9.x Revenue Streams — Working Papers — Sheet 3

Design Computation

Deflection = 26.150 × 5.813 = 152.002 mm   (permissible 202.027) — safe

Moment = 89.882 × 3.211 = 288.626 kNm   (permissible 369.960) — safe

Bearing pressure = 51.461 × 1.682 = 86.538 kPa   (permissible 133.796) — safe

Axial load = 25.618 × 4.671 = 119.665 kN   (permissible 171.792) — safe

Moment = 7.339 × 5.109 = 37.493 kNm   (permissible 53.178) — safe

The space programme has been developed from the functional brief agreed with the user department. Departmental areas, circulation and the gross-to-carpet ratio are tabulated in the area statement enclosed at the Annexure.

9.x Revenue Streams — Working Papers — Sheet 4

Building services comprise electrical distribution, water supply and sanitation, HVAC for the identified conditioned areas, medical gas pipeline and vertical transportation. Connected load and demand factors are given in the load schedule.

The structural design conforms to IS 456 for reinforced concrete and IS 1893 for earthquake resistant design. The site falls in seismic zone V and the importance factor applicable to a hospital building has been adopted.

Barrier-free access has been provided in accordance with the Harmonised Guidelines and Standards for Universal Accessibility. Ramps, lifts, signage and accessible toilets are provided on every floor of the block.

Quantities have been computed from the approved architectural and structural drawings and cross-checked against the detailed measurement sheets enclosed at the Annexure. Rates are as per the schedule of rates in force with lead and lift adjustments.

Reference: drawing no. PH-925, revision B.

9.x Revenue Streams — Working Papers — Sheet 5

Equipment Schedule

Item	Department	Nos	Unit rate (Rs)	Amount (Rs)
Patient monitor	Ward	1	868,707	868,707
X-ray unit	CSSD	9	1,016,450	9,148,050
Autoclave	CSSD	23	352,229	8,101,267
Autoclave	OT	9	1,698,070	15,282,630
Autoclave	CSSD	40	2,028,092	81,123,680
Patient monitor	ICU	13	2,169,927	28,209,051
Hospital bed	Ward	30	259,532	7,785,960
OT table	Radiology	31	2,387,127	74,000,937
Patient monitor	Radiology	17	271,512	4,615,704
Autoclave	ICU	37	1,460,771	54,048,527
Ultrasound	ICU	39	125,580	4,897,620
Patient monitor	Casualty	13	2,185,233	28,408,029
Hospital bed	CSSD	28	134,962	3,778,936
OT table	CSSD	13	1,872,901	24,347,713
Ultrasound	Casualty	5	1,206,640	6,033,200
Hospital bed	ICU	3	1,146,579	3,439,737
OT table	OT	27	1,575,024	42,525,648
Ventilator	OT	33	1,284,907	42,401,931
X-ray unit	Radiology	19	1,326,251	25,198,769
Patient monitor	Casualty	6	262,228	1,573,368
X-ray unit	ICU	19	294,228	5,590,332
Ventilator	CSSD	20	498,118	9,962,360
Autoclave	Radiology	36	2,131,427	76,731,372
X-ray unit	CSSD	23	72,623	1,670,329

The structural design conforms to IS 456 for reinforced concrete and IS 1893 for earthquake resistant design. The site falls in seismic zone V and the importance factor applicable to a hospital building has been adopted.

9.x Revenue Streams — Working Papers — Sheet 6

Design Computation

Shear = 30.176 × 6.993 = 211.032 kN   (permissible 240.498) — safe

Moment = 21.683 × 7.359 = 159.565 kNm   (permissible 236.868) — safe

Axial load = 39.882 × 1.873 = 74.708 kN   (permissible 109.780) — safe

Axial load = 31.100 × 3.722 = 115.751 kN   (permissible 165.907) — safe

Deflection = 24.794 × 10.377 = 257.283 mm   (permissible 280.722) — safe

Bearing pressure = 12.673 × 10.961 = 138.903 kPa   (permissible 249.330) — safe

Shear = 64.539 × 5.492 = 354.438 kN   (permissible 492.771) — safe

Bearing pressure = 41.903 × 7.720 = 323.475 kPa   (permissible 404.789) — revise section

Barrier-free access has been provided in accordance with the Harmonised Guidelines and Standards for Universal Accessibility. Ramps, lifts, signage and accessible toilets are provided on every floor of the block.

9.x Revenue Streams — Working Papers — Sheet 7

Building services comprise electrical distribution, water supply and sanitation, HVAC for the identified conditioned areas, medical gas pipeline and vertical transportation. Connected load and demand factors are given in the load schedule.

Building services comprise electrical distribution, water supply and sanitation, HVAC for the identified conditioned areas, medical gas pipeline and vertical transportation. Connected load and demand factors are given in the load schedule.

The space programme has been developed from the functional brief agreed with the user department. Departmental areas, circulation and the gross-to-carpet ratio are tabulated in the area statement enclosed at the Annexure.

Soil investigation was carried out by rotary drilling at the locations shown on the layout plan. The safe bearing capacity adopted for design is based on the lower of the shear and settlement criteria as reported in the geotechnical report.

Reference: drawing no. AR-780, revision A.

9.x Revenue Streams — Working Papers — Sheet 8

Departmental Area Statement

Department	Floor	Units	Area per unit (sqm)	Total (sqm)
Diagnostics	Third	17	78.34	1331.72
Operation Theatre	Ground	7	86.99	608.95
Administration	Third	5	54.59	272.93
Pharmacy	First	8	84.13	673.06
Pharmacy	Third	5	70.80	354.02
ICU	Ground	10	40.96	409.63
IPD Ward	Ground	3	78.35	235.05
Operation Theatre	First	7	82.28	575.98
Administration	First	4	53.20	212.80
Diagnostics	Third	8	115.11	920.89
Pharmacy	First	9	105.36	948.23
Administration	Ground	18	95.50	1719.00
Administration	Third	9	35.98	323.82
Diagnostics	Third	7	43.07	301.47
Pharmacy	First	18	18.07	325.30
OPD	Third	3	15.10	45.29
OPD	First	17	52.27	888.53
Administration	First	6	120.99	725.91
ICU	Second	14	126.67	1773.40
Diagnostics	Ground	14	65.58	918.12
ICU	Ground	14	83.05	1162.64

The space programme has been developed from the functional brief agreed with the user department. Departmental areas, circulation and the gross-to-carpet ratio are tabulated in the area statement enclosed at the Annexure.

10 COST BENEFIT ANALYSIS AND INVESTMENT CRITERIA

The appraisal has been carried out over a 30 year evaluation period at a discount rate of 12 per cent.

Investment criterion	Value
Internal Rate of Return	13.8 per cent
Net Present Value at 12 per cent	Rs. 74.20 crore
Benefit Cost Ratio	1.44
Payback period	12.1 years

The Internal Rate of Return of the project works out to 13.8 per cent, against the threshold of 12 per cent applied to projects of this class. The project is therefore viable and is recommended for sanction.

The year-wise cash flow statement on which this computation rests is enclosed at the annexure.

10.x Cost Benefit Analysis And Investment Criteria — Working Papers — Sheet 1

The structural design conforms to IS 456 for reinforced concrete and IS 1893 for earthquake resistant design. The site falls in seismic zone V and the importance factor applicable to a hospital building has been adopted.

Fire safety provisions follow Part 4 of the National Building Code. Means of egress, travel distances, staircase widths and the fire fighting installations have been designed for the occupancy classification of the block.

Soil investigation was carried out by rotary drilling at the locations shown on the layout plan. The safe bearing capacity adopted for design is based on the lower of the shear and settlement criteria as reported in the geotechnical report.

Quantities have been computed from the approved architectural and structural drawings and cross-checked against the detailed measurement sheets enclosed at the Annexure. Rates are as per the schedule of rates in force with lead and lift adjustments.

Reference: drawing no. EL-886, revision A.

10.x Cost Benefit Analysis And Investment Criteria — Working Papers — Sheet 2

Bore Log Summary

Depth (m)	Strata	N value	Sample
26.6	Clayey silt	35	DS
29.1	Coarse sand	40	DS
4.8	Clayey silt	47	DS
15.7	Silty clay	17	DS
4.4	Weathered rock	22	UDS
18.9	Silty clay	39	DS
11.1	Coarse sand	52	DS
13.9	Silty clay	32	UDS
8.0	Gravelly sand	45	UDS
29.3	Weathered rock	9	SPT
0.7	Medium sand	19	SPT
12.2	Medium sand	14	UDS
25.4	Coarse sand	21	UDS
2.9	Coarse sand	36	UDS
23.2	Medium sand	45	UDS
29.0	Clayey silt	59	DS
27.2	Gravelly sand	52	DS
26.2	Medium sand	12	DS
9.8	Weathered rock	12	SPT
23.3	Silty clay	54	DS
25.7	Medium sand	11	UDS

The space programme has been developed from the functional brief agreed with the user department. Departmental areas, circulation and the gross-to-carpet ratio are tabulated in the area statement enclosed at the Annexure.

10.x Cost Benefit Analysis And Investment Criteria — Working Papers — Sheet 3

Design Computation

Moment = 68.842 × 8.271 = 569.399 kNm   (permissible 1066.006) — safe

Axial load = 59.459 × 7.650 = 454.886 kN   (permissible 749.052) — safe

Shear = 70.688 × 9.009 = 636.847 kN   (permissible 1047.006) — revise section

Bearing pressure = 1.636 × 3.735 = 6.112 kPa   (permissible 6.600) — safe

Bearing pressure = 28.121 × 10.775 = 303.000 kPa   (permissible 326.589) — safe

The structural design conforms to IS 456 for reinforced concrete and IS 1893 for earthquake resistant design. The site falls in seismic zone V and the importance factor applicable to a hospital building has been adopted.

10.x Cost Benefit Analysis And Investment Criteria — Working Papers — Sheet 4

Barrier-free access has been provided in accordance with the Harmonised Guidelines and Standards for Universal Accessibility. Ramps, lifts, signage and accessible toilets are provided on every floor of the block.

Quantities have been computed from the approved architectural and structural drawings and cross-checked against the detailed measurement sheets enclosed at the Annexure. Rates are as per the schedule of rates in force with lead and lift adjustments.

Soil investigation was carried out by rotary drilling at the locations shown on the layout plan. The safe bearing capacity adopted for design is based on the lower of the shear and settlement criteria as reported in the geotechnical report.

Quantities have been computed from the approved architectural and structural drawings and cross-checked against the detailed measurement sheets enclosed at the Annexure. Rates are as per the schedule of rates in force with lead and lift adjustments.

Reference: drawing no. PH-533, revision C.

10.x Cost Benefit Analysis And Investment Criteria — Working Papers — Sheet 5

Equipment Schedule

Item	Department	Nos	Unit rate (Rs)	Amount (Rs)
Defibrillator	CSSD	25	1,444,025	36,100,625
Ultrasound	Ward	5	2,288,268	11,441,340
Defibrillator	ICU	8	1,439,554	11,516,432
Defibrillator	CSSD	2	629,761	1,259,522
Autoclave	OT	8	1,169,746	9,357,968
Hospital bed	CSSD	38	1,056,476	40,146,088
Ventilator	Radiology	7	1,234,993	8,644,951
Hospital bed	Casualty	20	74,305	1,486,100
Ventilator	Casualty	7	1,921,814	13,452,698
Patient monitor	CSSD	2	634,134	1,268,268
Patient monitor	ICU	35	1,819,280	63,674,800
Autoclave	ICU	29	1,489,685	43,200,865
Defibrillator	OT	4	1,089,158	4,356,632
Ventilator	Casualty	36	2,284,575	82,244,700
Patient monitor	CSSD	11	1,499,203	16,491,233
X-ray unit	Casualty	17	1,037,084	17,630,428
OT table	ICU	36	2,223,537	80,047,332
Ultrasound	OT	2	314,003	628,006
X-ray unit	Radiology	35	552,403	19,334,105
Autoclave	Ward	36	730,221	26,287,956
Ventilator	Radiology	39	2,091,798	81,580,122

The space programme has been developed from the functional brief agreed with the user department. Departmental areas, circulation and the gross-to-carpet ratio are tabulated in the area statement enclosed at the Annexure.

10.x Cost Benefit Analysis And Investment Criteria — Working Papers — Sheet 6

Design Computation

Axial load = 26.863 × 11.367 = 305.363 kN (permissible 514.886) — safe

Bearing pressure = 33.894 × 5.639 = 191.126 kPa (permissible 319.476) — safe

Axial load = 11.643 × 9.564 = 111.360 kN (permissible 164.467) — safe

Axial load = 89.764 × 9.961 = 894.152 kN (permissible 1238.730) — safe

Axial load = 76.432 × 7.476 = 571.433 kN (permissible 911.601) — safe

Axial load = 66.069 × 11.965 = 790.518 kN (permissible 1469.663) — safe

Moment = 18.782 × 11.651 = 218.818 kNm (permissible 274.441) — safe

Axial load = 81.477 × 0.667 = 54.343 kN (permissible 101.108) — safe

The structural design conforms to IS 456 for reinforced concrete and IS 1893 for earthquake resistant design. The site falls in seismic zone V and the importance factor applicable to a hospital building has been adopted.

10.x Cost Benefit Analysis And Investment Criteria — Working Papers — Sheet 7

Consultations were held with the head of each clinical department on adjacency, patient flow and equipment layout. Minutes of the consultation meetings are placed at the Annexure to this chapter.

Soil investigation was carried out by rotary drilling at the locations shown on the layout plan. The safe bearing capacity adopted for design is based on the lower of the shear and settlement criteria as reported in the geotechnical report.

The structural design conforms to IS 456 for reinforced concrete and IS 1893 for earthquake resistant design. The site falls in seismic zone V and the importance factor applicable to a hospital building has been adopted.

Consultations were held with the head of each clinical department on adjacency, patient flow and equipment layout. Minutes of the consultation meetings are placed at the Annexure to this chapter.

Reference: drawing no. PH-151, revision B.

10.x Cost Benefit Analysis And Investment Criteria — Working Papers — Sheet 8

Bore Log Summary

Depth (m)	Strata	N value	Sample
6.0	Gravelly sand	31	SPT
12.0	Coarse sand	31	SPT
26.6	Medium sand	43	UDS
27.4	Gravelly sand	44	DS
21.7	Coarse sand	40	UDS
28.3	Weathered rock	25	SPT
25.4	Weathered rock	13	SPT
20.6	Gravelly sand	57	UDS
26.5	Silty clay	21	SPT
25.5	Coarse sand	55	UDS
15.1	Clayey silt	41	SPT
16.2	Coarse sand	16	DS
29.4	Gravelly sand	22	SPT
12.7	Clayey silt	53	SPT
17.6	Clayey silt	57	SPT
23.0	Silty clay	21	SPT
0.6	Silty clay	58	SPT

The structural design conforms to IS 456 for reinforced concrete and IS 1893 for earthquake resistant design. The site falls in seismic zone V and the importance factor applicable to a hospital building has been adopted.

10.x Cost Benefit Analysis And Investment Criteria — Working Papers — Sheet 9

Design Computation

Axial load = 11.994 × 4.933 = 59.170 kN   (permissible 67.277) — safe

Deflection = 29.865 × 9.394 = 280.537 mm   (permissible 336.578) — safe

Deflection = 62.875 × 10.897 = 685.146 mm   (permissible 818.927) — safe

Bearing pressure = 84.497 × 1.499 = 126.645 kPa   (permissible 179.646) — revise section

Moment = 8.657 × 6.768 = 58.591 kNm   (permissible 69.900) — safe

Shear = 12.626 × 9.268 = 117.026 kN   (permissible 164.616) — safe

Deflection = 34.661 × 2.850 = 98.799 mm   (permissible 105.932) — safe

Axial load = 27.853 × 1.445 = 40.260 kN   (permissible 69.816) — safe

Fire safety provisions follow Part 4 of the National Building Code. Means of egress, travel distances, staircase widths and the fire fighting installations have been designed for the occupancy classification of the block.

10.x Cost Benefit Analysis And Investment Criteria — Working Papers — Sheet 10

Barrier-free access has been provided in accordance with the Harmonised Guidelines and Standards for Universal Accessibility. Ramps, lifts, signage and accessible toilets are provided on every floor of the block.

The space programme has been developed from the functional brief agreed with the user department. Departmental areas, circulation and the gross-to-carpet ratio are tabulated in the area statement enclosed at the Annexure.

Fire safety provisions follow Part 4 of the National Building Code. Means of egress, travel distances, staircase widths and the fire fighting installations have been designed for the occupancy classification of the block.

Soil investigation was carried out by rotary drilling at the locations shown on the layout plan. The safe bearing capacity adopted for design is based on the lower of the shear and settlement criteria as reported in the geotechnical report.

Reference: drawing no. AR-669, revision B.

10.x Cost Benefit Analysis And Investment Criteria — Working Papers — Sheet 11

Detailed Measurement Sheet

Item	L (m)	B (m)	D (m)	Nos	Quantity
Item 118	23.39	4.60	1.86	6	31.616
Item 718	12.00	0.91	0.96	13	324.132
Item 380	20.41	1.25	0.37	2	227.580
Item 408	11.81	1.76	1.10	15	327.115
Item 275	33.20	4.31	2.52	23	301.718
Item 870	33.09	5.23	2.90	13	302.835
Item 559	35.72	1.45	2.55	3	138.013
Item 569	29.24	5.10	0.90	13	258.370
Item 810	36.07	4.75	0.41	13	398.667
Item 191	2.08	5.78	2.64	17	114.469
Item 725	32.85	0.88	0.84	8	161.900
Item 471	14.50	0.81	0.84	7	389.878
Item 204	12.66	0.36	1.98	16	205.867
Item 346	33.49	4.84	2.19	19	273.948
Item 453	36.24	0.42	0.63	11	321.797
Item 405	17.56	0.84	2.65	6	256.758

Barrier-free access has been provided in accordance with the Harmonised Guidelines and Standards for Universal Accessibility. Ramps, lifts, signage and accessible toilets are provided on every floor of the block.

10.x Cost Benefit Analysis And Investment Criteria — Working Papers — Sheet 12

Design Computation

Moment = 71.527 × 7.043 = 503.744 kNm   (permissible 924.036) — revise section

Axial load = 86.701 × 3.743 = 324.533 kN   (permissible 474.065) — safe

Deflection = 19.706 × 1.946 = 38.344 mm   (permissible 40.781) — safe

Moment = 36.206 × 0.688 = 24.927 kNm   (permissible 29.159) — safe

Axial load = 17.847 × 9.794 = 174.801 kN   (permissible 297.523) — safe

Deflection = 39.297 × 9.229 = 362.684 mm   (permissible 511.380) — safe

Axial load = 79.051 × 9.497 = 750.724 kN   (permissible 950.272) — safe

The space programme has been developed from the functional brief agreed with the user department. Departmental areas, circulation and the gross-to-carpet ratio are tabulated in the area statement enclosed at the Annexure.

11A Supplementary Notes — Sheet 1

Fire safety provisions follow Part 4 of the National Building Code. Means of egress, travel distances, staircase widths and the fire fighting installations have been designed for the occupancy classification of the block.

The structural design conforms to IS 456 for reinforced concrete and IS 1893 for earthquake resistant design. The site falls in seismic zone V and the importance factor applicable to a hospital building has been adopted.

Barrier-free access has been provided in accordance with the Harmonised Guidelines and Standards for Universal Accessibility. Ramps, lifts, signage and accessible toilets are provided on every floor of the block.

Soil investigation was carried out by rotary drilling at the locations shown on the layout plan. The safe bearing capacity adopted for design is based on the lower of the shear and settlement criteria as reported in the geotechnical report.

Reference: drawing no. AR-538, revision C.

11A Supplementary Notes — Sheet 2

Departmental Area Statement

Department	Floor	Units	Area per unit (sqm)	Total (sqm)
Diagnostics	First	9	132.41	1191.66
Administration	Second	16	86.64	1386.18
OPD	Second	14	108.00	1512.00
OPD	Third	7	28.72	201.02
OPD	First	14	127.37	1783.14
Administration	Third	7	136.40	954.81
Operation Theatre	Ground	18	132.66	2387.89
IPD Ward	First	1	33.70	33.70
Staff Quarters	Second	6	119.39	716.35
ICU	First	13	89.87	1168.26
Operation Theatre	Third	8	65.63	525.01
OPD	Ground	10	81.00	809.95
OPD	Ground	15	123.69	1855.34
IPD Ward	Second	9	90.96	818.63
Diagnostics	Third	17	33.56	570.49
Operation Theatre	Third	6	56.65	339.90
Pharmacy	First	3	134.19	402.58
ICU	Ground	9	35.55	319.94

The space programme has been developed from the functional brief agreed with the user department. Departmental areas, circulation and the gross-to-carpet ratio are tabulated in the area statement enclosed at the Annexure.

11A Supplementary Notes — Sheet 3

Design Computation

Shear = 64.266 × 10.008 = 643.147 kN   (permissible 1197.529) — safe

Shear = 48.179 × 1.165 = 56.141 kN   (permissible 64.539) — safe

Bearing pressure = 81.853 × 5.685 = 465.364 kPa   (permissible 775.708) — safe

Shear = 80.025 × 10.267 = 821.579 kN   (permissible 1170.852) — safe

Axial load = 60.535 × 2.890 = 174.916 kN   (permissible 313.587) — safe

Moment = 2.819 × 8.193 = 23.094 kNm   (permissible 39.847) — safe

Axial load = 53.947 × 3.763 = 203.021 kN   (permissible 232.292) — safe

Shear = 1.030 × 3.405 = 3.507 kN   (permissible 6.306) — safe

The space programme has been developed from the functional brief agreed with the user department. Departmental areas, circulation and the gross-to-carpet ratio are tabulated in the area statement enclosed at the Annexure.

11A Supplementary Notes — Sheet 4

Barrier-free access has been provided in accordance with the Harmonised Guidelines and Standards for Universal Accessibility. Ramps, lifts, signage and accessible toilets are provided on every floor of the block.

Consultations were held with the head of each clinical department on adjacency, patient flow and equipment layout. Minutes of the consultation meetings are placed at the Annexure to this chapter.

The space programme has been developed from the functional brief agreed with the user department. Departmental areas, circulation and the gross-to-carpet ratio are tabulated in the area statement enclosed at the Annexure.

Building services comprise electrical distribution, water supply and sanitation, HVAC for the identified conditioned areas, medical gas pipeline and vertical transportation. Connected load and demand factors are given in the load schedule.

Reference: drawing no. PH-982, revision A.

11A Supplementary Notes — Sheet 5

Electrical Load Schedule

Panel	Description	Connected load (kW)	Demand factor	Demand (kVA)
DB-34	Lighting	129.98	0.46	67.05
DB-28	Sterilisation	175.53	0.44	86.33
DB-08	Sterilisation	59.23	0.59	38.51
DB-23	Lighting	75.93	0.57	47.71
DB-39	Lighting	87.55	0.41	40.06
DB-36	Medical equipment	74.74	0.50	41.51
DB-03	Medical equipment	47.13	0.64	33.36
DB-29	Lighting	73.64	0.44	35.79
DB-37	Pumps	100.73	0.65	72.51
DB-24	Lifts	160.17	0.60	107.17
DB-10	Pumps	26.00	0.85	24.52
DB-36	Medical equipment	113.22	0.50	63.12
DB-06	Lifts	43.13	0.79	37.76
DB-05	Power outlets	153.56	0.55	93.57
DB-33	Pumps	78.80	0.72	63.32
DB-15	HVAC	137.44	0.40	61.28
DB-14	Lifts	82.60	0.80	73.43
DB-38	Lighting	67.55	0.71	53.66
DB-37	Sterilisation	119.51	0.68	90.35
DB-25	HVAC	158.18	0.92	161.09
DB-38	Lighting	52.69	0.56	32.58
DB-14	Medical equipment	15.01	0.58	9.63
DB-11	Lighting	6.20	0.47	3.21
DB-29	Lifts	166.51	0.68	125.95
DB-08	Medical equipment	14.19	0.95	14.91

Barrier-free access has been provided in accordance with the Harmonised Guidelines and Standards for Universal Accessibility. Ramps, lifts, signage and accessible toilets are provided on every floor of the block.

11A Supplementary Notes — Sheet 6

Design Computation

Axial load = 55.649 × 3.263 = 181.571 kN   (permissible 341.723) — safe

Bearing pressure = 55.925 × 3.417 = 191.073 kPa   (permissible 215.270) — safe

Shear = 35.599 × 5.565 = 198.122 kN   (permissible 307.347) — safe

Shear = 10.372 × 3.923 = 40.694 kN   (permissible 49.942) — revise section

Deflection = 37.077 × 4.012 = 148.736 mm   (permissible 246.202) — safe

Moment = 48.227 × 11.400 = 549.764 kNm   (permissible 832.231) — safe

Bearing pressure = 77.972 × 1.783 = 139.050 kPa   (permissible 243.445) — revise section

Shear = 80.824 × 9.448 = 763.647 kN   (permissible 878.959) — safe

The space programme has been developed from the functional brief agreed with the user department. Departmental areas, circulation and the gross-to-carpet ratio are tabulated in the area statement enclosed at the Annexure.

12 PROJECT MANAGEMENT ORGANISATION

Implementation is by the Assam Health Infrastructure Development Society through a project management unit headed by a Superintending Engineer, with two Executive Engineers and an architect on deputation. Progress is reviewed fortnightly by the Mission Director and monthly by the Commissioner of Health.

12.x Project Management Organisation — Working Papers — Sheet 1

Building services comprise electrical distribution, water supply and sanitation, HVAC for the identified conditioned areas, medical gas pipeline and vertical transportation. Connected load and demand factors are given in the load schedule.

Building services comprise electrical distribution, water supply and sanitation, HVAC for the identified conditioned areas, medical gas pipeline and vertical transportation. Connected load and demand factors are given in the load schedule.

Consultations were held with the head of each clinical department on adjacency, patient flow and equipment layout. Minutes of the consultation meetings are placed at the Annexure to this chapter.

Soil investigation was carried out by rotary drilling at the locations shown on the layout plan. The safe bearing capacity adopted for design is based on the lower of the shear and settlement criteria as reported in the geotechnical report.

Reference: drawing no. EL-911, revision A.

12.x Project Management Organisation — Working Papers — Sheet 2

Departmental Area Statement

Department	Floor	Units	Area per unit (sqm)	Total (sqm)
Operation Theatre	Ground	11	59.95	659.47
Diagnostics	Second	11	50.30	553.25
IPD Ward	Third	10	39.85	398.45
ICU	Ground	7	134.66	942.61
Staff Quarters	Third	4	12.55	50.21
Administration	Third	13	115.06	1495.72
Administration	Ground	18	134.58	2422.51
Pharmacy	Second	1	15.81	15.81
Administration	Second	7	29.80	208.60
Pharmacy	Third	16	111.35	1781.55
Administration	Second	9	47.99	431.89
Pharmacy	Ground	17	123.02	2091.30
ICU	Second	11	82.77	910.42
Diagnostics	First	11	138.04	1518.49
IPD Ward	Ground	18	120.37	2166.61
IPD Ward	First	11	42.29	465.18
OPD	First	5	65.04	325.20
Staff Quarters	Ground	15	136.86	2052.93
ICU	First	4	87.94	351.75
ICU	Third	18	58.59	1054.63

Building services comprise electrical distribution, water supply and sanitation, HVAC for the identified conditioned areas, medical gas pipeline and vertical transportation. Connected load and demand factors are given in the load schedule.

12.x Project Management Organisation — Working Papers — Sheet 3

Design Computation

Bearing pressure = 31.864 × 2.503 = 79.761 kPa   (permissible 126.563) — safe

Bearing pressure = 17.461 × 2.278 = 39.776 kPa   (permissible 70.169) — safe

Bearing pressure = 61.671 × 5.718 = 352.632 kPa   (permissible 439.979) — safe

Moment = 63.620 × 6.231 = 396.429 kNm   (permissible 544.759) — safe

Shear = 61.516 × 8.781 = 540.163 kN   (permissible 802.778) — safe

Moment = 12.235 × 11.522 = 140.968 kNm   (permissible 168.475) — revise section

The space programme has been developed from the functional brief agreed with the user department. Departmental areas, circulation and the gross-to-carpet ratio are tabulated in the area statement enclosed at the Annexure.

12.x Project Management Organisation — Working Papers — Sheet 4

Quantities have been computed from the approved architectural and structural drawings and cross-checked against the detailed measurement sheets enclosed at the Annexure. Rates are as per the schedule of rates in force with lead and lift adjustments.

Building services comprise electrical distribution, water supply and sanitation, HVAC for the identified conditioned areas, medical gas pipeline and vertical transportation. Connected load and demand factors are given in the load schedule.

Soil investigation was carried out by rotary drilling at the locations shown on the layout plan. The safe bearing capacity adopted for design is based on the lower of the shear and settlement criteria as reported in the geotechnical report.

Quantities have been computed from the approved architectural and structural drawings and cross-checked against the detailed measurement sheets enclosed at the Annexure. Rates are as per the schedule of rates in force with lead and lift adjustments.

Reference: drawing no. PH-190, revision A.

12.x Project Management Organisation — Working Papers — Sheet 5

Departmental Area Statement

Department	Floor	Units	Area per unit (sqm)	Total (sqm)
Staff Quarters	Third	13	44.73	581.46
Administration	Second	17	47.36	805.19
Pharmacy	First	14	116.14	1625.95
Pharmacy	Second	7	26.58	186.04
Diagnostics	First	9	34.10	306.92
Operation Theatre	Second	10	24.04	240.36
OPD	First	18	127.42	2293.54
OPD	First	17	100.73	1712.49
OPD	Second	8	124.91	999.26
Administration	Third	3	133.22	399.65
Administration	Third	8	109.26	874.11
Staff Quarters	Ground	18	45.42	817.49
OPD	Ground	13	138.71	1803.29
OPD	First	3	88.45	265.34
OPD	First	10	94.94	949.37
ICU	Third	14	100.80	1411.27
IPD Ward	First	18	83.70	1506.59
Administration	Ground	3	48.89	146.68
OPD	First	13	107.96	1403.43
ICU	Third	16	30.03	480.41

The structural design conforms to IS 456 for reinforced concrete and IS 1893 for earthquake resistant design. The site falls in seismic zone V and the importance factor applicable to a hospital building has been adopted.

12.x Project Management Organisation — Working Papers — Sheet 6

Design Computation

Moment = 1.208 × 11.835 = 14.296 kNm   (permissible 26.475) — safe

Bearing pressure = 7.210 × 11.254 = 81.140 kPa   (permissible 137.428) — safe

Shear = 62.768 × 10.940 = 686.667 kN   (permissible 920.871) — safe

Shear = 75.801 × 7.720 = 585.163 kN   (permissible 850.787) — safe

Moment = 61.189 × 1.533 = 93.812 kNm   (permissible 173.136) — revise section

Bearing pressure = 5.882 × 5.978 = 35.160 kPa   (permissible 40.883) — safe

Axial load = 73.877 × 2.088 = 154.266 kN   (permissible 242.995) — safe

The structural design conforms to IS 456 for reinforced concrete and IS 1893 for earthquake resistant design. The site falls in seismic zone V and the importance factor applicable to a hospital building has been adopted.

12.x Project Management Organisation — Working Papers — Sheet 7

Consultations were held with the head of each clinical department on adjacency, patient flow and equipment layout. Minutes of the consultation meetings are placed at the Annexure to this chapter.

Fire safety provisions follow Part 4 of the National Building Code. Means of egress, travel distances, staircase widths and the fire fighting installations have been designed for the occupancy classification of the block.

Quantities have been computed from the approved architectural and structural drawings and cross-checked against the detailed measurement sheets enclosed at the Annexure. Rates are as per the schedule of rates in force with lead and lift adjustments.

Consultations were held with the head of each clinical department on adjacency, patient flow and equipment layout. Minutes of the consultation meetings are placed at the Annexure to this chapter.

Reference: drawing no. AR-489, revision A.

12.x Project Management Organisation — Working Papers — Sheet 8

Room Data Sheet

Room	Department	Carpet area (sqm)	Occupancy	Floor finish
Room 448	Pathology	27.72	21	Antistatic vinyl
Room 366	Paediatrics	18.55	3	Antistatic vinyl
Room 351	Pathology	49.25	13	Epoxy screed
Room 808	Paediatrics	67.40	27	Antistatic vinyl
Room 840	Casualty	47.07	16	Kota stone
Room 415	Radiology	16.01	8	Ceramic tile
Room 440	Surgery	85.48	29	Vitrified tile
Room 445	Casualty	33.03	26	Kota stone
Room 634	Pathology	87.68	6	Kota stone
Room 668	Pathology	60.47	23	Ceramic tile
Room 124	Pathology	77.08	20	Vitrified tile
Room 563	Paediatrics	40.04	12	Ceramic tile
Room 233	Pathology	34.36	18	Epoxy screed
Room 450	Obstetrics	25.65	22	Kota stone
Room 599	Pathology	79.79	3	Ceramic tile
Room 528	General Medicine	49.05	10	Epoxy screed
Room 266	Pathology	75.56	29	Ceramic tile
Room 720	General Medicine	85.79	4	Vitrified tile
Room 882	Paediatrics	27.43	5	Epoxy screed
Room 722	Pathology	66.30	28	Vitrified tile
Room 859	Obstetrics	85.72	30	Epoxy screed
Room 720	Casualty	50.50	5	Ceramic tile

Fire safety provisions follow Part 4 of the National Building Code. Means of egress, travel distances, staircase widths and the fire fighting installations have been designed for the occupancy classification of the block.

13 CONTRACT MANAGEMENT STRATEGY

The work is proposed on an item rate basis in three packages — civil, electromechanical, and medical gas and HVAC — using the standard bidding document of the Public Works Department. A price adjustment clause is proposed for cement and steel.

13.x Contract Management Strategy — Working Papers — Sheet 1

Soil investigation was carried out by rotary drilling at the locations shown on the layout plan. The safe bearing capacity adopted for design is based on the lower of the shear and settlement criteria as reported in the geotechnical report.

Building services comprise electrical distribution, water supply and sanitation, HVAC for the identified conditioned areas, medical gas pipeline and vertical transportation. Connected load and demand factors are given in the load schedule.

Barrier-free access has been provided in accordance with the Harmonised Guidelines and Standards for Universal Accessibility. Ramps, lifts, signage and accessible toilets are provided on every floor of the block.

The structural design conforms to IS 456 for reinforced concrete and IS 1893 for earthquake resistant design. The site falls in seismic zone V and the importance factor applicable to a hospital building has been adopted.

Reference: drawing no. ST-438, revision B.

13.x Contract Management Strategy — Working Papers — Sheet 2

Departmental Area Statement

Department	Floor	Units	Area per unit (sqm)	Total (sqm)
Administration	First	14	19.47	272.51
Pharmacy	Third	1	72.24	72.24
Operation Theatre	Third	8	31.11	248.89
ICU	Third	4	53.01	212.06
Pharmacy	Ground	17	74.53	1266.95
Pharmacy	Third	14	14.84	207.75
OPD	First	12	59.66	715.94
Administration	Ground	1	100.96	100.96
Operation Theatre	Ground	18	31.31	563.56
Administration	First	7	103.14	721.95
IPD Ward	Ground	16	130.92	2094.70
ICU	First	6	42.15	252.92
Pharmacy	Third	1	113.00	113.00
Pharmacy	Third	13	113.34	1473.38
Diagnostics	Third	13	53.60	696.85
OPD	Second	13	61.63	801.21
ICU	Third	14	104.86	1468.00
Administration	Ground	5	95.40	477.00
OPD	Ground	3	93.04	279.11
Staff Quarters	Second	17	84.36	1434.08
Operation Theatre	Second	16	51.55	824.73
Diagnostics	First	5	52.59	262.95
IPD Ward	Ground	6	124.53	747.18
Administration	Second	9	64.78	582.98
ICU	Second	1	57.84	57.84
ICU	Second	2	19.95	39.90

The structural design conforms to IS 456 for reinforced concrete and IS 1893 for earthquake resistant design. The site falls in seismic zone V and the importance factor applicable to a hospital building has been adopted.

13.x Contract Management Strategy — Working Papers — Sheet 3

Design Computation

Moment = 65.913 × 9.687 = 638.527 kNm   (permissible 674.027) — safe

Axial load = 52.133 × 9.921 = 517.236 kN   (permissible 571.277) — safe

Shear = 67.533 × 6.543 = 441.889 kN   (permissible 753.845) — safe

Moment = 34.468 × 4.345 = 149.772 kNm   (permissible 199.621) — safe

Deflection = 74.984 × 5.530 = 414.653 mm   (permissible 748.509) — safe

Moment = 55.803 × 5.485 = 306.105 kNm   (permissible 563.807) — safe

Building services comprise electrical distribution, water supply and sanitation, HVAC for the identified conditioned areas, medical gas pipeline and vertical transportation. Connected load and demand factors are given in the load schedule.

13.x Contract Management Strategy — Working Papers — Sheet 4

Quantities have been computed from the approved architectural and structural drawings and cross-checked against the detailed measurement sheets enclosed at the Annexure. Rates are as per the schedule of rates in force with lead and lift adjustments.

Barrier-free access has been provided in accordance with the Harmonised Guidelines and Standards for Universal Accessibility. Ramps, lifts, signage and accessible toilets are provided on every floor of the block.

Fire safety provisions follow Part 4 of the National Building Code. Means of egress, travel distances, staircase widths and the fire fighting installations have been designed for the occupancy classification of the block.

Fire safety provisions follow Part 4 of the National Building Code. Means of egress, travel distances, staircase widths and the fire fighting installations have been designed for the occupancy classification of the block.

Reference: drawing no. AR-773, revision A.

13.x Contract Management Strategy — Working Papers — Sheet 5

Room Data Sheet

Room	Department	Carpet area (sqm)	Occupancy	Floor finish
Room 708	Obstetrics	64.72	9	Antistatic vinyl
Room 857	General Medicine	30.24	18	Ceramic tile
Room 187	Obstetrics	37.07	25	Ceramic tile
Room 355	Casualty	43.56	17	Vitrified tile
Room 127	Pathology	33.98	15	Epoxy screed
Room 727	Radiology	22.96	29	Antistatic vinyl
Room 545	Radiology	11.51	23	Kota stone
Room 358	Radiology	29.21	6	Kota stone
Room 777	Obstetrics	15.54	3	Ceramic tile
Room 145	Radiology	68.28	27	Kota stone
Room 187	Surgery	51.34	6	Epoxy screed
Room 877	Surgery	54.63	20	Vitrified tile
Room 317	Radiology	42.01	27	Ceramic tile
Room 752	Surgery	36.34	18	Ceramic tile
Room 631	Casualty	62.19	22	Vitrified tile
Room 191	Paediatrics	27.26	27	Vitrified tile
Room 486	General Medicine	9.25	21	Vitrified tile
Room 555	Casualty	28.08	3	Kota stone
Room 823	Paediatrics	52.89	30	Antistatic vinyl
Room 309	Pathology	39.18	27	Vitrified tile
Room 843	Obstetrics	27.78	8	Antistatic vinyl
Room 836	Surgery	43.63	1	Epoxy screed

Consultations were held with the head of each clinical department on adjacency, patient flow and equipment layout. Minutes of the consultation meetings are placed at the Annexure to this chapter.

13.x Contract Management Strategy — Working Papers — Sheet 6

Design Computation

Axial load = 26.106 × 6.801 = 177.546 kN   (permissible 317.822) — safe

Axial load = 38.569 × 5.812 = 224.151 kN   (permissible 258.911) — safe

Deflection = 5.684 × 9.922 = 56.391 mm   (permissible 104.466) — safe

Shear = 62.807 × 11.238 = 705.828 kN   (permissible 788.514) — safe

Deflection = 16.400 × 1.443 = 23.671 mm   (permissible 38.232) — safe

Bearing pressure = 41.864 × 10.307 = 431.484 kPa   (permissible 669.870) — safe

Deflection = 80.988 × 7.274 = 589.104 mm   (permissible 631.743) — safe

Bearing pressure = 18.377 × 3.155 = 57.987 kPa   (permissible 79.133) — safe

The space programme has been developed from the functional brief agreed with the user department. Departmental areas, circulation and the gross-to-carpet ratio are tabulated in the area statement enclosed at the Annexure.

13.x Contract Management Strategy — Working Papers — Sheet 7

The space programme has been developed from the functional brief agreed with the user department. Departmental areas, circulation and the gross-to-carpet ratio are tabulated in the area statement enclosed at the Annexure.

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The structural design conforms to IS 456 for reinforced concrete and IS 1893 for earthquake resistant design. The site falls in seismic zone V and the importance factor applicable to a hospital building has been adopted.

Quantities have been computed from the approved architectural and structural drawings and cross-checked against the detailed measurement sheets enclosed at the Annexure. Rates are as per the schedule of rates in force with lead and lift adjustments.

Reference: drawing no. ST-165, revision A.

13.x Contract Management Strategy — Working Papers — Sheet 8

Detailed Measurement Sheet

Item	L (m)	B (m)	D (m)	Nos	Quantity
Item 843	4.81	3.46	0.46	24	109.081
Item 624	36.16	2.69	0.94	7	308.768
Item 135	18.57	4.53	1.36	5	106.485
Item 100	6.71	5.96	0.48	21	68.586
Item 139	24.74	3.00	1.28	9	136.175
Item 473	10.81	4.05	0.88	13	314.507
Item 888	18.17	0.51	2.01	7	231.094
Item 101	28.11	0.48	2.84	21	195.374
Item 229	11.02	4.20	0.42	8	359.528
Item 888	37.25	3.57	2.80	10	94.475
Item 780	24.20	2.18	2.62	2	7.997
Item 259	25.33	1.90	1.03	2	6.549
Item 746	17.69	3.35	0.61	23	77.018
Item 846	33.07	0.37	2.18	3	196.958
Item 490	24.20	4.00	0.20	10	299.247
Item 174	32.93	3.42	1.47	10	52.936
Item 138	2.98	1.56	1.20	4	163.206
Item 195	17.74	1.87	0.59	13	384.798
Item 306	37.22	0.58	1.34	17	375.289
Item 438	20.54	1.68	0.65	8	39.093
Item 590	23.04	5.08	2.67	9	391.267

Building services comprise electrical distribution, water supply and sanitation, HVAC for the identified conditioned areas, medical gas pipeline and vertical transportation. Connected load and demand factors are given in the load schedule.

14 IMPLEMENTATION SCHEDULE AND WBS

The zero date is the date of issue of the work order. The implementation period is 28 months.

WBS	Milestone	Month from zero date
1.0	Award and mobilisation	M0 – M2
2.0	Site preparation complete	M4
3.0	Substructure / foundation complete	M9
4.0	Superstructure complete	M18
5.0	Services and finishes	M25
6.0	Testing and commissioning	M28

14.x Implementation Schedule And Wbs — Working Papers — Sheet 1

Building services comprise electrical distribution, water supply and sanitation, HVAC for the identified conditioned areas, medical gas pipeline and vertical transportation. Connected load and demand factors are given in the load schedule.

Fire safety provisions follow Part 4 of the National Building Code. Means of egress, travel distances, staircase widths and the fire fighting installations have been designed for the occupancy classification of the block.

Soil investigation was carried out by rotary drilling at the locations shown on the layout plan. The safe bearing capacity adopted for design is based on the lower of the shear and settlement criteria as reported in the geotechnical report.

Quantities have been computed from the approved architectural and structural drawings and cross-checked against the detailed measurement sheets enclosed at the Annexure. Rates are as per the schedule of rates in force with lead and lift adjustments.

Reference: drawing no. ST-569, revision C.

14.x Implementation Schedule And Wbs — Working Papers — Sheet 2

Detailed Measurement Sheet

Item	L (m)	B (m)	D (m)	Nos	Quantity
Item 264	6.03	2.87	2.92	18	159.722
Item 843	25.88	1.11	1.12	16	30.577
Item 222	21.82	2.87	0.43	20	364.867
Item 388	16.04	5.61	0.31	11	159.084
Item 251	28.07	2.48	1.48	20	9.801
Item 300	34.38	2.66	1.04	16	245.663
Item 103	27.33	3.83	1.49	16	360.473
Item 717	28.33	5.05	0.34	21	385.514
Item 209	31.52	5.73	1.99	3	295.677
Item 357	32.15	5.44	2.67	8	341.367
Item 510	32.74	4.78	0.88	6	281.147
Item 821	16.44	2.24	2.29	13	304.285
Item 433	4.04	2.56	2.65	17	227.313
Item 524	26.13	1.27	0.34	6	165.532
Item 630	7.52	3.80	0.45	4	66.599
Item 794	19.00	4.89	1.34	12	339.881
Item 172	13.47	0.80	2.90	21	216.630
Item 560	39.44	1.70	1.78	1	24.277
Item 620	39.05	0.61	0.71	10	287.804
Item 478	32.97	5.53	1.90	17	317.891
Item 558	23.19	4.48	1.02	11	75.291
Item 884	19.84	0.51	0.48	22	19.933
Item 573	5.29	0.43	1.13	7	130.066
Item 380	12.21	0.31	0.50	12	155.336
Item 471	28.78	4.07	2.73	19	247.826

Building services comprise electrical distribution, water supply and sanitation, HVAC for the identified conditioned areas, medical gas pipeline and vertical transportation. Connected load and demand factors are given in the load schedule.

14.x Implementation Schedule And Wbs — Working Papers — Sheet 3

Design Computation

Bearing pressure = 82.579 × 4.131 = 341.172 kPa   (permissible 443.204) — safe

Axial load = 2.335 × 10.309 = 24.067 kN   (permissible 26.849) — safe

Axial load = 86.619 × 9.219 = 798.528 kN   (permissible 1090.994) — revise section

Axial load = 63.280 × 3.746 = 237.029 kN   (permissible 338.042) — safe

Shear = 80.614 × 11.572 = 932.843 kN   (permissible 1727.987) — safe

Moment = 86.174 × 7.975 = 687.201 kNm   (permissible 1165.733) — safe

Bearing pressure = 39.789 × 2.428 = 96.622 kPa   (permissible 176.694) — revise section

Barrier-free access has been provided in accordance with the Harmonised Guidelines and Standards for Universal Accessibility. Ramps, lifts, signage and accessible toilets are provided on every floor of the block.

14.x Implementation Schedule And Wbs — Working Papers — Sheet 4

Building services comprise electrical distribution, water supply and sanitation, HVAC for the identified conditioned areas, medical gas pipeline and vertical transportation. Connected load and demand factors are given in the load schedule.

Fire safety provisions follow Part 4 of the National Building Code. Means of egress, travel distances, staircase widths and the fire fighting installations have been designed for the occupancy classification of the block.

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Soil investigation was carried out by rotary drilling at the locations shown on the layout plan. The safe bearing capacity adopted for design is based on the lower of the shear and settlement criteria as reported in the geotechnical report.

Reference: drawing no. ST-734, revision A.

14.x Implementation Schedule And Wbs — Working Papers — Sheet 5

Equipment Schedule

Item	Department	Nos	Unit rate (Rs)	Amount (Rs)
OT table	Ward	9	2,320,227	20,882,043
OT table	CSSD	20	752,953	15,059,060
X-ray unit	Ward	32	275,389	8,812,448
Ventilator	ICU	25	1,985,558	49,638,950
Autoclave	Casualty	3	135,392	406,176
Autoclave	ICU	11	200,087	2,200,957
Ultrasound	Ward	32	997,319	31,914,208
Patient monitor	Casualty	28	1,295,468	36,273,104
Hospital bed	ICU	32	2,129,737	68,151,584
Ventilator	Radiology	31	539,704	16,730,824
OT table	Radiology	7	1,742,648	12,198,536
Ultrasound	Ward	39	744,741	29,044,899
Defibrillator	Ward	1	589,539	589,539
Autoclave	Casualty	23	1,826,812	42,016,676
Hospital bed	OT	28	1,868,815	52,326,820
Autoclave	Ward	8	1,167,884	9,343,072
Autoclave	OT	17	1,940,922	32,995,674
Defibrillator	CSSD	35	1,181,811	41,363,385
Ventilator	OT	14	1,028,430	14,398,020
Defibrillator	Ward	14	1,534,970	21,489,580
Autoclave	ICU	20	636,425	12,728,500
Hospital bed	OT	25	413,609	10,340,225
X-ray unit	ICU	5	1,452,611	7,263,055

The space programme has been developed from the functional brief agreed with the user department. Departmental areas, circulation and the gross-to-carpet ratio are tabulated in the area statement enclosed at the Annexure.

14.x Implementation Schedule And Wbs — Working Papers — Sheet 6

Design Computation

Axial load = 54.440 × 10.226 = 556.695 kN   (permissible 641.286) — revise section

Shear = 58.475 × 5.951 = 347.995 kN   (permissible 532.600) — safe

Shear = 59.806 × 8.987 = 537.488 kN   (permissible 703.753) — safe

Bearing pressure = 83.272 × 0.587 = 48.868 kPa   (permissible 79.593) — safe

Moment = 67.899 × 7.049 = 478.626 kNm   (permissible 755.035) — safe

Bearing pressure = 3.809 × 1.043 = 3.972 kPa   (permissible 5.400) — safe

Soil investigation was carried out by rotary drilling at the locations shown on the layout plan. The safe bearing capacity adopted for design is based on the lower of the shear and settlement criteria as reported in the geotechnical report.

14.x Implementation Schedule And Wbs — Working Papers — Sheet 7

Soil investigation was carried out by rotary drilling at the locations shown on the layout plan. The safe bearing capacity adopted for design is based on the lower of the shear and settlement criteria as reported in the geotechnical report.

Building services comprise electrical distribution, water supply and sanitation, HVAC for the identified conditioned areas, medical gas pipeline and vertical transportation. Connected load and demand factors are given in the load schedule.

The space programme has been developed from the functional brief agreed with the user department. Departmental areas, circulation and the gross-to-carpet ratio are tabulated in the area statement enclosed at the Annexure.

Building services comprise electrical distribution, water supply and sanitation, HVAC for the identified conditioned areas, medical gas pipeline and vertical transportation. Connected load and demand factors are given in the load schedule.

Reference: drawing no. PH-323, revision B.

14.x Implementation Schedule And Wbs — Working Papers — Sheet 8

Room Data Sheet

Room	Department	Carpet area (sqm)	Occupancy	Floor finish
Room 722	Casualty	71.41	10	Epoxy screed
Room 109	Casualty	25.53	26	Vitrified tile
Room 138	Paediatrics	75.39	10	Vitrified tile
Room 121	Pathology	90.92	13	Ceramic tile
Room 141	Casualty	9.89	11	Kota stone
Room 195	Radiology	83.91	20	Epoxy screed
Room 194	Pathology	14.51	2	Epoxy screed
Room 602	Casualty	45.38	20	Epoxy screed
Room 319	Paediatrics	48.80	13	Antistatic vinyl
Room 217	Surgery	85.02	19	Antistatic vinyl
Room 233	General Medicine	50.66	17	Antistatic vinyl
Room 401	General Medicine	52.67	25	Vitrified tile
Room 152	Paediatrics	52.94	18	Ceramic tile
Room 670	General Medicine	32.73	2	Vitrified tile
Room 349	General Medicine	77.73	5	Ceramic tile
Room 617	General Medicine	9.80	26	Vitrified tile
Room 618	General Medicine	60.45	14	Ceramic tile
Room 329	Radiology	24.00	27	Kota stone

Barrier-free access has been provided in accordance with the Harmonised Guidelines and Standards for Universal Accessibility. Ramps, lifts, signage and accessible toilets are provided on every floor of the block.

14.x Implementation Schedule And Wbs — Working Papers — Sheet 9

Design Computation

Deflection = 79.334 × 9.760 = 774.283 mm (permissible 1018.502) — safe

Shear = 66.350 × 3.934 = 261.038 kN (permissible 410.507) — safe

Deflection = 55.142 × 7.371 = 406.475 mm (permissible 646.085) — safe

Shear = 17.698 × 8.720 = 154.331 kN (permissible 229.707) — safe

Moment = 30.998 × 4.572 = 141.737 kNm (permissible 217.931) — safe

Soil investigation was carried out by rotary drilling at the locations shown on the layout plan. The safe bearing capacity adopted for design is based on the lower of the shear and settlement criteria as reported in the geotechnical report.

14.x Implementation Schedule And Wbs — Working Papers — Sheet 10

Quantities have been computed from the approved architectural and structural drawings and cross-checked against the detailed measurement sheets enclosed at the Annexure. Rates are as per the schedule of rates in force with lead and lift adjustments.

The structural design conforms to IS 456 for reinforced concrete and IS 1893 for earthquake resistant design. The site falls in seismic zone V and the importance factor applicable to a hospital building has been adopted.

Quantities have been computed from the approved architectural and structural drawings and cross-checked against the detailed measurement sheets enclosed at the Annexure. Rates are as per the schedule of rates in force with lead and lift adjustments.

Quantities have been computed from the approved architectural and structural drawings and cross-checked against the detailed measurement sheets enclosed at the Annexure. Rates are as per the schedule of rates in force with lead and lift adjustments.

Reference: drawing no. PH-409, revision C.

15 STATUTORY CLEARANCES

Sl.	Clearance	Authority	Status
1	Consent to establish	State Pollution Control Board	Application under process
2	Building permit / layout approval	Local body	Obtained, 3 March 2026
3	Fire safety clearance	Directorate of Fire Services	Obtained, 19 March 2026
4	Ground water abstraction	CGWA	Not applicable

Consent to establish from the State Pollution Control Board is yet to be obtained. The application was submitted and is under process.

15.x Statutory Clearances — Working Papers — Sheet 1

The space programme has been developed from the functional brief agreed with the user department. Departmental areas, circulation and the gross-to-carpet ratio are tabulated in the area statement enclosed at the Annexure.

Fire safety provisions follow Part 4 of the National Building Code. Means of egress, travel distances, staircase widths and the fire fighting installations have been designed for the occupancy classification of the block.

Soil investigation was carried out by rotary drilling at the locations shown on the layout plan. The safe bearing capacity adopted for design is based on the lower of the shear and settlement criteria as reported in the geotechnical report.

Barrier-free access has been provided in accordance with the Harmonised Guidelines and Standards for Universal Accessibility. Ramps, lifts, signage and accessible toilets are provided on every floor of the block.

Reference: drawing no. EL-210, revision A.

15.x Statutory Clearances — Working Papers — Sheet 2

Electrical Load Schedule

Panel	Description	Connected load (kW)	Demand factor	Demand (kVA)
DB-04	Power outlets	102.09	0.59	66.64
DB-16	Power outlets	26.04	0.71	20.50
DB-06	Medical equipment	2.30	0.64	1.63
DB-29	Lighting	69.21	0.40	31.04
DB-10	Power outlets	163.16	0.45	81.77
DB-28	Medical equipment	142.63	0.41	65.24
DB-03	Sterilisation	86.89	0.66	63.54
DB-31	Power outlets	56.31	0.69	42.98
DB-37	Sterilisation	148.19	0.75	123.71
DB-33	Lifts	172.98	0.41	78.21
DB-39	Sterilisation	123.90	0.75	103.40
DB-24	Medical equipment	88.53	0.77	75.51
DB-12	Pumps	7.67	0.81	6.94
DB-17	Power outlets	147.99	0.90	148.41
DB-05	Pumps	61.61	0.70	48.25
DB-22	Lighting	83.30	0.72	66.20
DB-24	Lighting	35.65	0.45	17.77
DB-26	Medical equipment	161.51	0.95	170.25
DB-04	Pumps	91.18	0.85	86.44
DB-15	Medical equipment	53.21	0.67	39.74
DB-35	Lifts	88.12	0.50	48.73
DB-01	Power outlets	69.39	0.65	50.40
DB-23	Pumps	83.23	0.41	38.06
DB-27	Lighting	50.40	0.64	35.78

Soil investigation was carried out by rotary drilling at the locations shown on the layout plan. The safe bearing capacity adopted for design is based on the lower of the shear and settlement criteria as reported in the geotechnical report.

15.x Statutory Clearances — Working Papers — Sheet 3

Design Computation

Moment = 46.419 × 8.602 = 399.283 kNm (permissible 566.271) — revise section

Bearing pressure = 48.686 × 1.252 = 60.975 kPa (permissible 65.534) — safe

Deflection = 86.063 × 0.783 = 67.401 mm (permissible 89.954) — safe

Bearing pressure = 57.882 × 11.648 = 674.235 kPa (permissible 744.766) — safe

Moment = 13.815 × 8.158 = 112.694 kNm (permissible 197.333) — safe

Barrier-free access has been provided in accordance with the Harmonised Guidelines and Standards for Universal Accessibility. Ramps, lifts, signage and accessible toilets are provided on every floor of the block.

15.x Statutory Clearances — Working Papers — Sheet 4

Fire safety provisions follow Part 4 of the National Building Code. Means of egress, travel distances, staircase widths and the fire fighting installations have been designed for the occupancy classification of the block.

The structural design conforms to IS 456 for reinforced concrete and IS 1893 for earthquake resistant design. The site falls in seismic zone V and the importance factor applicable to a hospital building has been adopted.

Fire safety provisions follow Part 4 of the National Building Code. Means of egress, travel distances, staircase widths and the fire fighting installations have been designed for the occupancy classification of the block.

Barrier-free access has been provided in accordance with the Harmonised Guidelines and Standards for Universal Accessibility. Ramps, lifts, signage and accessible toilets are provided on every floor of the block.

Reference: drawing no. EL-229, revision C.

15.x Statutory Clearances — Working Papers — Sheet 5

Bore Log Summary

Depth (m)	Strata	N value	Sample
4.0	Medium sand	55	DS
11.9	Silty clay	51	UDS
13.9	Silty clay	26	SPT
26.2	Gravelly sand	50	DS
20.2	Coarse sand	31	UDS
5.2	Coarse sand	59	DS
15.7	Medium sand	25	DS
3.6	Gravelly sand	27	UDS
26.9	Coarse sand	9	SPT
27.9	Medium sand	17	SPT
8.3	Silty clay	36	UDS
28.7	Clayey silt	16	UDS
6.2	Clayey silt	18	UDS
18.2	Silty clay	25	DS
16.4	Clayey silt	12	SPT
15.0	Weathered rock	53	UDS
8.1	Clayey silt	59	SPT
10.5	Medium sand	57	DS
27.8	Clayey silt	44	UDS
27.0	Clayey silt	42	SPT
10.6	Clayey silt	11	DS
2.0	Weathered rock	27	DS
25.7	Coarse sand	26	UDS
23.2	Coarse sand	55	UDS

The structural design conforms to IS 456 for reinforced concrete and IS 1893 for earthquake resistant design. The site falls in seismic zone V and the importance factor applicable to a hospital building has been adopted.

15.x Statutory Clearances — Working Papers — Sheet 6

Design Computation

Bearing pressure = 87.974 × 1.124 = 98.911 kPa   (permissible 135.413) — safe

Bearing pressure = 83.274 × 10.333 = 860.508 kPa   (permissible 1173.078) — safe

Deflection = 11.887 × 3.479 = 41.353 mm   (permissible 54.747) — safe

Moment = 5.593 × 4.906 = 27.436 kNm   (permissible 39.665) — safe

Bearing pressure = 63.864 × 3.215 = 205.324 kPa   (permissible 261.999) — safe

Bearing pressure = 51.865 × 3.282 = 170.201 kPa   (permissible 251.191) — safe

Shear = 25.477 × 11.476 = 292.381 kN   (permissible 337.653) — safe

Deflection = 45.185 × 1.930 = 87.203 mm   (permissible 94.732) — safe

The structural design conforms to IS 456 for reinforced concrete and IS 1893 for earthquake resistant design. The site falls in seismic zone V and the importance factor applicable to a hospital building has been adopted.

15.x Statutory Clearances — Working Papers — Sheet 7

Quantities have been computed from the approved architectural and structural drawings and cross-checked against the detailed measurement sheets enclosed at the Annexure. Rates are as per the schedule of rates in force with lead and lift adjustments.

Quantities have been computed from the approved architectural and structural drawings and cross-checked against the detailed measurement sheets enclosed at the Annexure. Rates are as per the schedule of rates in force with lead and lift adjustments.

Building services comprise electrical distribution, water supply and sanitation, HVAC for the identified conditioned areas, medical gas pipeline and vertical transportation. Connected load and demand factors are given in the load schedule.

The space programme has been developed from the functional brief agreed with the user department. Departmental areas, circulation and the gross-to-carpet ratio are tabulated in the area statement enclosed at the Annexure.

Reference: drawing no. PH-616, revision C.

15.x Statutory Clearances — Working Papers — Sheet 8

Bore Log Summary

Depth (m)	Strata	N value	Sample
27.1	Silty clay	24	DS
11.9	Gravelly sand	60	UDS
26.2	Coarse sand	14	SPT
11.0	Silty clay	42	UDS
15.2	Medium sand	16	UDS
8.4	Medium sand	52	DS
27.8	Medium sand	23	SPT
26.0	Weathered rock	18	SPT
16.1	Silty clay	32	DS
16.6	Coarse sand	55	UDS
15.9	Silty clay	44	SPT
12.5	Weathered rock	11	UDS
17.3	Gravelly sand	31	DS
11.0	Weathered rock	12	UDS
27.9	Medium sand	6	UDS
27.3	Silty clay	29	UDS
0.8	Gravelly sand	24	DS
1.4	Medium sand	50	SPT
21.1	Medium sand	23	UDS
8.8	Gravelly sand	56	SPT
6.8	Silty clay	51	SPT
23.7	Weathered rock	18	SPT
7.1	Gravelly sand	31	SPT
17.0	Weathered rock	8	SPT
26.1	Coarse sand	12	SPT

Soil investigation was carried out by rotary drilling at the locations shown on the layout plan. The safe bearing capacity adopted for design is based on the lower of the shear and settlement criteria as reported in the geotechnical report.

15.x Statutory Clearances — Working Papers — Sheet 9

Design Computation

Axial load = 11.858 × 11.147 = 132.186 kN   (permissible 200.795) — revise section

Shear = 47.358 × 3.750 = 177.571 kN   (permissible 245.403) — revise section

Moment = 46.258 × 10.668 = 493.493 kNm   (permissible 766.838) — safe

Bearing pressure = 8.367 × 8.157 = 68.250 kPa   (permissible 82.093) — safe

Moment = 43.290 × 10.276 = 444.833 kNm   (permissible 528.542) — safe

Shear = 30.461 × 1.539 = 46.888 kN   (permissible 67.470) — safe

Barrier-free access has been provided in accordance with the Harmonised Guidelines and Standards for Universal Accessibility. Ramps, lifts, signage and accessible toilets are provided on every floor of the block.

15.x Statutory Clearances — Working Papers — Sheet 10

The space programme has been developed from the functional brief agreed with the user department. Departmental areas, circulation and the gross-to-carpet ratio are tabulated in the area statement enclosed at the Annexure.

The space programme has been developed from the functional brief agreed with the user department. Departmental areas, circulation and the gross-to-carpet ratio are tabulated in the area statement enclosed at the Annexure.

The structural design conforms to IS 456 for reinforced concrete and IS 1893 for earthquake resistant design. The site falls in seismic zone V and the importance factor applicable to a hospital building has been adopted.

Soil investigation was carried out by rotary drilling at the locations shown on the layout plan. The safe bearing capacity adopted for design is based on the lower of the shear and settlement criteria as reported in the geotechnical report.

Reference: drawing no. PH-812, revision A.

16A Supplementary Notes — Sheet 1

Fire safety provisions follow Part 4 of the National Building Code. Means of egress, travel distances, staircase widths and the fire fighting installations have been designed for the occupancy classification of the block.

The space programme has been developed from the functional brief agreed with the user department. Departmental areas, circulation and the gross-to-carpet ratio are tabulated in the area statement enclosed at the Annexure.

The structural design conforms to IS 456 for reinforced concrete and IS 1893 for earthquake resistant design. The site falls in seismic zone V and the importance factor applicable to a hospital building has been adopted.

The space programme has been developed from the functional brief agreed with the user department. Departmental areas, circulation and the gross-to-carpet ratio are tabulated in the area statement enclosed at the Annexure.

Reference: drawing no. PH-386, revision B.

16A Supplementary Notes — Sheet 2

Room Data Sheet

Room	Department	Carpet area (sqm)	Occupancy	Floor finish
Room 269	Radiology	89.35	27	Epoxy screed
Room 122	Pathology	48.33	14	Vitrified tile
Room 334	Surgery	17.41	11	Antistatic vinyl
Room 245	Radiology	24.87	28	Kota stone
Room 650	Paediatrics	71.65	6	Vitrified tile
Room 565	Obstetrics	51.58	22	Vitrified tile
Room 494	Pathology	53.40	30	Vitrified tile
Room 721	Radiology	16.46	24	Ceramic tile
Room 493	Casualty	29.64	6	Epoxy screed
Room 841	Pathology	12.17	21	Epoxy screed
Room 752	Radiology	61.99	12	Ceramic tile
Room 888	Radiology	86.24	6	Vitrified tile
Room 643	General Medicine	75.00	2	Antistatic vinyl
Room 645	Pathology	17.36	20	Epoxy screed
Room 583	Pathology	48.21	19	Epoxy screed
Room 756	Obstetrics	69.37	16	Antistatic vinyl
Room 511	Obstetrics	22.17	1	Kota stone
Room 626	Paediatrics	48.19	5	Epoxy screed
Room 875	Obstetrics	50.61	11	Kota stone
Room 174	Pathology	76.64	27	Kota stone
Room 500	General Medicine	18.40	7	Ceramic tile

Building services comprise electrical distribution, water supply and sanitation, HVAC for the identified conditioned areas, medical gas pipeline and vertical transportation. Connected load and demand factors are given in the load schedule.

16A Supplementary Notes — Sheet 3

Design Computation

Moment = 87.061 × 3.005 = 261.657 kNm (permissible 293.000) — revise section

Shear = 66.390 × 1.671 = 110.963 kN (permissible 202.829) — safe

Bearing pressure = 16.036 × 10.330 = 165.656 kPa (permissible 285.055) — safe

Moment = 65.982 × 7.950 = 524.541 kNm (permissible 757.786) — safe

Shear = 7.553 × 5.365 = 40.520 kN (permissible 54.972) — safe

Bearing pressure = 39.864 × 9.237 = 368.216 kPa (permissible 693.170) — revise section

Deflection = 56.989 × 8.973 = 511.360 mm (permissible 775.401) — safe

Shear = 40.290 × 11.397 = 459.161 kN (permissible 837.989) — safe

Quantities have been computed from the approved architectural and structural drawings and cross-checked against the detailed measurement sheets enclosed at the Annexure. Rates are as per the schedule of rates in force with lead and lift adjustments.

16A Supplementary Notes — Sheet 4

Soil investigation was carried out by rotary drilling at the locations shown on the layout plan. The safe bearing capacity adopted for design is based on the lower of the shear and settlement criteria as reported in the geotechnical report.

Soil investigation was carried out by rotary drilling at the locations shown on the layout plan. The safe bearing capacity adopted for design is based on the lower of the shear and settlement criteria as reported in the geotechnical report.

Consultations were held with the head of each clinical department on adjacency, patient flow and equipment layout. Minutes of the consultation meetings are placed at the Annexure to this chapter.

Soil investigation was carried out by rotary drilling at the locations shown on the layout plan. The safe bearing capacity adopted for design is based on the lower of the shear and settlement criteria as reported in the geotechnical report.

Reference: drawing no. EL-427, revision B.

16A Supplementary Notes — Sheet 5

Electrical Load Schedule

Panel	Description	Connected load (kW)	Demand factor	Demand (kVA)
DB-01	Pumps	13.44	0.92	13.80
DB-39	Power outlets	135.19	0.61	90.98
DB-29	Lighting	162.66	0.58	105.45
DB-33	Sterilisation	167.35	0.81	150.43
DB-18	Pumps	179.41	0.56	111.94
DB-18	Lighting	121.65	0.47	63.55
DB-15	Lifts	147.10	0.47	76.06
DB-27	Lighting	127.81	0.45	63.92
DB-37	Sterilisation	71.03	0.66	52.44
DB-39	Pumps	170.21	0.65	123.54
DB-07	Sterilisation	35.73	0.75	29.86
DB-33	Pumps	29.77	0.45	14.86
DB-06	Lifts	171.01	0.53	100.60
DB-16	Lighting	35.00	0.74	28.65
DB-33	Lifts	109.13	0.46	55.60
DB-25	Power outlets	138.88	0.47	72.80
DB-04	HVAC	159.61	0.41	72.12
DB-21	Lighting	34.96	0.69	26.97

Fire safety provisions follow Part 4 of the National Building Code. Means of egress, travel distances, staircase widths and the fire fighting installations have been designed for the occupancy classification of the block.

16A Supplementary Notes — Sheet 6

Design Computation

Axial load = 16.202 × 1.374 = 22.269 kN   (permissible 26.195) — revise section

Moment = 13.129 × 3.933 = 51.632 kNm   (permissible 70.811) — safe

Axial load = 41.454 × 11.782 = 488.428 kN   (permissible 875.584) — safe

Deflection = 77.441 × 2.085 = 161.447 mm   (permissible 255.110) — safe

Deflection = 88.245 × 2.932 = 258.751 mm   (permissible 474.482) — safe

The structural design conforms to IS 456 for reinforced concrete and IS 1893 for earthquake resistant design. The site falls in seismic zone V and the importance factor applicable to a hospital building has been adopted.

17 OPERATIONS AND MAINTENANCE PLAN

The completed asset will be handed over to Assam Health Infrastructure Development Society for operation and maintenance.

The annual recurring requirement for operation and maintenance is estimated at Rs. 9.80 crore, to be met from the maintenance budget head of the department. The Finance Department has conveyed concurrence to the inclusion of this recurring provision.

A routine maintenance schedule covering inspection, servicing and periodic replacement is placed at the end of this chapter.

17.x Operations And Maintenance Plan — Working Papers — Sheet 1

Building services comprise electrical distribution, water supply and sanitation, HVAC for the identified conditioned areas, medical gas pipeline and vertical transportation. Connected load and demand factors are given in the load schedule.

Building services comprise electrical distribution, water supply and sanitation, HVAC for the identified conditioned areas, medical gas pipeline and vertical transportation. Connected load and demand factors are given in the load schedule.

Fire safety provisions follow Part 4 of the National Building Code. Means of egress, travel distances, staircase widths and the fire fighting installations have been designed for the occupancy classification of the block.

The structural design conforms to IS 456 for reinforced concrete and IS 1893 for earthquake resistant design. The site falls in seismic zone V and the importance factor applicable to a hospital building has been adopted.

Reference: drawing no. EL-254, revision C.

17.x Operations And Maintenance Plan — Working Papers — Sheet 2

Bore Log Summary

Depth (m)	Strata	N value	Sample
19.1	Silty clay	49	DS
5.1	Clayey silt	46	SPT
19.3	Coarse sand	22	UDS
15.0	Medium sand	59	SPT
28.0	Silty clay	21	DS
18.4	Clayey silt	58	SPT
1.3	Clayey silt	44	DS
23.2	Medium sand	58	UDS
2.4	Weathered rock	54	DS
23.7	Weathered rock	33	DS
15.9	Weathered rock	16	DS
4.0	Gravelly sand	27	DS
19.5	Medium sand	57	UDS
13.1	Clayey silt	19	DS
10.3	Weathered rock	53	DS
29.0	Weathered rock	42	DS
15.6	Gravelly sand	56	SPT
16.1	Gravelly sand	46	SPT
16.0	Silty clay	10	UDS
13.0	Medium sand	33	DS
25.5	Weathered rock	26	DS
12.9	Gravelly sand	22	UDS
15.5	Clayey silt	27	DS
9.9	Clayey silt	28	UDS
4.1	Clayey silt	15	UDS

Soil investigation was carried out by rotary drilling at the locations shown on the layout plan. The safe bearing capacity adopted for design is based on the lower of the shear and settlement criteria as reported in the geotechnical report.

17.x Operations And Maintenance Plan — Working Papers — Sheet 3

Design Computation

Axial load = 53.133 × 2.638 = 140.150 kN   (permissible 229.070) — safe

Deflection = 59.696 × 4.136 = 246.920 mm   (permissible 354.605) — safe

Deflection = 60.494 × 5.146 = 311.306 mm   (permissible 479.075) — safe

Axial load = 47.903 × 9.680 = 463.721 kN   (permissible 581.928) — safe

Bearing pressure = 87.905 × 5.959 = 523.821 kPa   (permissible 897.094) — safe

Moment = 46.374 × 4.441 = 205.931 kNm   (permissible 349.462) — safe

Deflection = 42.079 × 2.235 = 94.058 mm   (permissible 137.761) — safe

Deflection = 32.212 × 5.279 = 170.048 mm   (permissible 188.458) — safe

The space programme has been developed from the functional brief agreed with the user department. Departmental areas, circulation and the gross-to-carpet ratio are tabulated in the area statement enclosed at the Annexure.

17.x Operations And Maintenance Plan — Working Papers — Sheet 4

The space programme has been developed from the functional brief agreed with the user department. Departmental areas, circulation and the gross-to-carpet ratio are tabulated in the area statement enclosed at the Annexure.

The structural design conforms to IS 456 for reinforced concrete and IS 1893 for earthquake resistant design. The site falls in seismic zone V and the importance factor applicable to a hospital building has been adopted.

Quantities have been computed from the approved architectural and structural drawings and cross-checked against the detailed measurement sheets enclosed at the Annexure. Rates are as per the schedule of rates in force with lead and lift adjustments.

Consultations were held with the head of each clinical department on adjacency, patient flow and equipment layout. Minutes of the consultation meetings are placed at the Annexure to this chapter.

Reference: drawing no. ST-119, revision C.

17.x Operations And Maintenance Plan — Working Papers — Sheet 5

Detailed Measurement Sheet

Item	L (m)	B (m)	D (m)	Nos	Quantity
Item 200	35.56	2.86	0.78	19	258.835
Item 638	5.16	1.46	2.35	8	37.694
Item 415	25.43	2.99	0.69	8	159.471
Item 585	34.66	3.37	2.67	1	197.856
Item 576	21.51	0.43	1.74	6	129.021
Item 868	36.08	0.62	1.78	21	175.234
Item 149	18.77	1.80	2.90	1	331.904
Item 204	4.87	5.44	2.56	13	265.225
Item 317	14.50	1.66	0.41	11	90.684
Item 671	9.99	3.11	2.75	18	383.148
Item 117	7.78	4.57	0.82	8	292.763
Item 231	9.34	1.53	1.10	22	207.631
Item 597	10.31	3.93	2.48	21	144.319
Item 891	28.27	2.29	1.50	1	275.654
Item 138	28.12	0.31	0.36	10	232.171
Item 192	14.02	4.48	0.63	11	377.824

Quantities have been computed from the approved architectural and structural drawings and cross-checked against the detailed measurement sheets enclosed at the Annexure. Rates are as per the schedule of rates in force with lead and lift adjustments.

17.x Operations And Maintenance Plan — Working Papers — Sheet 6

Design Computation

Moment = 16.025 × 5.921 = 94.891 kNm   (permissible 169.467) — safe

Moment = 4.295 × 9.806 = 42.119 kNm   (permissible 67.759) — revise section

Shear = 20.872 × 0.401 = 8.373 kN   (permissible 13.841) — safe

Axial load = 78.538 × 9.303 = 730.653 kN   (permissible 886.589) — safe

Bearing pressure = 45.274 × 9.925 = 449.358 kPa   (permissible 805.178) — safe

Deflection = 29.043 × 6.630 = 192.556 mm   (permissible 205.812) — safe

Axial load = 88.087 × 10.729 = 945.058 kN   (permissible 1004.543) — safe

Soil investigation was carried out by rotary drilling at the locations shown on the layout plan. The safe bearing capacity adopted for design is based on the lower of the shear and settlement criteria as reported in the geotechnical report.

17.x Operations And Maintenance Plan — Working Papers — Sheet 7

Soil investigation was carried out by rotary drilling at the locations shown on the layout plan. The safe bearing capacity adopted for design is based on the lower of the shear and settlement criteria as reported in the geotechnical report.

The space programme has been developed from the functional brief agreed with the user department. Departmental areas, circulation and the gross-to-carpet ratio are tabulated in the area statement enclosed at the Annexure.

Consultations were held with the head of each clinical department on adjacency, patient flow and equipment layout. Minutes of the consultation meetings are placed at the Annexure to this chapter.

Quantities have been computed from the approved architectural and structural drawings and cross-checked against the detailed measurement sheets enclosed at the Annexure. Rates are as per the schedule of rates in force with lead and lift adjustments.

Reference: drawing no. AR-370, revision A.

17.x Operations And Maintenance Plan — Working Papers — Sheet 8

Departmental Area Statement

Department	Floor	Units	Area per unit (sqm)	Total (sqm)
Operation Theatre	Third	12	27.14	325.67
Pharmacy	First	18	96.70	1740.55
Diagnostics	First	3	89.36	268.09
IPD Ward	Ground	6	25.05	150.29
Administration	Third	4	44.95	179.80
IPD Ward	Third	10	102.92	1029.25
IPD Ward	Ground	12	53.73	644.76
Staff Quarters	Third	9	23.02	207.22
Operation Theatre	Third	8	100.48	803.80
Administration	Second	4	48.81	195.23
OPD	Second	16	127.41	2038.62
IPD Ward	Ground	5	67.24	336.21
Staff Quarters	Second	14	131.55	1841.76
Staff Quarters	First	15	102.34	1535.12
ICU	Second	3	135.61	406.83
Staff Quarters	Third	14	47.29	662.01
Staff Quarters	First	4	81.30	325.20
OPD	Third	1	19.83	19.83
Staff Quarters	First	5	93.09	465.45
Pharmacy	Third	5	125.93	629.63
Pharmacy	First	18	18.75	337.46
Operation Theatre	Third	16	69.62	1113.92
Operation Theatre	First	3	136.62	409.86
Operation Theatre	First	13	97.85	1272.01

The space programme has been developed from the functional brief agreed with the user department. Departmental areas, circulation and the gross-to-carpet ratio are tabulated in the area statement enclosed at the Annexure.

17.x Operations And Maintenance Plan — Working Papers — Sheet 9

Design Computation

Deflection = 17.854 × 11.644 = 207.886 mm   (permissible 314.820) — safe

Moment = 29.127 × 9.762 = 284.324 kNm   (permissible 379.366) — safe

Shear = 16.399 × 0.939 = 15.392 kN   (permissible 17.702) — safe

Bearing pressure = 57.410 × 0.561 = 32.222 kPa   (permissible 53.925) — safe

Shear = 66.156 × 9.679 = 640.321 kN   (permissible 930.140) — safe

Moment = 27.428 × 11.597 = 318.069 kNm   (permissible 524.128) — safe

Barrier-free access has been provided in accordance with the Harmonised Guidelines and Standards for Universal Accessibility. Ramps, lifts, signage and accessible toilets are provided on every floor of the block.

17.x Operations And Maintenance Plan — Working Papers — Sheet 10

The structural design conforms to IS 456 for reinforced concrete and IS 1893 for earthquake resistant design. The site falls in seismic zone V and the importance factor applicable to a hospital building has been adopted.

Fire safety provisions follow Part 4 of the National Building Code. Means of egress, travel distances, staircase widths and the fire fighting installations have been designed for the occupancy classification of the block.

Building services comprise electrical distribution, water supply and sanitation, HVAC for the identified conditioned areas, medical gas pipeline and vertical transportation. Connected load and demand factors are given in the load schedule.

Barrier-free access has been provided in accordance with the Harmonised Guidelines and Standards for Universal Accessibility. Ramps, lifts, signage and accessible toilets are provided on every floor of the block.

Reference: drawing no. ST-841, revision A.

ANNEXURE I — SITE PLAN AND KEY MAP

Site plan showing the proposed block against the existing hospital campus. Drawing SP-001.

ANNEXURE I — Sheet 1

Soil investigation was carried out by rotary drilling at the locations shown on the layout plan. The safe bearing capacity adopted for design is based on the lower of the shear and settlement criteria as reported in the geotechnical report.

Quantities have been computed from the approved architectural and structural drawings and cross-checked against the detailed measurement sheets enclosed at the Annexure. Rates are as per the schedule of rates in force with lead and lift adjustments.

Barrier-free access has been provided in accordance with the Harmonised Guidelines and Standards for Universal Accessibility. Ramps, lifts, signage and accessible toilets are provided on every floor of the block.

The space programme has been developed from the functional brief agreed with the user department. Departmental areas, circulation and the gross-to-carpet ratio are tabulated in the area statement enclosed at the Annexure.

Reference: drawing no. EL-434, revision C.

ANNEXURE I — Sheet 2

Detailed Measurement Sheet

Item	L (m)	B (m)	D (m)	Nos	Quantity
Item 581	25.59	1.84	1.55	18	244.399
Item 242	37.10	5.91	0.37	8	86.815
Item 811	37.09	0.38	0.50	18	107.481
Item 285	26.86	3.05	2.21	19	356.797
Item 878	26.45	2.62	1.71	18	273.430
Item 400	20.44	4.64	1.07	12	357.336
Item 485	9.13	2.64	2.19	14	352.243
Item 376	2.51	2.66	2.56	8	202.830
Item 357	27.29	3.88	0.50	9	137.596
Item 296	13.47	0.50	0.70	14	237.564
Item 684	33.61	5.09	0.40	17	349.529
Item 623	8.79	0.74	0.77	18	291.507
Item 600	19.21	0.49	2.27	15	234.627
Item 627	15.50	4.79	1.35	6	390.852
Item 265	19.41	0.43	0.80	1	354.372
Item 544	29.01	4.74	2.79	21	343.702
Item 448	9.80	3.19	2.96	14	240.115
Item 257	33.31	2.60	1.48	14	395.588
Item 785	31.87	5.34	0.89	6	383.151
Item 141	21.35	4.60	0.38	5	218.067
Item 497	32.40	2.00	2.40	17	125.218
Item 144	28.34	0.47	0.46	10	214.323
Item 236	27.89	5.97	2.05	12	191.272
Item 539	2.57	5.36	2.18	13	342.562
Item 811	7.50	4.31	2.06	17	324.078
Item 376	5.84	3.07	0.50	10	103.238

Consultations were held with the head of each clinical department on adjacency, patient flow and equipment layout. Minutes of the consultation meetings are placed at the Annexure to this chapter.

ANNEXURE I — Sheet 3

Design Computation

Moment = 86.635 × 10.864 = 941.216 kNm   (permissible 1492.483) — safe

Moment = 52.989 × 11.211 = 594.060 kNm   (permissible 624.171) — safe

Axial load = 24.557 × 10.332 = 253.716 kN   (permissible 306.511) — safe

Deflection = 77.864 × 4.861 = 378.470 mm   (permissible 498.533) — safe

Axial load = 33.011 × 3.951 = 130.430 kN   (permissible 204.874) — safe

Bearing pressure = 21.888 × 0.858 = 18.770 kPa   (permissible 25.110) — revise section

Barrier-free access has been provided in accordance with the Harmonised Guidelines and Standards for Universal Accessibility. Ramps, lifts, signage and accessible toilets are provided on every floor of the block.

ANNEXURE I — Sheet 4

The space programme has been developed from the functional brief agreed with the user department. Departmental areas, circulation and the gross-to-carpet ratio are tabulated in the area statement enclosed at the Annexure.

The structural design conforms to IS 456 for reinforced concrete and IS 1893 for earthquake resistant design. The site falls in seismic zone V and the importance factor applicable to a hospital building has been adopted.

The structural design conforms to IS 456 for reinforced concrete and IS 1893 for earthquake resistant design. The site falls in seismic zone V and the importance factor applicable to a hospital building has been adopted.

Building services comprise electrical distribution, water supply and sanitation, HVAC for the identified conditioned areas, medical gas pipeline and vertical transportation. Connected load and demand factors are given in the load schedule.

Reference: drawing no. AR-563, revision C.

ANNEXURE I — Sheet 5

Equipment Schedule

Item	Department	Nos	Unit rate (Rs)	Amount (Rs)
X-ray unit	OT	8	1,893,917	15,151,336
Patient monitor	Casualty	6	73,409	440,454
Autoclave	Casualty	21	1,018,023	21,378,483
Patient monitor	Ward	23	2,049,869	47,146,987
Defibrillator	Ward	17	1,633,357	27,767,069
Hospital bed	CSSD	29	506,971	14,702,159
X-ray unit	Casualty	29	654,820	18,989,780
Patient monitor	Casualty	38	1,658,179	63,010,802
Autoclave	ICU	6	1,091,735	6,550,410
Ventilator	OT	26	1,520,976	39,545,376
Ultrasound	CSSD	24	708,944	17,014,656
Defibrillator	OT	40	673,658	26,946,320
Ventilator	CSSD	37	2,217,936	82,063,632
Ultrasound	Radiology	11	359,768	3,957,448
OT table	OT	10	2,281,951	22,819,510
Ventilator	Radiology	37	462,642	17,117,754
Patient monitor	OT	4	2,186,471	8,745,884
X-ray unit	Casualty	5	1,367,123	6,835,615
Ventilator	CSSD	40	2,046,621	81,864,840
Defibrillator	ICU	26	1,276,422	33,186,972
X-ray unit	Ward	18	106,146	1,910,628

The structural design conforms to IS 456 for reinforced concrete and IS 1893 for earthquake resistant design. The site falls in seismic zone V and the importance factor applicable to a hospital building has been adopted.

ANNEXURE I — Sheet 6

Design Computation

Moment = 56.392 × 9.336 = 526.483 kNm   (permissible 842.018) — revise section

Bearing pressure = 89.505 × 9.892 = 885.423 kPa   (permissible 1235.013) — safe

Deflection = 56.597 × 7.473 = 422.939 mm   (permissible 628.982) — safe

Shear = 23.041 × 0.428 = 9.871 kN   (permissible 17.640) — safe

Bearing pressure = 34.173 × 1.279 = 43.704 kPa   (permissible 60.441) — safe

Moment = 45.119 × 0.772 = 34.846 kNm   (permissible 39.842) — safe

Shear = 1.758 × 10.911 = 19.180 kN   (permissible 24.967) — safe

Consultations were held with the head of each clinical department on adjacency, patient flow and equipment layout. Minutes of the consultation meetings are placed at the Annexure to this chapter.

ANNEXURE I — Sheet 7

The space programme has been developed from the functional brief agreed with the user department. Departmental areas, circulation and the gross-to-carpet ratio are tabulated in the area statement enclosed at the Annexure.

The space programme has been developed from the functional brief agreed with the user department. Departmental areas, circulation and the gross-to-carpet ratio are tabulated in the area statement enclosed at the Annexure.

Barrier-free access has been provided in accordance with the Harmonised Guidelines and Standards for Universal Accessibility. Ramps, lifts, signage and accessible toilets are provided on every floor of the block.

The space programme has been developed from the functional brief agreed with the user department. Departmental areas, circulation and the gross-to-carpet ratio are tabulated in the area statement enclosed at the Annexure.

Reference: drawing no. EL-185, revision A.

ANNEXURE I — Sheet 8

Bore Log Summary

Depth (m)	Strata	N value	Sample
8.4	Medium sand	9	SPT
7.8	Weathered rock	35	SPT
17.6	Clayey silt	44	DS
16.8	Medium sand	15	SPT
8.5	Gravelly sand	34	UDS
26.0	Gravelly sand	20	UDS
3.3	Medium sand	18	DS
28.0	Silty clay	15	DS
5.8	Gravelly sand	40	DS
10.2	Medium sand	35	SPT
6.9	Medium sand	36	SPT
23.3	Gravelly sand	30	SPT
29.7	Gravelly sand	60	DS
24.0	Gravelly sand	54	SPT
4.2	Medium sand	31	UDS
26.7	Gravelly sand	6	UDS
27.3	Medium sand	48	SPT
5.6	Medium sand	27	UDS
25.6	Gravelly sand	38	UDS
2.1	Gravelly sand	29	SPT
1.9	Gravelly sand	37	SPT
14.5	Silty clay	52	SPT

The structural design conforms to IS 456 for reinforced concrete and IS 1893 for earthquake resistant design. The site falls in seismic zone V and the importance factor applicable to a hospital building has been adopted.

ANNEXURE II — ARCHITECTURAL DRAWINGS

Floor plans, elevations and sections. Drawings AR-010 to AR-042.

ANNEXURE II — Sheet 1

Barrier-free access has been provided in accordance with the Harmonised Guidelines and Standards for Universal Accessibility. Ramps, lifts, signage and accessible toilets are provided on every floor of the block.

Consultations were held with the head of each clinical department on adjacency, patient flow and equipment layout. Minutes of the consultation meetings are placed at the Annexure to this chapter.

Barrier-free access has been provided in accordance with the Harmonised Guidelines and Standards for Universal Accessibility. Ramps, lifts, signage and accessible toilets are provided on every floor of the block.

Building services comprise electrical distribution, water supply and sanitation, HVAC for the identified conditioned areas, medical gas pipeline and vertical transportation. Connected load and demand factors are given in the load schedule.

Reference: drawing no. ST-792, revision B.

ANNEXURE II — Sheet 2

Room Data Sheet

Room	Department	Carpet area (sqm)	Occupancy	Floor finish
Room 553	Casualty	46.46	29	Ceramic tile
Room 433	Surgery	28.71	21	Ceramic tile
Room 872	Casualty	62.35	7	Epoxy screed
Room 434	Radiology	19.66	15	Vitrified tile
Room 634	Surgery	71.49	15	Epoxy screed
Room 604	Paediatrics	89.34	30	Antistatic vinyl
Room 741	Casualty	17.29	13	Kota stone
Room 111	Surgery	26.68	7	Vitrified tile
Room 466	Surgery	81.92	23	Antistatic vinyl
Room 546	Surgery	34.58	21	Kota stone
Room 296	Paediatrics	33.24	28	Epoxy screed
Room 884	Surgery	89.93	25	Vitrified tile
Room 727	Surgery	74.79	13	Ceramic tile
Room 880	Pathology	37.73	14	Antistatic vinyl
Room 346	Pathology	81.91	29	Kota stone
Room 213	Surgery	57.74	1	Vitrified tile
Room 781	Pathology	41.36	11	Vitrified tile
Room 394	Obstetrics	23.02	28	Ceramic tile
Room 793	Radiology	88.54	10	Antistatic vinyl
Room 673	Surgery	64.12	15	Antistatic vinyl
Room 550	General Medicine	81.83	6	Kota stone
Room 843	Obstetrics	45.88	24	Ceramic tile

The structural design conforms to IS 456 for reinforced concrete and IS 1893 for earthquake resistant design. The site falls in seismic zone V and the importance factor applicable to a hospital building has been adopted.

ANNEXURE II — Sheet 3

Design Computation

Shear = 45.487 × 2.578 = 117.260 kN   (permissible 186.620) — safe

Moment = 72.655 × 10.404 = 755.882 kNm   (permissible 1112.552) — revise section

Moment = 18.335 × 8.303 = 152.238 kNm   (permissible 186.513) — safe

Bearing pressure = 4.231 × 6.667 = 28.210 kPa   (permissible 41.601) — safe

Bearing pressure = 21.848 × 8.126 = 177.538 kPa   (permissible 328.425) — safe

Building services comprise electrical distribution, water supply and sanitation, HVAC for the identified conditioned areas, medical gas pipeline and vertical transportation. Connected load and demand factors are given in the load schedule.

ANNEXURE II — Sheet 4

Consultations were held with the head of each clinical department on adjacency, patient flow and equipment layout. Minutes of the consultation meetings are placed at the Annexure to this chapter.

Consultations were held with the head of each clinical department on adjacency, patient flow and equipment layout. Minutes of the consultation meetings are placed at the Annexure to this chapter.

Building services comprise electrical distribution, water supply and sanitation, HVAC for the identified conditioned areas, medical gas pipeline and vertical transportation. Connected load and demand factors are given in the load schedule.

Soil investigation was carried out by rotary drilling at the locations shown on the layout plan. The safe bearing capacity adopted for design is based on the lower of the shear and settlement criteria as reported in the geotechnical report.

Reference: drawing no. ST-120, revision A.

ANNEXURE II — Sheet 5

Electrical Load Schedule

Panel	Description	Connected load (kW)	Demand factor	Demand (kVA)
DB-39	Medical equipment	117.17	0.68	88.32
DB-26	Medical equipment	116.26	0.89	115.46
DB-20	Power outlets	153.79	0.92	156.73
DB-36	Medical equipment	46.46	0.46	23.49
DB-32	Lifts	132.80	0.71	104.59
DB-16	Pumps	97.94	0.47	50.65
DB-36	Sterilisation	165.52	0.76	140.45
DB-18	Sterilisation	23.37	0.74	19.16
DB-40	Lifts	41.26	0.73	33.60
DB-36	Power outlets	28.43	0.93	29.42
DB-16	Pumps	142.10	0.59	92.67
DB-06	HVAC	113.16	0.52	65.17
DB-13	Sterilisation	123.66	0.75	102.86
DB-29	Lighting	37.28	0.51	21.04
DB-19	HVAC	126.81	0.78	109.97
DB-10	Pumps	37.93	0.93	39.18

Barrier-free access has been provided in accordance with the Harmonised Guidelines and Standards for Universal Accessibility. Ramps, lifts, signage and accessible toilets are provided on every floor of the block.

ANNEXURE II — Sheet 6

Design Computation

Bearing pressure = 2.908 × 3.891 = 11.316 kPa   (permissible 14.958) — safe

Bearing pressure = 35.392 × 8.043 = 284.652 kPa   (permissible 387.607) — safe

Deflection = 12.198 × 3.696 = 45.082 mm   (permissible 63.678) — safe

Shear = 17.728 × 9.703 = 172.025 kN   (permissible 240.944) — safe

Shear = 31.848 × 11.077 = 352.793 kN   (permissible 456.080) — safe

Consultations were held with the head of each clinical department on adjacency, patient flow and equipment layout. Minutes of the consultation meetings are placed at the Annexure to this chapter.

ANNEXURE II — Sheet 7

Building services comprise electrical distribution, water supply and sanitation, HVAC for the identified conditioned areas, medical gas pipeline and vertical transportation. Connected load and demand factors are given in the load schedule.

Building services comprise electrical distribution, water supply and sanitation, HVAC for the identified conditioned areas, medical gas pipeline and vertical transportation. Connected load and demand factors are given in the load schedule.

The space programme has been developed from the functional brief agreed with the user department. Departmental areas, circulation and the gross-to-carpet ratio are tabulated in the area statement enclosed at the Annexure.

Quantities have been computed from the approved architectural and structural drawings and cross-checked against the detailed measurement sheets enclosed at the Annexure. Rates are as per the schedule of rates in force with lead and lift adjustments.

Reference: drawing no. EL-852, revision A.

ANNEXURE II — Sheet 8

Room Data Sheet

Room	Department	Carpet area (sqm)	Occupancy	Floor finish
Room 786	Obstetrics	83.32	8	Kota stone
Room 417	Surgery	39.32	10	Antistatic vinyl
Room 557	Pathology	73.34	13	Kota stone
Room 543	Casualty	89.14	19	Kota stone
Room 653	Pathology	25.05	24	Vitrified tile
Room 477	Surgery	68.45	23	Antistatic vinyl
Room 781	Obstetrics	23.55	2	Antistatic vinyl
Room 529	Pathology	30.49	8	Vitrified tile
Room 168	Casualty	16.61	3	Vitrified tile
Room 871	Obstetrics	35.20	11	Ceramic tile
Room 394	Surgery	83.33	26	Vitrified tile
Room 846	Paediatrics	83.24	23	Antistatic vinyl
Room 163	Surgery	37.42	14	Ceramic tile
Room 304	Pathology	37.46	5	Antistatic vinyl
Room 470	Obstetrics	81.65	15	Vitrified tile
Room 616	Surgery	10.20	5	Ceramic tile
Room 341	Radiology	78.58	18	Ceramic tile
Room 734	Paediatrics	64.06	10	Epoxy screed
Room 372	Surgery	11.53	12	Ceramic tile
Room 314	Paediatrics	49.55	2	Epoxy screed
Room 162	Pathology	91.38	30	Ceramic tile
Room 539	Radiology	48.50	21	Antistatic vinyl
Room 231	Radiology	72.88	5	Ceramic tile
Room 878	Casualty	73.83	28	Vitrified tile

Soil investigation was carried out by rotary drilling at the locations shown on the layout plan. The safe bearing capacity adopted for design is based on the lower of the shear and settlement criteria as reported in the geotechnical report.

ANNEXURE II — Sheet 9

Design Computation

Axial load = 38.202 × 1.158 = 44.244 kN   (permissible 53.095) — safe

Axial load = 65.538 × 1.656 = 108.543 kN   (permissible 193.509) — revise section

Bearing pressure = 7.943 × 6.740 = 53.535 kPa   (permissible 99.509) — safe

Deflection = 80.322 × 6.374 = 511.975 mm   (permissible 566.350) — safe

Bearing pressure = 27.971 × 0.596 = 16.658 kPa   (permissible 20.888) — safe

Deflection = 69.734 × 2.102 = 146.573 mm   (permissible 195.339) — safe

Deflection = 5.614 × 4.932 = 27.686 mm   (permissible 43.948) — safe

Quantities have been computed from the approved architectural and structural drawings and cross-checked against the detailed measurement sheets enclosed at the Annexure. Rates are as per the schedule of rates in force with lead and lift adjustments.

ANNEXURE II — Sheet 10

Fire safety provisions follow Part 4 of the National Building Code. Means of egress, travel distances, staircase widths and the fire fighting installations have been designed for the occupancy classification of the block.

The structural design conforms to IS 456 for reinforced concrete and IS 1893 for earthquake resistant design. The site falls in seismic zone V and the importance factor applicable to a hospital building has been adopted.

Quantities have been computed from the approved architectural and structural drawings and cross-checked against the detailed measurement sheets enclosed at the Annexure. Rates are as per the schedule of rates in force with lead and lift adjustments.

The space programme has been developed from the functional brief agreed with the user department. Departmental areas, circulation and the gross-to-carpet ratio are tabulated in the area statement enclosed at the Annexure.

Reference: drawing no. ST-365, revision C.

ANNEXURE II — Sheet 11

Room Data Sheet

Room	Department	Carpet area (sqm)	Occupancy	Floor finish
Room 693	Paediatrics	63.41	23	Ceramic tile
Room 715	Radiology	24.78	9	Antistatic vinyl
Room 194	Radiology	79.44	13	Epoxy screed
Room 138	Radiology	54.44	10	Antistatic vinyl
Room 893	Obstetrics	13.07	22	Vitrified tile
Room 129	General Medicine	59.49	13	Ceramic tile
Room 250	Paediatrics	48.33	10	Ceramic tile
Room 583	Radiology	60.22	4	Ceramic tile
Room 580	General Medicine	44.03	30	Antistatic vinyl
Room 377	Surgery	39.10	15	Epoxy screed
Room 426	Pathology	87.52	28	Antistatic vinyl
Room 878	Casualty	69.44	22	Epoxy screed
Room 116	Paediatrics	66.80	6	Epoxy screed
Room 836	Pathology	18.53	19	Vitrified tile
Room 398	Pathology	49.23	2	Vitrified tile
Room 899	Casualty	87.53	29	Antistatic vinyl
Room 456	Casualty	42.41	19	Antistatic vinyl
Room 882	Obstetrics	42.62	6	Antistatic vinyl
Room 678	Paediatrics	28.06	30	Ceramic tile

The structural design conforms to IS 456 for reinforced concrete and IS 1893 for earthquake resistant design. The site falls in seismic zone V and the importance factor applicable to a hospital building has been adopted.

ANNEXURE II — Sheet 12

Design Computation

Deflection = 39.231 × 10.478 = 411.063 mm   (permissible 579.656) — safe

Axial load = 71.561 × 9.658 = 691.105 kN   (permissible 1197.556) — safe

Deflection = 89.817 × 10.244 = 920.132 mm   (permissible 1420.964) — safe

Bearing pressure = 79.815 × 1.470 = 117.355 kPa   (permissible 206.069) — safe

Deflection = 60.552 × 7.337 = 444.249 mm   (permissible 708.152) — safe

Bearing pressure = 55.092 × 2.224 = 122.500 kPa   (permissible 206.454) — safe

Moment = 40.804 × 2.564 = 104.615 kNm   (permissible 168.172) — safe

Shear = 87.208 × 5.216 = 454.902 kN   (permissible 815.164) — safe

Quantities have been computed from the approved architectural and structural drawings and cross-checked against the detailed measurement sheets enclosed at the Annexure. Rates are as per the schedule of rates in force with lead and lift adjustments.

ANNEXURE II — Sheet 13

Quantities have been computed from the approved architectural and structural drawings and cross-checked against the detailed measurement sheets enclosed at the Annexure. Rates are as per the schedule of rates in force with lead and lift adjustments.

Building services comprise electrical distribution, water supply and sanitation, HVAC for the identified conditioned areas, medical gas pipeline and vertical transportation. Connected load and demand factors are given in the load schedule.

Fire safety provisions follow Part 4 of the National Building Code. Means of egress, travel distances, staircase widths and the fire fighting installations have been designed for the occupancy classification of the block.

Building services comprise electrical distribution, water supply and sanitation, HVAC for the identified conditioned areas, medical gas pipeline and vertical transportation. Connected load and demand factors are given in the load schedule.

Reference: drawing no. EL-308, revision A.

ANNEXURE II — Sheet 14

Bore Log Summary

Depth (m)	Strata	N value	Sample
28.8	Weathered rock	15	DS
2.7	Clayey silt	14	DS
12.1	Weathered rock	27	UDS
23.5	Silty clay	11	SPT
29.8	Gravelly sand	25	SPT
6.0	Medium sand	14	DS
29.2	Gravelly sand	53	DS
21.6	Coarse sand	42	DS
7.1	Coarse sand	60	UDS
27.8	Clayey silt	14	DS
1.0	Medium sand	39	SPT
1.0	Weathered rock	8	DS
25.9	Weathered rock	38	SPT
24.1	Silty clay	7	SPT
22.8	Coarse sand	43	DS
5.1	Weathered rock	21	SPT
25.4	Clayey silt	50	SPT
23.2	Clayey silt	46	SPT

Building services comprise electrical distribution, water supply and sanitation, HVAC for the identified conditioned areas, medical gas pipeline and vertical transportation. Connected load and demand factors are given in the load schedule.

ANNEXURE II — Sheet 15

Design Computation

Deflection = 29.682 × 2.639 = 78.339 mm   (permissible 127.323) — safe

Moment = 2.539 × 1.776 = 4.509 kNm   (permissible 7.082) — safe

Axial load = 25.033 × 2.726 = 68.228 kN   (permissible 85.980) — safe

Shear = 66.288 × 1.548 = 102.596 kN   (permissible 163.287) — safe

Deflection = 40.533 × 2.591 = 105.017 mm   (permissible 195.157) — safe

Moment = 17.413 × 9.838 = 171.312 kNm   (permissible 208.997) — safe

Shear = 83.469 × 7.243 = 604.563 kN   (permissible 753.419) — safe

Deflection = 5.191 × 10.660 = 55.342 mm   (permissible 103.112) — safe

Barrier-free access has been provided in accordance with the Harmonised Guidelines and Standards for Universal Accessibility. Ramps, lifts, signage and accessible toilets are provided on every floor of the block.

ANNEXURE II — Sheet 16

Barrier-free access has been provided in accordance with the Harmonised Guidelines and Standards for Universal Accessibility. Ramps, lifts, signage and accessible toilets are provided on every floor of the block.

The structural design conforms to IS 456 for reinforced concrete and IS 1893 for earthquake resistant design. The site falls in seismic zone V and the importance factor applicable to a hospital building has been adopted.

Building services comprise electrical distribution, water supply and sanitation, HVAC for the identified conditioned areas, medical gas pipeline and vertical transportation. Connected load and demand factors are given in the load schedule.

Consultations were held with the head of each clinical department on adjacency, patient flow and equipment layout. Minutes of the consultation meetings are placed at the Annexure to this chapter.

Reference: drawing no. EL-161, revision B.

ANNEXURE II — Sheet 17

Bore Log Summary

Depth (m)	Strata	N value	Sample
2.4	Medium sand	7	DS
11.1	Weathered rock	51	UDS
9.6	Clayey silt	6	SPT
12.0	Weathered rock	56	DS
27.5	Silty clay	10	UDS
20.4	Medium sand	10	SPT
8.1	Weathered rock	35	SPT
27.5	Gravelly sand	20	SPT
14.3	Silty clay	31	DS
14.6	Gravelly sand	55	SPT
12.8	Weathered rock	32	DS
4.2	Silty clay	38	SPT
24.3	Clayey silt	50	DS
10.0	Silty clay	56	SPT
24.3	Gravelly sand	42	DS
4.9	Clayey silt	42	SPT
2.5	Coarse sand	57	UDS
29.9	Gravelly sand	17	DS
26.0	Gravelly sand	37	SPT
16.5	Silty clay	4	DS
20.2	Clayey silt	16	DS

The structural design conforms to IS 456 for reinforced concrete and IS 1893 for earthquake resistant design. The site falls in seismic zone V and the importance factor applicable to a hospital building has been adopted.

ANNEXURE II — Sheet 18

Design Computation

Bearing pressure = 26.642 × 7.037 = 187.478 kPa   (permissible 352.493) — safe

Moment = 66.217 × 11.153 = 738.531 kNm   (permissible 1124.724) — safe

Moment = 25.740 × 8.167 = 210.213 kNm   (permissible 284.167) — safe

Axial load = 83.593 × 10.482 = 876.264 kN   (permissible 1631.423) — safe

Axial load = 42.422 × 1.966 = 83.397 kN   (permissible 133.009) — safe

Deflection = 64.943 × 7.511 = 487.785 mm   (permissible 912.242) — safe

Building services comprise electrical distribution, water supply and sanitation, HVAC for the identified conditioned areas, medical gas pipeline and vertical transportation. Connected load and demand factors are given in the load schedule.

ANNEXURE III — STRUCTURAL DRAWINGS

Foundation layout, framing plans and reinforcement details. Drawings ST-050 to ST-088.

ANNEXURE III — Sheet 1

Soil investigation was carried out by rotary drilling at the locations shown on the layout plan. The safe bearing capacity adopted for design is based on the lower of the shear and settlement criteria as reported in the geotechnical report.

The space programme has been developed from the functional brief agreed with the user department. Departmental areas, circulation and the gross-to-carpet ratio are tabulated in the area statement enclosed at the Annexure.

Consultations were held with the head of each clinical department on adjacency, patient flow and equipment layout. Minutes of the consultation meetings are placed at the Annexure to this chapter.

Building services comprise electrical distribution, water supply and sanitation, HVAC for the identified conditioned areas, medical gas pipeline and vertical transportation. Connected load and demand factors are given in the load schedule.

Reference: drawing no. AR-898, revision C.

ANNEXURE III — Sheet 2

Bore Log Summary

Depth (m)	Strata	N value	Sample
7.5	Silty clay	53	UDS
10.7	Silty clay	51	UDS
4.0	Clayey silt	51	SPT
17.4	Clayey silt	28	SPT
6.5	Gravelly sand	10	SPT
28.3	Weathered rock	42	SPT
8.1	Medium sand	11	UDS
27.4	Weathered rock	55	DS
4.6	Clayey silt	59	SPT
2.5	Gravelly sand	57	SPT
8.8	Weathered rock	53	SPT
19.2	Silty clay	56	UDS
5.6	Silty clay	44	UDS
2.9	Silty clay	8	SPT
6.9	Coarse sand	54	SPT
14.9	Gravelly sand	36	UDS
1.9	Medium sand	30	DS
21.5	Gravelly sand	6	SPT
15.7	Silty clay	25	UDS
27.6	Coarse sand	49	UDS
20.9	Clayey silt	7	SPT
21.5	Weathered rock	22	UDS
13.5	Silty clay	12	DS
14.8	Silty clay	45	UDS
26.4	Weathered rock	8	UDS

Soil investigation was carried out by rotary drilling at the locations shown on the layout plan. The safe bearing capacity adopted for design is based on the lower of the shear and settlement criteria as reported in the geotechnical report.

ANNEXURE III — Sheet 3

Design Computation

Moment = 21.050 × 1.633 = 34.384 kNm   (permissible 52.388) — safe

Axial load = 14.160 × 11.692 = 165.556 kN   (permissible 306.158) — safe

Deflection = 7.955 × 2.960 = 23.542 mm   (permissible 28.363) — safe

Axial load = 37.690 × 11.231 = 423.281 kN   (permissible 599.094) — safe

Moment = 37.429 × 6.658 = 249.214 kNm   (permissible 443.991) — safe

Axial load = 65.991 × 10.898 = 719.196 kN   (permissible 952.890) — safe

Soil investigation was carried out by rotary drilling at the locations shown on the layout plan. The safe bearing capacity adopted for design is based on the lower of the shear and settlement criteria as reported in the geotechnical report.

ANNEXURE III — Sheet 4

Fire safety provisions follow Part 4 of the National Building Code. Means of egress, travel distances, staircase widths and the fire fighting installations have been designed for the occupancy classification of the block.

Building services comprise electrical distribution, water supply and sanitation, HVAC for the identified conditioned areas, medical gas pipeline and vertical transportation. Connected load and demand factors are given in the load schedule.

Building services comprise electrical distribution, water supply and sanitation, HVAC for the identified conditioned areas, medical gas pipeline and vertical transportation. Connected load and demand factors are given in the load schedule.

The structural design conforms to IS 456 for reinforced concrete and IS 1893 for earthquake resistant design. The site falls in seismic zone V and the importance factor applicable to a hospital building has been adopted.

Reference: drawing no. AR-410, revision A.

ANNEXURE III — Sheet 5

Room Data Sheet

Room	Department	Carpet area (sqm)	Occupancy	Floor finish
Room 812	Obstetrics	85.18	21	Antistatic vinyl
Room 100	Obstetrics	18.39	9	Ceramic tile
Room 623	Obstetrics	63.62	13	Ceramic tile
Room 607	General Medicine	17.95	13	Epoxy screed
Room 138	Pathology	9.61	14	Epoxy screed
Room 475	Obstetrics	72.63	30	Epoxy screed
Room 532	Paediatrics	61.81	23	Kota stone
Room 268	Surgery	59.20	4	Ceramic tile
Room 151	Casualty	38.66	5	Kota stone
Room 563	Pathology	56.18	5	Ceramic tile
Room 561	Paediatrics	88.88	27	Antistatic vinyl
Room 727	Surgery	87.18	15	Vitrified tile
Room 488	Surgery	37.60	30	Ceramic tile
Room 408	Casualty	30.95	21	Ceramic tile
Room 158	Radiology	50.38	23	Vitrified tile
Room 559	General Medicine	31.34	9	Ceramic tile
Room 497	Surgery	9.19	17	Epoxy screed
Room 588	Obstetrics	31.18	15	Vitrified tile
Room 588	Surgery	57.16	18	Kota stone
Room 449	Paediatrics	59.20	10	Vitrified tile
Room 373	Surgery	19.85	7	Epoxy screed
Room 421	Obstetrics	27.38	2	Ceramic tile
Room 877	Surgery	51.24	5	Kota stone

Consultations were held with the head of each clinical department on adjacency, patient flow and equipment layout. Minutes of the consultation meetings are placed at the Annexure to this chapter.

ANNEXURE III — Sheet 6

Design Computation

Deflection = 61.204 × 1.698 = 103.913 mm (permissible 122.666) — safe

Moment = 57.135 × 2.352 = 134.377 kNm (permissible 247.525) — safe

Moment = 81.742 × 10.808 = 883.500 kNm (permissible 936.766) — safe

Deflection = 81.910 × 11.573 = 947.909 mm (permissible 1429.890) — safe

Deflection = 38.302 × 4.322 = 165.557 mm (permissible 272.060) — safe

Shear = 37.339 × 10.584 = 395.191 kN (permissible 735.577) — safe

Soil investigation was carried out by rotary drilling at the locations shown on the layout plan. The safe bearing capacity adopted for design is based on the lower of the shear and settlement criteria as reported in the geotechnical report.

ANNEXURE III — Sheet 7

Barrier-free access has been provided in accordance with the Harmonised Guidelines and Standards for Universal Accessibility. Ramps, lifts, signage and accessible toilets are provided on every floor of the block.

Quantities have been computed from the approved architectural and structural drawings and cross-checked against the detailed measurement sheets enclosed at the Annexure. Rates are as per the schedule of rates in force with lead and lift adjustments.

The structural design conforms to IS 456 for reinforced concrete and IS 1893 for earthquake resistant design. The site falls in seismic zone V and the importance factor applicable to a hospital building has been adopted.

The space programme has been developed from the functional brief agreed with the user department. Departmental areas, circulation and the gross-to-carpet ratio are tabulated in the area statement enclosed at the Annexure.

Reference: drawing no. ST-113, revision A.

ANNEXURE III — Sheet 8

Electrical Load Schedule

Panel	Description	Connected load (kW)	Demand factor	Demand (kVA)
DB-27	Lifts	35.04	0.66	25.53
DB-23	Lifts	34.47	0.47	18.12
DB-40	Pumps	16.84	0.68	12.67
DB-07	HVAC	175.46	0.44	86.38
DB-10	Lifts	110.00	0.49	59.37
DB-24	Power outlets	106.53	0.78	92.47
DB-27	Lighting	26.49	0.41	11.94
DB-06	Lifts	138.96	0.46	70.40
DB-02	Medical equipment	74.84	0.81	67.55
DB-12	Power outlets	154.87	0.88	151.20
DB-05	Lighting	172.51	0.49	93.98
DB-39	Pumps	115.75	0.78	99.75
DB-22	Lighting	9.07	0.68	6.85
DB-34	Medical equipment	156.07	0.54	93.69
DB-30	Power outlets	43.50	0.76	36.57
DB-01	Lifts	58.51	0.50	32.37
DB-04	Sterilisation	5.80	0.54	3.48
DB-06	Sterilisation	25.19	0.90	25.33
DB-31	HVAC	78.32	0.64	55.66
DB-31	Medical equipment	125.31	0.75	104.06

The structural design conforms to IS 456 for reinforced concrete and IS 1893 for earthquake resistant design. The site falls in seismic zone V and the importance factor applicable to a hospital building has been adopted.

ANNEXURE III — Sheet 9

Design Computation

Moment = 44.300 × 10.498 = 465.043 kNm   (permissible 693.766) — safe

Bearing pressure = 5.693 × 3.454 = 19.664 kPa   (permissible 31.049) — safe

Moment = 10.699 × 8.261 = 88.393 kNm   (permissible 97.576) — safe

Deflection = 73.272 × 2.542 = 186.251 mm   (permissible 331.607) — safe

Axial load = 64.045 × 7.438 = 476.367 kN   (permissible 813.261) — safe

Shear = 47.218 × 3.657 = 172.682 kN   (permissible 231.894) — safe

Bearing pressure = 52.524 × 1.768 = 92.882 kPa   (permissible 137.204) — safe

Bearing pressure = 53.554 × 1.798 = 96.302 kPa   (permissible 182.930) — safe

Quantities have been computed from the approved architectural and structural drawings and cross-checked against the detailed measurement sheets enclosed at the Annexure. Rates are as per the schedule of rates in force with lead and lift adjustments.

ANNEXURE III — Sheet 10

Consultations were held with the head of each clinical department on adjacency, patient flow and equipment layout. Minutes of the consultation meetings are placed at the Annexure to this chapter.

Soil investigation was carried out by rotary drilling at the locations shown on the layout plan. The safe bearing capacity adopted for design is based on the lower of the shear and settlement criteria as reported in the geotechnical report.

Fire safety provisions follow Part 4 of the National Building Code. Means of egress, travel distances, staircase widths and the fire fighting installations have been designed for the occupancy classification of the block.

Building services comprise electrical distribution, water supply and sanitation, HVAC for the identified conditioned areas, medical gas pipeline and vertical transportation. Connected load and demand factors are given in the load schedule.

Reference: drawing no. PH-158, revision C.

ANNEXURE III — Sheet 11

Departmental Area Statement

Department	Floor	Units	Area per unit (sqm)	Total (sqm)
ICU	Second	5	82.11	410.53
Administration	Ground	11	30.44	334.80
Administration	Third	12	108.12	1297.47
ICU	Ground	7	23.34	163.40
IPD Ward	Third	12	110.74	1328.91
Diagnostics	Third	16	72.82	1165.15
IPD Ward	Ground	15	113.03	1695.39
ICU	Second	4	80.10	320.41
Pharmacy	Third	5	104.61	523.03
ICU	First	7	113.20	792.43
Pharmacy	Second	12	135.62	1627.43
OPD	Third	13	86.30	1121.88
Operation Theatre	Ground	5	85.89	429.43
Administration	Second	7	53.40	373.79
Operation Theatre	Third	10	130.87	1308.74
OPD	First	3	113.61	340.83
Diagnostics	First	3	65.89	197.66
IPD Ward	Second	7	137.10	959.69
ICU	Third	2	67.89	135.79
Diagnostics	Ground	8	38.66	309.28
Administration	Second	11	137.84	1516.28
ICU	First	7	127.45	892.14

Fire safety provisions follow Part 4 of the National Building Code. Means of egress, travel distances, staircase widths and the fire fighting installations have been designed for the occupancy classification of the block.

ANNEXURE III — Sheet 12

Design Computation

Bearing pressure = 85.315 × 7.340 = 626.240 kPa   (permissible 749.191) — safe

Deflection = 14.766 × 0.496 = 7.331 mm   (permissible 13.646) — safe

Deflection = 11.053 × 10.786 = 119.216 mm   (permissible 151.939) — safe

Deflection = 15.799 × 9.515 = 150.335 mm   (permissible 183.361) — safe

Shear = 77.745 × 7.469 = 580.641 kN   (permissible 961.059) — safe

Shear = 63.293 × 8.919 = 564.498 kN   (permissible 951.413) — safe

Moment = 69.878 × 3.055 = 213.476 kNm   (permissible 236.499) — safe

Building services comprise electrical distribution, water supply and sanitation, HVAC for the identified conditioned areas, medical gas pipeline and vertical transportation. Connected load and demand factors are given in the load schedule.

ANNEXURE III — Sheet 13

Fire safety provisions follow Part 4 of the National Building Code. Means of egress, travel distances, staircase widths and the fire fighting installations have been designed for the occupancy classification of the block.

Barrier-free access has been provided in accordance with the Harmonised Guidelines and Standards for Universal Accessibility. Ramps, lifts, signage and accessible toilets are provided on every floor of the block.

Quantities have been computed from the approved architectural and structural drawings and cross-checked against the detailed measurement sheets enclosed at the Annexure. Rates are as per the schedule of rates in force with lead and lift adjustments.

Soil investigation was carried out by rotary drilling at the locations shown on the layout plan. The safe bearing capacity adopted for design is based on the lower of the shear and settlement criteria as reported in the geotechnical report.

Reference: drawing no. PH-886, revision C.

ANNEXURE III — Sheet 14

Equipment Schedule

Item	Department	Nos	Unit rate (Rs)	Amount (Rs)
Autoclave	ICU	31	2,262,363	70,133,253
X-ray unit	Radiology	12	1,979,442	23,753,304
Hospital bed	ICU	11	2,362,853	25,991,383
Patient monitor	ICU	2	2,290,278	4,580,556
X-ray unit	OT	4	1,949,199	7,796,796
Ventilator	Casualty	28	1,926,948	53,954,544
OT table	ICU	31	865,266	26,823,246
OT table	OT	26	987,149	25,665,874
Hospital bed	Radiology	40	2,367,078	94,683,120
Ventilator	ICU	1	288,145	288,145
Defibrillator	OT	39	1,590,292	62,021,388
X-ray unit	Radiology	18	1,504,104	27,073,872
Ventilator	ICU	25	107,138	2,678,450
X-ray unit	Radiology	13	132,469	1,722,097
X-ray unit	Casualty	15	1,704,560	25,568,400
Patient monitor	Ward	33	1,295,462	42,750,246
Ultrasound	CSSD	9	223,237	2,009,133

The structural design conforms to IS 456 for reinforced concrete and IS 1893 for earthquake resistant design. The site falls in seismic zone V and the importance factor applicable to a hospital building has been adopted.

ANNEXURE III — Sheet 15

Design Computation

Shear = 83.994 × 2.899 = 243.519 kN (permissible 361.199) — safe

Shear = 34.703 × 11.377 = 394.825 kN (permissible 657.904) — safe

Axial load = 38.581 × 4.679 = 180.540 kN (permissible 211.168) — safe

Deflection = 69.311 × 11.677 = 809.333 mm (permissible 1306.831) — safe

Axial load = 3.015 × 0.834 = 2.515 kN (permissible 3.339) — safe

Moment = 6.048 × 10.401 = 62.908 kNm (permissible 115.960) — safe

Bearing pressure = 42.048 × 8.207 = 345.071 kPa (permissible 407.990) — safe

Barrier-free access has been provided in accordance with the Harmonised Guidelines and Standards for Universal Accessibility. Ramps, lifts, signage and accessible toilets are provided on every floor of the block.

ANNEXURE III — Sheet 16

The space programme has been developed from the functional brief agreed with the user department. Departmental areas, circulation and the gross-to-carpet ratio are tabulated in the area statement enclosed at the Annexure.

The structural design conforms to IS 456 for reinforced concrete and IS 1893 for earthquake resistant design. The site falls in seismic zone V and the importance factor applicable to a hospital building has been adopted.

Quantities have been computed from the approved architectural and structural drawings and cross-checked against the detailed measurement sheets enclosed at the Annexure. Rates are as per the schedule of rates in force with lead and lift adjustments.

Quantities have been computed from the approved architectural and structural drawings and cross-checked against the detailed measurement sheets enclosed at the Annexure. Rates are as per the schedule of rates in force with lead and lift adjustments.

Reference: drawing no. ST-838, revision C.

ANNEXURE III — Sheet 17

Bore Log Summary

Depth (m)	Strata	N value	Sample
25.1	Coarse sand	17	UDS
21.8	Gravelly sand	16	DS
11.0	Silty clay	13	UDS
1.7	Medium sand	60	SPT
23.7	Coarse sand	24	DS
10.0	Gravelly sand	11	UDS
16.6	Weathered rock	4	DS
8.8	Clayey silt	55	UDS
5.2	Weathered rock	38	UDS
18.1	Clayey silt	28	DS
16.1	Medium sand	46	UDS
13.8	Weathered rock	11	DS
15.8	Clayey silt	50	DS
7.8	Coarse sand	24	SPT
21.2	Medium sand	7	DS
13.5	Gravelly sand	9	DS
4.2	Silty clay	25	UDS

Soil investigation was carried out by rotary drilling at the locations shown on the layout plan. The safe bearing capacity adopted for design is based on the lower of the shear and settlement criteria as reported in the geotechnical report.

ANNEXURE III — Sheet 18

Design Computation

Shear = 70.328 × 9.757 = 686.208 kN   (permissible 832.625) — safe

Deflection = 5.718 × 8.948 = 51.162 mm   (permissible 94.434) — safe

Deflection = 20.040 × 4.173 = 83.629 mm   (permissible 105.387) — safe

Deflection = 72.704 × 10.368 = 753.801 mm   (permissible 1287.178) — safe

Deflection = 63.601 × 3.510 = 223.213 mm   (permissible 301.544) — safe

Axial load = 1.027 × 8.599 = 8.831 kN   (permissible 11.991) — safe

Bearing pressure = 31.371 × 4.514 = 141.605 kPa   (permissible 230.762) — safe

Bearing pressure = 42.356 × 11.210 = 474.810 kPa   (permissible 731.315) — safe

The space programme has been developed from the functional brief agreed with the user department. Departmental areas, circulation and the gross-to-carpet ratio are tabulated in the area statement enclosed at the Annexure.

ANNEXURE IV — DETAILED ESTIMATE

Item-wise estimate with quantities, rates and amounts, supported by measurement sheets.

ANNEXURE IV — Room Data Sheet, page 1

Room	Department	Carpet area (sqm)	Occupancy	Floor finish
Room 187	Pathology	26.38	14	Epoxy screed
Room 440	General Medicine	66.80	6	Kota stone
Room 522	General Medicine	45.41	15	Ceramic tile
Room 467	Casualty	78.11	7	Antistatic vinyl
Room 401	Casualty	28.68	6	Kota stone
Room 560	Radiology	26.06	26	Kota stone
Room 237	Radiology	52.31	24	Ceramic tile
Room 135	Casualty	87.38	19	Ceramic tile
Room 558	Radiology	42.57	29	Ceramic tile
Room 165	Radiology	45.92	11	Antistatic vinyl
Room 575	Paediatrics	85.90	27	Epoxy screed
Room 897	Pathology	79.74	5	Epoxy screed
Room 617	Casualty	91.87	13	Epoxy screed
Room 471	Obstetrics	42.81	24	Vitrified tile
Room 737	Casualty	33.19	29	Vitrified tile
Room 240	Pathology	84.99	24	Epoxy screed
Room 844	Casualty	60.15	15	Ceramic tile
Room 805	Casualty	69.89	19	Ceramic tile
Room 475	Paediatrics	34.76	14	Vitrified tile
Room 513	Paediatrics	52.98	12	Epoxy screed
Room 389	Casualty	26.78	18	Epoxy screed
Room 302	Casualty	77.56	25	Epoxy screed
Room 672	Pathology	22.78	12	Vitrified tile
Room 311	General Medicine	26.36	15	Kota stone
Room 300	Casualty	61.72	15	Epoxy screed
Room 864	Surgery	32.55	26	Kota stone
Room 869	Casualty	89.70	9	Ceramic tile
Room 877	Radiology	10.52	24	Epoxy screed

ANNEXURE IV — Departmental Area Statement, page 2

Department	Floor	Units	Area per unit (sqm)	Total (sqm)
Staff Quarters	Ground	5	43.49	217.43
OPD	First	3	76.22	228.65
Diagnostics	Second	10	25.81	258.14
Pharmacy	Second	15	35.36	530.42
OPD	Second	11	80.53	885.87
OPD	Second	6	55.99	335.95
Staff Quarters	Second	6	57.19	343.15
Administration	First	16	84.29	1348.64
Staff Quarters	First	12	79.43	953.19
Pharmacy	Ground	3	56.88	170.64
OPD	Second	4	18.97	75.88
IPD Ward	Ground	13	37.24	484.17
Administration	Ground	8	89.85	718.83
Diagnostics	Ground	7	133.06	931.39
Staff Quarters	First	10	61.42	614.15
Pharmacy	First	17	139.51	2371.65
OPD	First	2	134.37	268.73
Diagnostics	Second	8	97.34	778.76
ICU	Second	13	108.24	1407.08
Staff Quarters	Third	10	36.37	363.67
OPD	Ground	12	31.65	379.75
Operation Theatre	First	10	87.99	879.88
ICU	Ground	13	42.01	546.13
OPD	Third	8	21.23	169.83
Pharmacy	Ground	13	129.21	1679.77
IPD Ward	Ground	8	90.91	727.27

ANNEXURE IV — Bore Log Summary, page 3

Depth (m)	Strata	N value	Sample
27.1	Clayey silt	36	UDS
27.6	Weathered rock	19	DS
26.0	Coarse sand	36	UDS
16.1	Clayey silt	19	DS
13.3	Gravelly sand	32	SPT
7.1	Gravelly sand	24	SPT
9.2	Gravelly sand	31	UDS
12.0	Medium sand	13	DS
8.6	Clayey silt	16	SPT
4.0	Weathered rock	35	UDS
18.3	Clayey silt	50	DS
21.7	Gravelly sand	11	SPT
21.3	Medium sand	46	UDS
12.5	Weathered rock	59	SPT
8.0	Coarse sand	60	UDS
25.7	Clayey silt	23	UDS
22.0	Clayey silt	4	DS
23.6	Clayey silt	37	DS
24.9	Gravelly sand	39	SPT
29.2	Coarse sand	42	UDS
27.6	Clayey silt	58	SPT
13.2	Gravelly sand	51	UDS
29.5	Silty clay	43	UDS
13.9	Weathered rock	29	DS
21.9	Weathered rock	12	DS
25.2	Gravelly sand	25	UDS
27.6	Gravelly sand	52	UDS
5.7	Clayey silt	19	UDS
20.7	Gravelly sand	34	DS

ANNEXURE IV — Electrical Load Schedule, page 4

Panel	Description	Connected load (kW)	Demand factor	Demand (kVA)
DB-14	HVAC	132.82	0.92	135.08
DB-30	Lifts	38.28	0.84	35.73
DB-16	Lifts	73.50	0.53	43.54
DB-11	HVAC	131.61	0.45	66.27
DB-31	Lighting	136.87	0.82	123.95
DB-37	Lighting	156.70	0.57	99.56
DB-22	Medical equipment	26.39	0.64	18.78
DB-33	Medical equipment	69.45	0.47	36.06
DB-34	Power outlets	53.07	0.59	34.83
DB-38	Medical equipment	107.68	0.48	57.98
DB-17	Pumps	98.94	0.55	60.97
DB-01	Power outlets	142.07	0.85	133.43
DB-32	Medical equipment	80.62	0.80	72.02
DB-03	Pumps	23.85	0.43	11.46
DB-19	Pumps	148.83	0.61	100.78
DB-37	Medical equipment	115.38	0.90	115.40
DB-20	Lifts	3.18	0.46	1.64
DB-02	Medical equipment	65.06	0.88	63.66
DB-30	Lifts	112.89	0.63	78.94
DB-29	Sterilisation	83.93	0.70	64.84
DB-27	HVAC	109.16	0.82	99.24
DB-37	Lifts	76.91	0.44	37.63
DB-19	Lighting	126.91	0.80	113.51
DB-39	Power outlets	171.50	0.55	103.89
DB-32	Medical equipment	93.15	0.73	75.14
DB-11	Pumps	169.26	0.89	166.89
DB-37	Medical equipment	169.02	0.55	102.83
DB-08	Power outlets	76.08	0.63	53.34

ANNEXURE IV — Equipment Schedule, page 5

Item	Department	Nos	Unit rate (Rs)	Amount (Rs)
Patient monitor	CSSD	30	1,040,958	31,228,740
Hospital bed	ICU	11	1,547,866	17,026,526
OT table	OT	23	1,740,223	40,025,129
OT table	ICU	18	611,176	11,001,168
X-ray unit	CSSD	10	2,017,612	20,176,120
Ultrasound	CSSD	14	1,202,392	16,833,488
OT table	ICU	27	1,209,198	32,648,346
Ultrasound	Ward	8	2,376,602	19,012,816
Ventilator	CSSD	10	182,637	1,826,370
Defibrillator	OT	6	1,706,984	10,241,904
OT table	Ward	31	952,961	29,541,791
Ultrasound	CSSD	1	1,423,670	1,423,670
Hospital bed	Ward	33	92,709	3,059,397
Ultrasound	OT	18	2,317,835	41,721,030
Defibrillator	OT	35	1,482,185	51,876,475
Ultrasound	Radiology	25	621,876	15,546,900
Defibrillator	Casualty	32	1,176,080	37,634,560
X-ray unit	CSSD	6	2,276,723	13,660,338
X-ray unit	OT	24	1,895,365	45,488,760
Patient monitor	Ward	10	2,173,616	21,736,160
Autoclave	Radiology	26	1,634,085	42,486,210
Defibrillator	OT	26	1,819,348	47,303,048
Ultrasound	OT	15	1,814,269	27,214,035
OT table	Casualty	26	398,939	10,372,414
Autoclave	OT	33	854,757	28,206,981
Ventilator	OT	10	2,306,793	23,067,930
Ultrasound	OT	12	2,153,937	25,847,244

ANNEXURE IV — Detailed Measurement Sheet, page 6

Item	L (m)	B (m)	D (m)	Nos	Quantity
Item 824	29.34	2.52	0.23	4	345.071
Item 220	22.70	3.38	1.90	8	320.118
Item 764	29.78	0.89	0.73	22	387.631
Item 156	36.56	4.27	1.81	2	298.651
Item 538	10.44	4.03	2.77	23	349.817
Item 256	21.92	1.90	2.05	21	190.064
Item 447	15.53	1.47	1.88	21	179.527
Item 810	36.67	5.58	0.36	13	361.006
Item 275	7.33	0.56	1.06	7	46.994
Item 897	6.74	5.10	2.68	22	170.745
Item 683	3.67	5.21	2.36	22	68.737
Item 729	6.84	1.93	0.55	23	292.989
Item 724	33.90	5.00	1.85	12	42.208
Item 119	20.59	4.60	0.55	22	187.813
Item 318	2.09	5.93	1.33	23	192.541
Item 287	22.42	3.32	0.23	13	366.463
Item 219	10.47	1.05	2.38	15	320.651
Item 757	6.73	5.96	2.79	15	195.104
Item 385	26.24	0.88	1.60	23	30.374
Item 716	9.03	2.03	1.37	14	366.341
Item 560	25.42	0.62	1.68	4	247.734
Item 540	6.39	2.05	1.95	13	145.913
Item 142	8.42	5.87	0.53	14	341.324
Item 742	5.50	1.39	1.76	17	206.373

ANNEXURE IV — Room Data Sheet, page 7

Room	Department	Carpet area (sqm)	Occupancy	Floor finish
Room 615	Casualty	79.95	20	Kota stone
Room 554	Casualty	64.56	13	Antistatic vinyl
Room 637	Radiology	39.72	22	Vitrified tile
Room 665	Obstetrics	33.45	5	Ceramic tile
Room 645	Surgery	33.62	28	Epoxy screed
Room 209	Casualty	27.98	10	Ceramic tile
Room 870	Surgery	16.34	13	Antistatic vinyl
Room 799	Radiology	64.63	23	Vitrified tile
Room 643	General Medicine	31.81	26	Epoxy screed
Room 816	General Medicine	86.59	8	Epoxy screed
Room 217	Obstetrics	37.72	24	Kota stone
Room 773	Pathology	73.27	15	Epoxy screed
Room 638	General Medicine	19.44	1	Ceramic tile
Room 429	Radiology	46.22	26	Epoxy screed
Room 711	General Medicine	49.08	14	Ceramic tile
Room 868	Obstetrics	60.73	8	Vitrified tile
Room 269	Pathology	14.91	6	Ceramic tile
Room 803	Casualty	9.23	23	Antistatic vinyl
Room 488	General Medicine	22.89	11	Vitrified tile
Room 717	General Medicine	29.16	22	Antistatic vinyl
Room 605	Pathology	20.43	5	Antistatic vinyl
Room 731	Obstetrics	62.53	23	Epoxy screed
Room 448	Radiology	52.69	13	Antistatic vinyl
Room 798	Paediatrics	21.48	6	Ceramic tile
Room 615	Surgery	38.98	24	Epoxy screed
Room 823	General Medicine	83.50	23	Kota stone
Room 653	Pathology	39.70	8	Kota stone
Room 610	Obstetrics	74.06	5	Kota stone
Room 158	Casualty	15.69	13	Antistatic vinyl

ANNEXURE IV — Departmental Area Statement, page 8

Department	Floor	Units	Area per unit (sqm)	Total (sqm)
IPD Ward	First	18	87.71	1578.70
OPD	Ground	5	38.32	191.62
IPD Ward	Ground	12	58.48	701.78
OPD	First	7	123.09	861.64
OPD	Second	5	133.25	666.27
IPD Ward	Third	5	36.23	181.15
IPD Ward	Second	11	99.70	1096.72
Pharmacy	Second	13	125.68	1633.85
IPD Ward	Ground	4	28.93	115.71
Staff Quarters	Ground	1	123.29	123.29
ICU	Ground	9	98.15	883.39
OPD	Ground	18	62.37	1122.57
IPD Ward	Ground	3	82.04	246.13
OPD	Ground	18	79.10	1423.72
Operation Theatre	Third	16	109.57	1753.14
Operation Theatre	First	2	112.33	224.66
ICU	First	11	113.14	1244.59
ICU	Third	15	85.89	1288.35
Diagnostics	First	9	37.69	339.19
ICU	Ground	8	72.63	581.05
OPD	Ground	16	92.35	1477.68
ICU	First	2	137.41	274.82
IPD Ward	Second	12	67.20	806.36
IPD Ward	Third	3	30.89	92.67
Pharmacy	Second	13	43.14	560.84
Diagnostics	Ground	10	22.81	228.07
Diagnostics	Ground	7	21.43	150.04
Operation Theatre	Third	6	103.08	618.50

ANNEXURE IV — Bore Log Summary, page 9

Depth (m)	Strata	N value	Sample
2.3	Medium sand	24	UDS
4.3	Weathered rock	7	DS
8.6	Gravelly sand	32	SPT
23.6	Clayey silt	19	DS
17.7	Medium sand	31	SPT
3.5	Medium sand	24	UDS
12.5	Gravelly sand	19	UDS
14.2	Clayey silt	41	DS
23.3	Medium sand	51	UDS
5.6	Coarse sand	28	UDS
20.8	Gravelly sand	58	SPT
9.7	Weathered rock	19	DS
2.8	Clayey silt	23	DS
24.4	Coarse sand	38	UDS
4.1	Weathered rock	47	UDS
20.6	Medium sand	48	DS
16.9	Medium sand	54	DS
11.1	Weathered rock	16	DS
3.7	Silty clay	22	UDS
9.8	Clayey silt	11	DS
5.1	Clayey silt	54	UDS
18.6	Weathered rock	58	DS
2.2	Silty clay	13	SPT
29.4	Coarse sand	40	SPT
21.6	Weathered rock	45	DS
8.9	Weathered rock	16	UDS
21.6	Coarse sand	54	UDS

ANNEXURE IV — Electrical Load Schedule, page 10

Panel	Description	Connected load (kW)	Demand factor	Demand (kVA)
DB-08	Lifts	5.91	0.53	3.49
DB-24	Power outlets	124.78	0.86	119.09
DB-07	Lighting	76.22	0.45	37.80
DB-05	Sterilisation	108.21	0.45	53.90
DB-37	Power outlets	73.99	0.51	42.29
DB-17	Lighting	17.36	0.90	17.32
DB-09	Medical equipment	14.82	0.54	8.85
DB-34	Power outlets	67.54	0.64	48.29
DB-40	Lifts	167.60	0.68	125.74
DB-34	Sterilisation	43.12	0.87	41.52
DB-37	HVAC	68.50	0.61	46.71
DB-02	Pumps	175.21	0.51	98.34
DB-14	Sterilisation	100.91	0.46	52.03
DB-36	Lifts	106.50	0.87	103.38
DB-30	HVAC	41.27	0.72	32.96
DB-23	Medical equipment	6.17	0.62	4.22
DB-12	HVAC	109.20	0.88	106.20
DB-26	Medical equipment	33.18	0.74	27.29
DB-35	Lighting	67.32	0.65	48.79
DB-01	Lighting	55.99	0.64	39.71
DB-04	HVAC	20.89	0.67	15.63
DB-14	Lifts	97.11	0.76	81.69
DB-33	Lifts	98.82	0.59	64.75
DB-31	Lifts	48.54	0.56	30.16

ANNEXURE IV — Equipment Schedule, page 11

Item	Department	Nos	Unit rate (Rs)	Amount (Rs)
OT table	CSSD	15	2,385,524	35,782,860
Ultrasound	OT	28	1,891,425	52,959,900
OT table	ICU	22	891,413	19,611,086
Ventilator	Casualty	11	1,311,359	14,424,949
OT table	CSSD	20	280,105	5,602,100
Ventilator	Radiology	14	969,065	13,566,910
Patient monitor	Ward	16	2,296,077	36,737,232
Ultrasound	ICU	40	885,012	35,400,480
Ventilator	Casualty	15	1,264,478	18,967,170
Autoclave	Radiology	31	1,138,714	35,300,134
Defibrillator	OT	2	288,473	576,946
Ventilator	ICU	32	1,915,286	61,289,152
OT table	Radiology	5	917,226	4,586,130
Patient monitor	ICU	27	2,087,650	56,366,550
Hospital bed	ICU	29	1,869,577	54,217,733
OT table	OT	29	915,437	26,547,673
Ultrasound	OT	1	251,885	251,885
Defibrillator	CSSD	35	1,741,904	60,966,640
Patient monitor	Radiology	14	2,367,000	33,138,000
Hospital bed	Casualty	11	915,841	10,074,251
Patient monitor	OT	35	435,956	15,258,460
Ultrasound	Ward	8	2,244,328	17,954,624
Patient monitor	OT	28	1,911,667	53,526,676
Patient monitor	Radiology	11	2,386,589	26,252,479
Ultrasound	ICU	18	1,099,687	19,794,366
Patient monitor	CSSD	38	567,524	21,565,912
X-ray unit	ICU	3	70,330	210,990

ANNEXURE IV — Detailed Measurement Sheet, page 12

Item	L (m)	B (m)	D (m)	Nos	Quantity
Item 704	28.13	3.50	0.53	3	287.869
Item 563	32.29	1.23	2.07	6	109.271
Item 728	13.99	5.82	0.79	1	312.969
Item 606	34.32	3.14	0.78	8	119.502
Item 718	3.11	5.87	0.26	13	371.173
Item 707	3.75	4.88	0.46	19	67.952
Item 648	28.19	2.93	1.59	9	108.267
Item 304	5.21	4.22	2.53	17	162.925
Item 725	38.85	5.40	1.00	23	200.478
Item 751	23.22	2.72	1.79	23	29.418
Item 871	20.20	4.50	1.59	19	393.371
Item 274	5.35	1.98	2.15	1	8.184
Item 192	26.80	2.75	2.56	18	64.400
Item 638	26.15	2.25	0.43	18	158.613
Item 117	12.06	0.62	0.89	22	297.564
Item 309	13.69	1.60	1.22	18	106.937
Item 460	5.11	5.35	2.15	9	231.354
Item 673	18.75	4.53	2.31	17	286.968
Item 722	37.33	1.86	1.79	1	42.295
Item 483	28.61	5.47	0.56	3	389.586
Item 554	6.40	2.50	0.81	20	399.964
Item 108	35.90	2.35	1.55	9	56.107
Item 688	26.41	3.31	0.85	9	61.244
Item 224	14.99	4.76	0.96	6	150.008
Item 391	19.00	2.50	0.73	1	335.433
Item 369	21.11	4.01	0.52	13	236.232
Item 275	4.43	3.72	0.67	13	288.729

ANNEXURE IV — Room Data Sheet, page 13

Room	Department	Carpet area (sqm)	Occupancy	Floor finish
Room 536	Obstetrics	67.67	19	Kota stone
Room 803	Surgery	69.91	29	Kota stone
Room 496	Casualty	89.16	8	Antistatic vinyl
Room 727	General Medicine	41.62	30	Ceramic tile
Room 198	Surgery	30.63	7	Antistatic vinyl
Room 575	Casualty	14.36	4	Kota stone
Room 805	Obstetrics	27.73	23	Kota stone
Room 232	Paediatrics	52.45	18	Kota stone
Room 168	Casualty	23.19	7	Kota stone
Room 601	Paediatrics	21.54	8	Ceramic tile
Room 182	Radiology	80.06	8	Antistatic vinyl
Room 469	Surgery	29.54	5	Kota stone
Room 371	Radiology	40.00	16	Antistatic vinyl
Room 557	Pathology	16.38	10	Vitrified tile
Room 244	General Medicine	28.79	3	Epoxy screed
Room 331	Pathology	14.07	2	Ceramic tile
Room 190	General Medicine	62.18	6	Epoxy screed
Room 370	Paediatrics	19.06	28	Kota stone
Room 348	Pathology	18.13	27	Ceramic tile
Room 570	Radiology	22.30	20	Antistatic vinyl
Room 517	Surgery	38.69	20	Kota stone
Room 813	Pathology	65.94	30	Kota stone
Room 451	General Medicine	77.02	22	Vitrified tile

ANNEXURE IV — Departmental Area Statement, page 14

Department	Floor	Units	Area per unit (sqm)	Total (sqm)
Administration	Third	7	80.50	563.50
IPD Ward	First	1	50.54	50.54
Diagnostics	Third	12	58.42	701.06
Staff Quarters	Ground	2	90.26	180.52
Operation Theatre	First	13	58.62	762.07
Administration	Second	18	134.19	2415.45
OPD	First	11	84.14	925.54
OPD	First	12	91.92	1103.09
Operation Theatre	Second	7	30.65	214.56
Operation Theatre	First	13	138.99	1806.92
OPD	First	1	123.31	123.31
Administration	Second	17	81.55	1386.27
Operation Theatre	Second	6	52.27	313.62
Pharmacy	First	10	39.52	395.25
IPD Ward	Third	5	56.88	284.39
Pharmacy	Second	18	117.22	2110.01
Operation Theatre	Second	9	136.48	1228.28
ICU	Second	14	56.21	786.89
IPD Ward	First	14	121.83	1705.64
Pharmacy	First	8	42.89	343.12
ICU	First	14	65.32	914.52
Diagnostics	Third	10	58.06	580.57
ICU	Ground	4	43.65	174.61
OPD	Second	16	87.71	1403.44
IPD Ward	Ground	16	65.18	1042.86
Administration	Third	3	109.44	328.33
IPD Ward	First	2	58.60	117.20
Staff Quarters	Third	2	113.99	227.97

ANNEXURE IV — Bore Log Summary, page 15

Depth (m)	Strata	N value	Sample
9.0	Silty clay	43	SPT
3.1	Weathered rock	49	UDS
10.2	Weathered rock	46	SPT
11.7	Silty clay	15	DS
17.4	Coarse sand	7	UDS
29.9	Weathered rock	30	SPT
6.7	Clayey silt	50	DS
27.7	Medium sand	5	UDS
25.3	Clayey silt	18	DS
18.3	Clayey silt	25	DS
29.7	Coarse sand	29	SPT
24.4	Coarse sand	42	SPT
10.9	Medium sand	21	DS
6.6	Clayey silt	15	SPT
5.6	Silty clay	30	DS
14.9	Medium sand	60	UDS
0.8	Coarse sand	41	SPT
7.2	Medium sand	59	UDS
7.6	Clayey silt	41	UDS
2.9	Clayey silt	6	UDS
24.8	Medium sand	12	UDS
14.9	Medium sand	48	DS
17.3	Weathered rock	50	UDS
12.8	Clayey silt	23	DS
23.8	Weathered rock	42	UDS
10.1	Coarse sand	36	DS
24.6	Gravelly sand	41	DS

ANNEXURE IV — Electrical Load Schedule, page 16

Panel	Description	Connected load (kW)	Demand factor	Demand (kVA)
DB-32	Power outlets	156.90	0.85	147.50
DB-15	Power outlets	154.09	0.44	74.98
DB-11	HVAC	75.09	0.78	65.39
DB-11	Pumps	132.91	0.40	59.41
DB-20	HVAC	77.83	0.72	62.14
DB-15	Sterilisation	172.65	0.94	180.74
DB-18	Medical equipment	57.12	0.80	50.61
DB-33	Lighting	169.40	0.41	77.48
DB-01	Pumps	114.90	0.57	73.24
DB-39	Pumps	11.77	0.91	11.91
DB-29	Lifts	35.10	0.54	21.04
DB-19	Power outlets	7.57	0.41	3.44
DB-26	Pumps	102.92	0.79	89.86
DB-20	Lifts	13.04	0.91	13.11
DB-11	Medical equipment	61.38	0.54	36.69
DB-09	Sterilisation	33.14	0.58	21.41
DB-14	Lifts	96.67	0.76	81.49
DB-21	Sterilisation	73.79	0.54	43.91
DB-18	Sterilisation	40.00	0.73	32.26
DB-09	HVAC	142.22	0.77	121.57
DB-20	HVAC	160.54	0.50	88.65
DB-22	HVAC	119.35	0.67	88.56
DB-27	Power outlets	35.38	0.57	22.23
DB-30	Pumps	176.40	0.68	132.90

ANNEXURE IV — Equipment Schedule, page 17

Item	Department	Nos	Unit rate (Rs)	Amount (Rs)
Autoclave	Casualty	25	1,555,203	38,880,075
X-ray unit	OT	18	2,003,573	36,064,314
Ventilator	Ward	39	2,090,498	81,529,422
Defibrillator	ICU	33	2,063,865	68,107,545
Ultrasound	OT	30	31,888	956,640
Hospital bed	OT	33	2,392,639	78,957,087
Autoclave	Casualty	37	1,990,421	73,645,577
Ventilator	Ward	3	779,015	2,337,045
Ultrasound	Ward	6	1,377,493	8,264,958
Ventilator	Casualty	6	2,158,910	12,953,460
Ultrasound	Radiology	13	255,824	3,325,712
Ventilator	ICU	25	824,180	20,604,500
OT table	Casualty	17	1,766,989	30,038,813
X-ray unit	CSSD	4	1,667,194	6,668,776
Patient monitor	ICU	36	402,818	14,501,448
X-ray unit	Casualty	18	1,219,630	21,953,340
Defibrillator	CSSD	17	1,897,500	32,257,500
Ultrasound	Radiology	11	202,094	2,223,034
Ventilator	CSSD	38	1,260,731	47,907,778
Autoclave	Casualty	8	1,282,516	10,260,128
X-ray unit	Ward	12	2,166,632	25,999,584
Hospital bed	Casualty	21	2,257,668	47,411,028
OT table	Ward	27	556,190	15,017,130
Hospital bed	ICU	25	1,122,101	28,052,525
OT table	Casualty	39	1,987,541	77,514,099
Ultrasound	CSSD	17	26,620	452,540
Autoclave	Ward	10	1,391,827	13,918,270
Defibrillator	Ward	15	707,762	10,616,430

ANNEXURE IV — Detailed Measurement Sheet, page 18

Item	L (m)	B (m)	D (m)	Nos	Quantity
Item 652	11.50	2.42	2.57	3	202.446
Item 760	11.77	0.68	1.25	18	209.156
Item 796	8.98	0.99	1.07	2	295.640
Item 737	12.10	1.60	2.38	5	9.185
Item 265	33.87	1.95	0.31	16	236.434
Item 201	12.56	0.43	0.63	10	145.495
Item 265	20.36	5.03	1.95	23	244.165
Item 654	37.23	2.88	1.55	5	50.490
Item 857	32.94	2.72	1.51	9	41.634
Item 228	14.49	4.80	2.11	22	92.282
Item 292	22.62	4.28	1.43	15	178.272
Item 154	7.63	0.83	0.65	18	122.981
Item 363	6.47	0.77	2.23	6	53.484
Item 572	24.75	1.79	1.25	4	147.037
Item 148	12.23	2.45	0.25	1	180.170
Item 784	25.31	2.75	1.80	22	95.903
Item 817	20.91	1.14	1.70	20	314.576
Item 189	14.69	4.53	1.07	16	46.982
Item 550	4.19	2.51	0.34	16	161.152
Item 493	5.24	4.46	1.43	14	348.605
Item 238	14.94	0.54	2.46	6	276.353
Item 297	2.18	1.37	2.76	11	64.752
Item 130	13.46	1.62	1.77	22	135.909
Item 347	22.84	4.27	2.39	7	237.492
Item 159	37.66	1.63	2.85	2	275.448
Item 306	9.55	5.89	1.30	9	95.647
Item 732	20.34	2.24	1.67	18	69.205
Item 249	16.35	1.88	2.81	17	106.719

ANNEXURE IV — Room Data Sheet, page 19

Room	Department	Carpet area (sqm)	Occupancy	Floor finish
Room 142	Paediatrics	27.28	2	Antistatic vinyl
Room 155	Radiology	19.70	8	Vitrified tile
Room 191	Obstetrics	21.19	22	Epoxy screed
Room 668	General Medicine	61.61	13	Kota stone
Room 534	Casualty	58.47	22	Kota stone
Room 844	Pathology	43.06	13	Kota stone
Room 883	Casualty	17.83	13	Kota stone
Room 133	Surgery	17.34	28	Antistatic vinyl
Room 734	Surgery	23.05	27	Ceramic tile
Room 371	Pathology	18.34	13	Antistatic vinyl
Room 624	Casualty	79.58	19	Antistatic vinyl
Room 183	Radiology	69.51	8	Ceramic tile
Room 206	Obstetrics	57.32	17	Epoxy screed
Room 549	Obstetrics	15.00	24	Vitrified tile
Room 633	Pathology	70.14	28	Kota stone
Room 653	Pathology	35.16	28	Ceramic tile
Room 677	Pathology	67.14	24	Epoxy screed
Room 222	General Medicine	52.68	25	Vitrified tile
Room 540	Paediatrics	15.47	27	Vitrified tile
Room 667	Radiology	83.19	29	Kota stone
Room 394	Casualty	40.80	28	Epoxy screed
Room 298	Casualty	74.93	18	Kota stone
Room 612	Casualty	42.34	4	Vitrified tile
Room 795	Pathology	59.17	9	Ceramic tile
Room 793	Casualty	76.00	14	Epoxy screed

ANNEXURE IV — Departmental Area Statement, page 20

Department	Floor	Units	Area per unit (sqm)	Total (sqm)
Operation Theatre	First	6	103.94	623.64
Administration	Second	14	70.09	981.28
Operation Theatre	First	18	122.27	2200.80
Staff Quarters	Ground	7	22.37	156.61
Operation Theatre	First	4	68.19	272.76
Diagnostics	First	2	44.06	88.11
Administration	Third	14	16.37	229.21
Staff Quarters	Ground	15	121.04	1815.64
Diagnostics	Ground	6	34.03	204.15
Administration	Ground	6	51.52	309.09
Pharmacy	Second	9	129.29	1163.61
ICU	Ground	2	43.88	87.75
Operation Theatre	Second	9	73.05	657.47
IPD Ward	Second	7	51.88	363.16
ICU	Ground	7	48.63	340.41
Pharmacy	First	10	139.86	1398.56
ICU	Second	5	22.89	114.44
Operation Theatre	Second	15	108.51	1627.62
Diagnostics	Second	16	39.59	633.48
Diagnostics	Third	13	95.98	1247.70
Staff Quarters	Ground	14	87.86	1230.04
IPD Ward	First	9	121.67	1095.00
Operation Theatre	Second	3	15.71	47.12
ICU	Third	10	12.10	120.99
Administration	First	8	41.97	335.72
Administration	Second	16	36.56	585.02

ANNEXURE IV — Bore Log Summary, page 21

Depth (m)	Strata	N value	Sample
29.2	Clayey silt	45	DS
15.4	Coarse sand	58	DS
1.5	Silty clay	17	SPT
11.1	Coarse sand	6	UDS
5.4	Coarse sand	34	SPT
17.1	Weathered rock	43	UDS
26.5	Gravelly sand	45	SPT
5.1	Gravelly sand	38	UDS
8.0	Clayey silt	12	UDS
21.6	Coarse sand	23	UDS
9.9	Silty clay	57	UDS
25.6	Medium sand	35	SPT
14.0	Weathered rock	57	UDS
21.2	Clayey silt	11	DS
20.2	Gravelly sand	31	SPT
1.6	Weathered rock	30	UDS
20.4	Weathered rock	47	UDS
21.8	Gravelly sand	34	SPT
19.9	Silty clay	33	DS
14.1	Clayey silt	7	SPT
14.5	Coarse sand	58	UDS
8.4	Weathered rock	51	UDS
2.2	Weathered rock	28	UDS
17.3	Weathered rock	20	SPT
26.7	Clayey silt	40	SPT
7.2	Gravelly sand	57	SPT
7.5	Weathered rock	55	UDS
9.5	Clayey silt	51	DS

ANNEXURE IV — Electrical Load Schedule, page 22

Panel	Description	Connected load (kW)	Demand factor	Demand (kVA)
DB-11	Lifts	43.15	0.45	21.78
DB-25	Pumps	61.89	0.56	38.67
DB-28	Sterilisation	41.00	0.88	39.94
DB-06	Power outlets	39.00	0.79	34.29
DB-35	Lifts	94.55	0.57	60.29
DB-12	Lighting	153.22	0.45	76.69
DB-26	Lighting	166.24	0.44	81.73
DB-35	Lighting	176.98	0.78	153.68
DB-33	Lighting	80.69	0.61	55.00
DB-01	Medical equipment	63.25	0.61	42.59
DB-09	Pumps	23.36	0.94	24.34
DB-39	Medical equipment	93.96	0.68	71.50
DB-31	Lighting	73.31	0.75	61.37
DB-09	Sterilisation	68.24	0.90	68.32
DB-27	Lighting	25.38	0.54	15.25
DB-29	Lifts	20.00	0.41	9.14
DB-25	Lifts	94.60	0.47	49.50
DB-16	Power outlets	85.91	0.76	73.01
DB-28	Power outlets	99.50	0.58	63.70
DB-20	Sterilisation	138.12	0.42	64.47
DB-19	HVAC	45.95	0.82	41.84
DB-01	Sterilisation	13.11	0.89	12.97
DB-16	Sterilisation	159.29	0.63	110.72
DB-05	Power outlets	35.15	0.72	28.10
DB-34	HVAC	30.34	0.82	27.73
DB-29	Power outlets	74.89	0.75	62.20
DB-04	Lifts	23.53	0.41	10.73
DB-36	Lighting	128.66	0.86	123.00
DB-22	Lifts	101.28	0.51	57.73
DB-36	Medical equipment	10.05	0.49	5.48

ANNEXURE IV — Equipment Schedule, page 23

Item	Department	Nos	Unit rate (Rs)	Amount (Rs)
Hospital bed	OT	1	1,698,310	1,698,310
Autoclave	ICU	34	24,252	824,568
Hospital bed	OT	19	819,911	15,578,309
Ultrasound	Radiology	23	773,973	17,801,379
Hospital bed	CSSD	37	1,329,962	49,208,594
Defibrillator	Casualty	10	250,296	2,502,960
OT table	ICU	34	815,413	27,724,042
Patient monitor	OT	24	1,558,690	37,408,560
Ventilator	Radiology	3	1,562,713	4,688,139
Hospital bed	Ward	26	159,354	4,143,204
Autoclave	Casualty	35	414,567	14,509,845
OT table	Ward	25	631,876	15,796,900
X-ray unit	Ward	10	530,179	5,301,790
Hospital bed	Ward	39	2,267,738	88,441,782
Hospital bed	ICU	22	205,194	4,514,268
Patient monitor	Ward	18	1,576,492	28,376,856
Hospital bed	Radiology	20	164,557	3,291,140
Patient monitor	Casualty	32	731,833	23,418,656
Defibrillator	CSSD	30	1,421,669	42,650,070
X-ray unit	Radiology	36	57,697	2,077,092
Hospital bed	Ward	29	2,067,289	59,951,381
Hospital bed	Casualty	15	2,133,059	31,995,885
Ventilator	ICU	5	172,871	864,355
Defibrillator	ICU	37	231,591	8,568,867
Autoclave	OT	2	1,746,111	3,492,222

ANNEXURE IV — Detailed Measurement Sheet, page 24

Item	L (m)	B (m)	D (m)	Nos	Quantity
Item 866	9.97	0.62	1.45	24	57.601
Item 699	8.78	1.14	2.88	24	339.342
Item 127	28.23	0.61	0.37	23	364.603
Item 173	39.55	4.18	2.47	19	209.993
Item 230	5.16	1.30	2.96	14	308.908
Item 734	27.92	3.49	1.40	6	382.683
Item 135	21.37	1.78	2.03	22	193.208
Item 687	9.84	3.84	1.19	22	267.028
Item 607	33.49	2.59	2.87	14	365.701
Item 740	32.25	2.12	2.75	23	173.602
Item 532	36.81	2.80	1.31	20	263.873
Item 829	9.87	1.68	0.43	5	180.030
Item 174	17.07	3.12	1.63	17	54.626
Item 675	38.21	1.53	1.44	20	75.075
Item 214	13.38	3.03	2.44	21	216.578
Item 183	14.58	1.81	1.60	8	108.923
Item 230	12.21	4.39	1.98	20	68.383
Item 267	27.66	5.40	2.45	9	333.648
Item 177	35.06	4.99	2.56	14	376.708
Item 675	26.53	2.53	0.70	12	56.270
Item 372	27.49	1.72	0.72	12	382.438
Item 185	4.88	4.37	0.41	2	378.628

ANNEXURE IV — Room Data Sheet, page 25

Room	Department	Carpet area (sqm)	Occupancy	Floor finish
Room 194	Casualty	46.20	27	Ceramic tile
Room 177	Casualty	11.73	7	Ceramic tile
Room 171	Casualty	19.58	18	Antistatic vinyl
Room 289	General Medicine	34.26	26	Vitrified tile
Room 409	Paediatrics	45.24	17	Kota stone
Room 898	Casualty	26.15	4	Vitrified tile
Room 718	Surgery	89.61	16	Kota stone
Room 765	Surgery	44.20	10	Antistatic vinyl
Room 462	Casualty	20.24	21	Kota stone
Room 752	General Medicine	72.17	21	Epoxy screed
Room 841	Obstetrics	25.93	7	Ceramic tile
Room 594	Obstetrics	15.15	18	Ceramic tile
Room 637	Obstetrics	17.34	12	Antistatic vinyl
Room 453	Radiology	59.89	8	Vitrified tile
Room 698	Casualty	84.88	28	Ceramic tile
Room 291	Radiology	25.78	7	Kota stone
Room 250	Casualty	82.00	22	Antistatic vinyl
Room 232	Casualty	57.00	1	Vitrified tile
Room 595	Surgery	35.21	10	Ceramic tile
Room 520	Casualty	66.94	6	Ceramic tile
Room 595	Obstetrics	17.83	24	Ceramic tile
Room 645	Obstetrics	87.31	13	Vitrified tile

ANNEXURE IV — Departmental Area Statement, page 26

Department	Floor	Units	Area per unit (sqm)	Total (sqm)
Operation Theatre	First	3	82.44	247.32
Administration	Ground	13	40.48	526.24
OPD	Ground	18	126.37	2274.68
Administration	Second	15	109.75	1646.28
OPD	Second	10	52.39	523.89
Staff Quarters	Third	4	94.82	379.28
Operation Theatre	Ground	10	86.84	868.45
Administration	Ground	3	30.56	91.68
Staff Quarters	Third	8	92.10	736.81
Operation Theatre	Second	7	130.61	914.28
Pharmacy	Ground	5	115.63	578.16
OPD	Third	4	13.74	54.95
Staff Quarters	Second	13	16.84	218.96
Administration	Second	8	42.88	343.06
Diagnostics	Ground	10	56.66	566.62
OPD	First	8	36.33	290.64
Pharmacy	Third	18	23.20	417.53
Staff Quarters	Third	6	87.45	524.69
Administration	Third	14	89.58	1254.11
Administration	First	1	79.04	79.04
Staff Quarters	Ground	17	26.86	456.55
Diagnostics	Second	8	112.59	900.68
Staff Quarters	Third	17	51.29	871.87
Operation Theatre	Ground	3	98.24	294.73
Pharmacy	Ground	11	133.65	1470.18

ANNEXURE IV — Bore Log Summary, page 27

Depth (m)	Strata	N value	Sample
23.5	Medium sand	35	UDS
28.6	Coarse sand	49	SPT
16.5	Weathered rock	11	SPT
16.8	Silty clay	51	SPT
3.6	Medium sand	36	SPT
21.3	Medium sand	28	DS
8.0	Coarse sand	48	DS
14.2	Medium sand	54	SPT
3.6	Gravelly sand	17	DS
12.9	Coarse sand	41	DS
6.0	Gravelly sand	58	DS
9.2	Gravelly sand	59	UDS
16.1	Clayey silt	12	SPT
21.2	Silty clay	25	UDS
3.5	Medium sand	39	DS
12.6	Weathered rock	31	SPT
9.4	Silty clay	35	UDS
15.2	Silty clay	50	UDS
22.2	Medium sand	27	DS
8.1	Weathered rock	41	UDS
17.0	Coarse sand	54	UDS
9.7	Medium sand	30	UDS
4.0	Coarse sand	49	SPT
9.1	Weathered rock	32	UDS
5.3	Weathered rock	26	UDS
8.3	Coarse sand	59	UDS
21.4	Weathered rock	39	UDS
26.8	Coarse sand	56	DS
4.9	Coarse sand	31	SPT
26.8	Coarse sand	46	DS

ANNEXURE IV — Electrical Load Schedule, page 28

Panel	Description	Connected load (kW)	Demand factor	Demand (kVA)
DB-04	Pumps	115.39	0.45	58.16
DB-37	HVAC	25.57	0.52	14.70
DB-35	Lifts	79.36	0.44	39.01
DB-08	Power outlets	152.38	0.62	105.41
DB-08	Pumps	99.25	0.69	75.59
DB-17	HVAC	155.02	0.91	157.41
DB-26	Lighting	67.64	0.93	69.56
DB-11	Sterilisation	136.75	0.77	117.35
DB-03	Power outlets	151.00	0.60	100.81
DB-19	Lighting	19.54	0.47	10.22
DB-23	Medical equipment	173.51	0.79	152.38
DB-25	Pumps	136.53	0.43	64.65
DB-31	HVAC	159.08	0.85	150.93
DB-27	Power outlets	48.64	0.95	51.26
DB-04	Lighting	136.52	0.42	63.81
DB-18	HVAC	42.16	0.69	32.45
DB-09	Lighting	112.03	0.85	105.40
DB-02	Pumps	84.44	0.70	65.46
DB-04	Pumps	113.85	0.85	107.40
DB-36	Medical equipment	24.42	0.83	22.40
DB-16	Lifts	17.64	0.59	11.66
DB-13	Sterilisation	110.00	0.67	81.70
DB-20	Medical equipment	97.55	0.83	90.22
DB-24	Power outlets	170.60	0.47	88.36
DB-08	Lifts	138.11	0.48	73.84
DB-05	HVAC	106.09	0.88	103.34
DB-14	Medical equipment	178.43	0.68	135.42

ANNEXURE IV — Equipment Schedule, page 29

Item	Department	Nos	Unit rate (Rs)	Amount (Rs)
Hospital bed	OT	33	63,227	2,086,491
OT table	ICU	8	1,672,718	13,381,744
Hospital bed	OT	1	2,053,280	2,053,280
Ventilator	Ward	18	864,767	15,565,806
Autoclave	ICU	4	1,967,278	7,869,112
OT table	Casualty	31	1,707,106	52,920,286
Ultrasound	ICU	25	1,581,248	39,531,200
Autoclave	Ward	2	585,224	1,170,448
OT table	Casualty	9	935,739	8,421,651
X-ray unit	OT	16	1,219,425	19,510,800
Ventilator	CSSD	13	1,656,481	21,534,253
Patient monitor	CSSD	32	1,893,891	60,604,512
OT table	ICU	37	1,799,018	66,563,666
X-ray unit	ICU	18	1,949,415	35,089,470
Defibrillator	ICU	1	1,392,832	1,392,832
Defibrillator	Radiology	38	2,280,299	86,651,362
Autoclave	CSSD	35	144,556	5,059,460
X-ray unit	ICU	12	1,491,264	17,895,168
Ventilator	ICU	29	1,071,880	31,084,520
Patient monitor	Radiology	6	251,208	1,507,248
Ventilator	CSSD	17	1,883,057	32,011,969
X-ray unit	CSSD	7	2,172,007	15,204,049
OT table	OT	22	1,368,102	30,098,244
Autoclave	OT	13	1,723,793	22,409,309
X-ray unit	Radiology	19	1,661,859	31,575,321

ANNEXURE IV — Detailed Measurement Sheet, page 30

Item	L (m)	B (m)	D (m)	Nos	Quantity
Item 665	15.16	1.87	2.36	15	199.449
Item 519	31.75	5.43	2.04	20	318.723
Item 283	31.80	5.75	0.88	14	355.955
Item 839	6.54	1.78	2.24	14	393.860
Item 571	29.74	3.24	1.57	24	174.245
Item 758	17.01	4.36	1.61	14	126.348
Item 468	29.46	2.36	2.10	17	229.554
Item 144	6.36	2.32	0.86	5	393.578
Item 842	25.18	3.25	1.98	9	366.673
Item 440	27.98	2.10	1.12	23	19.102
Item 465	28.73	5.27	2.17	22	300.580
Item 478	30.78	3.73	2.66	13	277.134
Item 210	38.85	5.44	2.56	16	126.147
Item 789	16.58	5.80	2.34	19	265.242
Item 370	28.46	4.67	0.41	6	347.505
Item 176	4.60	0.51	0.21	6	168.470
Item 899	7.32	5.76	2.22	9	77.925
Item 887	25.41	0.54	0.91	11	234.583
Item 342	33.83	0.91	1.08	9	314.943
Item 308	36.58	5.69	0.27	20	373.686
Item 541	37.71	3.87	2.07	17	157.386
Item 872	9.79	1.03	2.84	9	160.437

ANNEXURE IV — Room Data Sheet, page 31

Room	Department	Carpet area (sqm)	Occupancy	Floor finish
Room 268	Paediatrics	29.89	14	Kota stone
Room 279	Casualty	43.34	8	Vitrified tile
Room 618	Surgery	76.20	21	Kota stone
Room 220	Surgery	51.34	5	Kota stone
Room 459	Obstetrics	76.41	17	Vitrified tile
Room 645	Radiology	38.13	27	Vitrified tile
Room 237	Surgery	66.20	20	Antistatic vinyl
Room 295	Pathology	30.19	21	Antistatic vinyl
Room 831	Pathology	41.21	27	Antistatic vinyl
Room 525	Paediatrics	83.29	20	Vitrified tile
Room 432	Pathology	44.95	1	Kota stone
Room 778	Pathology	25.36	7	Vitrified tile
Room 760	Radiology	56.17	30	Kota stone
Room 708	Paediatrics	47.15	22	Ceramic tile
Room 154	Paediatrics	64.22	8	Kota stone
Room 398	Pathology	72.78	27	Ceramic tile
Room 802	Casualty	58.95	23	Epoxy screed
Room 878	Paediatrics	46.51	13	Vitrified tile
Room 361	Obstetrics	31.32	26	Epoxy screed
Room 473	Paediatrics	83.37	25	Epoxy screed
Room 671	Obstetrics	22.85	20	Antistatic vinyl
Room 373	Obstetrics	68.55	16	Kota stone

ANNEXURE IV — Departmental Area Statement, page 32

Department	Floor	Units	Area per unit (sqm)	Total (sqm)
Operation Theatre	Ground	5	24.34	121.71
Staff Quarters	Ground	2	13.04	26.08
IPD Ward	Ground	17	80.77	1373.06
ICU	Second	1	45.75	45.75
Operation Theatre	Ground	11	48.11	529.19
Pharmacy	Ground	18	54.66	983.87
Diagnostics	First	17	26.08	443.40
IPD Ward	Second	7	100.95	706.62
Diagnostics	First	1	19.78	19.78
Diagnostics	Second	5	49.94	249.69
IPD Ward	First	9	120.54	1084.85
OPD	Second	4	76.51	306.04
Administration	Second	9	94.39	849.48
IPD Ward	First	11	101.35	1114.85
OPD	Second	4	22.51	90.03
Administration	Ground	5	139.18	695.92
ICU	Third	5	123.41	617.03
OPD	Second	12	24.28	291.33
Operation Theatre	First	5	87.31	436.57
Administration	Second	7	69.43	486.00
OPD	First	3	42.51	127.52
Pharmacy	Third	14	60.93	853.07
Staff Quarters	First	13	15.09	196.21
Diagnostics	Second	3	56.55	169.65
Pharmacy	Second	14	23.27	325.73
OPD	First	1	98.69	98.69
OPD	Third	7	79.44	556.09

ANNEXURE IV — Bore Log Summary, page 33

Depth (m)	Strata	N value	Sample
26.0	Silty clay	57	SPT
4.9	Silty clay	8	SPT
27.4	Gravelly sand	6	DS
2.8	Coarse sand	60	SPT
29.1	Silty clay	5	SPT
29.8	Gravelly sand	12	DS
2.5	Clayey silt	42	DS
7.5	Weathered rock	34	SPT
10.0	Weathered rock	46	DS
11.3	Medium sand	59	DS
4.7	Weathered rock	39	SPT
3.3	Clayey silt	20	SPT
16.4	Clayey silt	44	UDS
3.3	Gravelly sand	30	DS
16.9	Clayey silt	52	SPT
14.9	Silty clay	58	DS
29.0	Silty clay	22	UDS
12.1	Silty clay	59	UDS
12.7	Coarse sand	9	DS
6.2	Silty clay	46	UDS
19.5	Silty clay	7	UDS
9.7	Clayey silt	30	SPT
25.1	Silty clay	59	SPT
20.8	Coarse sand	53	UDS
12.0	Coarse sand	17	UDS
26.6	Medium sand	54	DS

ANNEXURE IV — Electrical Load Schedule, page 34

Panel	Description	Connected load (kW)	Demand factor	Demand (kVA)
DB-02	Lighting	41.49	0.43	19.82
DB-33	Pumps	40.13	0.64	28.53
DB-35	HVAC	127.20	0.82	115.85
DB-17	Sterilisation	86.36	0.55	53.02
DB-04	Power outlets	86.91	0.87	84.03
DB-01	Sterilisation	91.69	0.46	47.22
DB-09	Sterilisation	77.75	0.92	79.65
DB-09	Sterilisation	44.11	0.81	39.56
DB-27	HVAC	103.67	0.45	52.35
DB-40	HVAC	48.79	0.81	44.10
DB-09	HVAC	134.59	0.85	126.95
DB-26	Lighting	136.54	0.82	124.78
DB-36	Sterilisation	120.24	0.44	58.66
DB-03	Power outlets	38.93	0.44	18.99
DB-14	Power outlets	54.18	0.64	38.61
DB-19	Lighting	119.96	0.56	74.68
DB-01	Medical equipment	48.30	0.85	45.41
DB-32	Sterilisation	26.23	0.80	23.30
DB-12	Lifts	153.93	0.75	127.75
DB-29	Pumps	151.23	0.53	89.58
DB-22	Sterilisation	130.07	0.74	107.31
DB-20	Medical equipment	16.87	0.52	9.82
DB-18	Lighting	66.64	0.84	62.04
DB-04	Medical equipment	80.64	0.50	45.07
DB-32	Medical equipment	63.79	0.82	57.97
DB-33	Sterilisation	116.85	0.84	108.73

ANNEXURE IV — Equipment Schedule, page 35

Item	Department	Nos	Unit rate (Rs)	Amount (Rs)
Autoclave	Radiology	2	2,187,840	4,375,680
Autoclave	Radiology	36	1,206,590	43,437,240
Ultrasound	ICU	3	109,310	327,930
Ventilator	Radiology	31	1,359,400	42,141,400
Defibrillator	Casualty	12	1,399,355	16,792,260
OT table	OT	14	2,315,389	32,415,446
Ventilator	Casualty	5	352,289	1,761,445
Ventilator	Radiology	20	2,159,478	43,189,560
Patient monitor	Radiology	24	1,300,907	31,221,768
Ventilator	ICU	5	1,487,087	7,435,435
Ultrasound	OT	13	2,008,941	26,116,233
Defibrillator	Radiology	25	1,439,126	35,978,150
Autoclave	ICU	14	1,756,665	24,593,310
Patient monitor	Ward	24	1,671,941	40,126,584
X-ray unit	Radiology	3	1,518,483	4,555,449
OT table	Casualty	1	2,015,376	2,015,376
Defibrillator	ICU	22	1,465,967	32,251,274
Ultrasound	ICU	16	794,670	12,714,720
Ultrasound	OT	13	1,847,542	24,018,046
Hospital bed	Radiology	4	139,702	558,808
Ultrasound	Ward	37	529,427	19,588,799
Patient monitor	Casualty	23	1,992,580	45,829,340
OT table	ICU	12	1,522,371	18,268,452

ANNEXURE IV — Detailed Measurement Sheet, page 36

Item	L (m)	B (m)	D (m)	Nos	Quantity
Item 338	34.64	0.92	2.10	23	117.843
Item 155	34.16	1.05	1.77	18	227.708
Item 796	6.66	1.93	1.98	6	368.147
Item 815	21.54	3.95	2.71	3	16.349
Item 481	30.07	5.07	2.48	8	338.906
Item 625	5.73	1.91	2.13	2	272.840
Item 683	32.10	1.06	2.74	18	208.362
Item 857	15.26	4.16	0.25	4	95.924
Item 432	17.07	4.71	1.87	23	137.753
Item 210	19.06	2.92	0.44	14	71.419
Item 754	16.68	2.50	2.26	10	40.122
Item 758	6.82	2.12	1.24	12	306.118
Item 428	6.75	5.08	0.91	2	97.464
Item 326	8.50	5.84	0.22	3	378.965
Item 849	19.74	4.56	2.56	14	166.277
Item 499	6.07	0.94	1.77	12	59.759
Item 864	36.86	2.98	2.74	7	52.893
Item 343	7.19	3.71	0.38	10	210.132
Item 198	5.30	1.27	2.73	19	11.013
Item 575	19.75	2.46	1.76	20	156.918
Item 705	2.27	4.66	2.50	24	75.512
Item 559	29.26	4.12	1.48	24	27.229
Item 191	26.18	4.31	1.18	4	290.673
Item 472	11.78	0.88	2.65	14	105.630
Item 863	32.11	4.08	1.84	2	327.702
Item 381	2.30	3.97	1.27	11	378.274

ANNEXURE IV — Room Data Sheet, page 37

Room	Department	Carpet area (sqm)	Occupancy	Floor finish
Room 198	Casualty	74.88	7	Antistatic vinyl
Room 466	Casualty	14.93	13	Vitrified tile
Room 293	General Medicine	60.97	25	Vitrified tile
Room 856	Paediatrics	36.26	22	Vitrified tile
Room 209	Pathology	19.12	18	Antistatic vinyl
Room 785	Casualty	21.89	21	Vitrified tile
Room 849	Radiology	38.16	18	Vitrified tile
Room 178	Paediatrics	20.84	1	Epoxy screed
Room 593	Obstetrics	65.95	2	Epoxy screed
Room 323	General Medicine	50.41	15	Antistatic vinyl
Room 871	Paediatrics	64.27	2	Antistatic vinyl
Room 507	Obstetrics	68.20	29	Antistatic vinyl
Room 464	Obstetrics	77.69	14	Vitrified tile
Room 435	General Medicine	28.53	5	Ceramic tile
Room 836	Pathology	55.38	17	Antistatic vinyl
Room 395	Pathology	12.65	30	Epoxy screed
Room 768	Pathology	47.28	23	Ceramic tile
Room 661	Obstetrics	29.91	25	Kota stone
Room 157	General Medicine	49.42	14	Epoxy screed
Room 669	Radiology	29.89	20	Kota stone
Room 756	Surgery	15.58	12	Kota stone
Room 565	Obstetrics	51.76	25	Epoxy screed
Room 729	Surgery	19.25	26	Vitrified tile
Room 179	General Medicine	49.18	27	Ceramic tile
Room 493	Obstetrics	33.25	21	Kota stone

ANNEXURE IV — Departmental Area Statement, page 38

Department	Floor	Units	Area per unit (sqm)	Total (sqm)
OPD	Third	15	46.48	697.20
Pharmacy	Second	4	89.90	359.61
Staff Quarters	First	11	111.04	1221.40
Diagnostics	First	16	53.60	857.63
Pharmacy	Third	4	43.19	172.76
IPD Ward	Third	11	80.08	880.86
Pharmacy	Ground	5	85.33	426.66
Diagnostics	First	3	75.58	226.73
Operation Theatre	Second	5	23.81	119.04
Operation Theatre	First	4	93.86	375.42
Staff Quarters	Third	13	89.94	1169.23
Operation Theatre	Ground	4	85.28	341.14
OPD	Second	2	71.38	142.76
OPD	Second	12	110.39	1324.72
ICU	Third	5	91.97	459.84
ICU	Ground	2	75.27	150.54
OPD	First	17	85.07	1446.18
Pharmacy	Ground	15	23.39	350.81
OPD	First	17	75.40	1281.85
ICU	First	16	17.62	281.94
Administration	Second	4	33.21	132.83
Operation Theatre	First	4	36.50	145.99
Diagnostics	Second	16	40.89	654.18
IPD Ward	Second	4	103.46	413.84
Pharmacy	Second	3	78.72	236.17
Administration	Second	15	40.33	604.94
IPD Ward	Second	11	65.94	725.33

ANNEXURE IV — Bore Log Summary, page 39

Depth (m)	Strata	N value	Sample
22.7	Clayey silt	7	UDS
19.2	Gravelly sand	9	SPT
6.4	Gravelly sand	6	SPT
12.5	Weathered rock	29	UDS
6.1	Weathered rock	5	SPT
8.5	Medium sand	47	DS
17.9	Weathered rock	9	DS
18.9	Medium sand	18	DS
11.6	Silty clay	9	SPT
13.8	Coarse sand	37	UDS
17.0	Gravelly sand	40	SPT
13.3	Medium sand	51	SPT
27.2	Silty clay	31	DS
24.3	Clayey silt	40	SPT
9.3	Coarse sand	13	DS
19.7	Silty clay	60	UDS
2.5	Weathered rock	20	SPT
8.1	Silty clay	49	DS
25.4	Weathered rock	59	SPT
7.3	Medium sand	10	UDS
13.8	Weathered rock	47	DS
25.7	Weathered rock	33	SPT
8.3	Silty clay	52	SPT
21.8	Weathered rock	50	SPT
13.4	Medium sand	56	DS

ANNEXURE IV — Electrical Load Schedule, page 40

Panel	Description	Connected load (kW)	Demand factor	Demand (kVA)
DB-03	Pumps	90.50	0.79	79.20
DB-38	Sterilisation	17.80	0.52	10.20
DB-38	Lifts	133.35	0.58	85.76
DB-06	Sterilisation	82.48	0.72	66.36
DB-26	Lifts	164.38	0.58	105.96
DB-18	Lifts	70.85	0.82	64.91
DB-03	HVAC	25.20	0.77	21.66
DB-10	HVAC	154.78	0.64	110.05
DB-11	HVAC	11.26	0.77	9.69
DB-27	Lighting	91.58	0.48	48.85
DB-28	Medical equipment	106.81	0.90	106.48
DB-20	HVAC	143.66	0.45	71.74
DB-15	Medical equipment	35.51	0.59	23.16
DB-37	Pumps	80.48	0.71	63.68
DB-21	Power outlets	21.33	0.64	15.14
DB-23	Sterilisation	128.68	0.63	90.46
DB-36	Power outlets	90.03	0.86	86.25
DB-20	Sterilisation	54.73	0.49	30.06
DB-19	Power outlets	37.67	0.70	29.35
DB-39	Sterilisation	178.15	0.90	178.66
DB-30	Sterilisation	87.43	0.42	40.34
DB-30	Power outlets	148.88	0.94	155.25
DB-09	HVAC	140.57	0.69	107.25
DB-25	Medical equipment	48.49	0.61	32.75
DB-03	Sterilisation	8.92	0.83	8.25
DB-36	Lighting	55.35	0.52	32.01
DB-12	Lifts	120.71	0.83	111.23
DB-36	Sterilisation	92.34	0.51	52.04
DB-09	HVAC	154.92	0.54	93.56
DB-12	Lighting	34.92	0.83	32.20

ANNEXURE IV — Equipment Schedule, page 41

Item	Department	Nos	Unit rate (Rs)	Amount (Rs)
X-ray unit	ICU	19	214,692	4,079,148
Hospital bed	CSSD	14	1,387,572	19,426,008
OT table	Casualty	2	446,566	893,132
Hospital bed	CSSD	6	1,426,861	8,561,166
Hospital bed	ICU	15	1,705,148	25,577,220
Defibrillator	CSSD	23	1,831,158	42,116,634
Ultrasound	CSSD	15	367,177	5,507,655
Hospital bed	ICU	39	72,729	2,836,431
Patient monitor	ICU	40	2,096,333	83,853,320
Patient monitor	Radiology	7	345,732	2,420,124
Patient monitor	Casualty	13	559,423	7,272,499
OT table	ICU	31	917,333	28,437,323
Hospital bed	Ward	34	1,012,464	34,423,776
Defibrillator	Casualty	31	1,499,908	46,497,148
Autoclave	CSSD	21	1,012,140	21,254,940
Defibrillator	OT	23	2,006,959	46,160,057
Autoclave	ICU	4	989,872	3,959,488
X-ray unit	ICU	14	2,236,969	31,317,566
OT table	CSSD	7	1,053,245	7,372,715
Hospital bed	CSSD	23	1,062,050	24,427,150
Patient monitor	Casualty	36	44,908	1,616,688
Autoclave	Radiology	17	2,013,859	34,235,603
Autoclave	Casualty	1	107,391	107,391
Defibrillator	Ward	37	147,521	5,458,277
Ventilator	CSSD	30	509,993	15,299,790
OT table	Casualty	36	1,503,570	54,128,520
Patient monitor	Radiology	14	1,767,987	24,751,818
Hospital bed	Ward	36	261,482	9,413,352
OT table	ICU	15	1,947,889	29,218,335

ANNEXURE IV — Detailed Measurement Sheet, page 42

Item	L (m)	B (m)	D (m)	Nos	Quantity
Item 540	10.80	0.43	0.56	6	333.371
Item 131	5.59	1.26	0.24	3	338.589
Item 699	31.17	2.77	2.38	10	283.041
Item 692	27.57	3.04	2.49	20	85.844
Item 885	18.29	4.72	2.72	4	353.768
Item 641	8.82	2.45	2.95	17	21.849
Item 436	14.57	2.48	2.47	21	61.274
Item 223	19.53	0.36	1.70	15	306.461
Item 539	17.38	1.78	2.87	6	109.337
Item 697	37.76	3.48	0.40	1	141.316
Item 724	34.63	1.81	2.28	11	218.808
Item 432	3.85	0.33	0.86	20	217.135
Item 105	24.83	5.64	2.69	2	252.852
Item 847	6.14	2.44	2.39	10	120.054
Item 380	36.69	3.36	1.97	22	7.827
Item 727	26.90	0.59	0.66	14	34.891
Item 847	4.33	1.75	2.61	6	364.127
Item 136	7.43	2.51	1.71	19	301.455
Item 863	37.94	5.84	1.25	19	183.171
Item 522	10.67	0.72	1.43	18	104.813
Item 124	7.23	4.10	1.92	24	147.585
Item 716	33.22	2.60	0.62	14	143.483
Item 113	12.26	5.87	1.54	6	253.629
Item 324	22.45	4.02	1.26	17	170.194
Item 230	26.80	5.54	1.87	5	52.522
Item 566	33.80	1.91	1.23	18	35.986
Item 377	21.92	3.28	1.69	8	81.158
Item 861	16.31	0.59	2.58	24	101.338

ANNEXURE IV — Room Data Sheet, page 43

Room	Department	Carpet area (sqm)	Occupancy	Floor finish
Room 780	Surgery	68.92	14	Antistatic vinyl
Room 199	Pathology	35.94	2	Antistatic vinyl
Room 331	Obstetrics	19.96	16	Epoxy screed
Room 700	Surgery	48.80	11	Epoxy screed
Room 368	Paediatrics	31.68	12	Epoxy screed
Room 278	Paediatrics	68.71	5	Epoxy screed
Room 263	Pathology	38.74	20	Epoxy screed
Room 840	General Medicine	64.56	9	Kota stone
Room 713	Radiology	9.32	14	Kota stone
Room 366	Obstetrics	28.93	18	Vitrified tile
Room 514	Obstetrics	19.17	20	Vitrified tile
Room 422	Obstetrics	68.37	30	Ceramic tile
Room 285	Obstetrics	60.27	15	Vitrified tile
Room 764	Casualty	78.35	4	Kota stone
Room 769	Paediatrics	23.71	27	Kota stone
Room 536	Casualty	9.60	5	Vitrified tile
Room 273	Radiology	82.33	24	Antistatic vinyl
Room 734	Pathology	38.83	28	Ceramic tile
Room 476	Casualty	87.78	23	Ceramic tile
Room 864	Paediatrics	76.64	19	Kota stone
Room 141	Paediatrics	44.82	19	Vitrified tile
Room 698	Radiology	17.65	11	Antistatic vinyl

ANNEXURE IV — Departmental Area Statement, page 44

Department	Floor	Units	Area per unit (sqm)	Total (sqm)
Staff Quarters	Ground	13	109.92	1428.96
Pharmacy	Ground	4	73.55	294.20
IPD Ward	Second	14	66.29	928.09
Administration	Ground	4	72.25	289.02
IPD Ward	First	16	13.32	213.17
Pharmacy	Second	4	133.17	532.69
OPD	Ground	18	64.45	1160.10
Operation Theatre	First	10	39.34	393.36
Diagnostics	Second	15	21.92	328.75
Staff Quarters	Ground	11	56.33	619.62
OPD	First	10	73.72	737.16
Pharmacy	First	15	119.24	1788.58
Operation Theatre	Third	6	62.49	374.95
Pharmacy	Ground	17	54.81	931.80
Administration	Ground	8	123.74	989.96
Diagnostics	First	9	19.82	178.35
ICU	Second	16	138.11	2209.75
Staff Quarters	Ground	1	118.51	118.51
ICU	First	3	92.60	277.80
Staff Quarters	Third	11	77.93	857.26
Diagnostics	Ground	13	123.63	1607.20
Operation Theatre	First	11	63.01	693.08

ANNEXURE IV — Bore Log Summary, page 45

Depth (m)	Strata	N value	Sample
20.1	Clayey silt	36	SPT
27.4	Coarse sand	26	UDS
28.1	Medium sand	48	SPT
12.4	Clayey silt	9	DS
17.8	Weathered rock	43	UDS
4.4	Medium sand	11	DS
19.9	Weathered rock	25	UDS
29.8	Gravelly sand	58	SPT
12.2	Gravelly sand	56	UDS
23.0	Silty clay	51	SPT
25.8	Silty clay	18	SPT
25.0	Silty clay	12	DS
5.7	Medium sand	27	UDS
26.8	Silty clay	21	UDS
6.4	Gravelly sand	40	UDS
3.2	Coarse sand	38	UDS
11.7	Medium sand	53	UDS
0.8	Weathered rock	6	SPT
16.6	Weathered rock	39	DS
17.9	Weathered rock	39	UDS
16.3	Gravelly sand	55	DS
17.4	Coarse sand	42	DS
24.7	Coarse sand	28	DS
26.1	Clayey silt	22	UDS
9.7	Medium sand	4	DS
26.8	Gravelly sand	27	SPT
29.3	Weathered rock	38	SPT
6.2	Clayey silt	55	UDS
11.9	Silty clay	48	DS

ANNEXURE IV-A — YEAR-WISE CASH FLOW STATEMENT

All figures in Rs. crore. Year 0 is the year of commencement.

Year	Capital cost	O&M cost	Gross benefit	Net cash flow
0	62.00	-	-	-62.00
1	74.50	-	-	-74.50
2	49.90	-	-	-49.90
3	-	3.91	35.31	31.40
4	-	3.91	35.17	31.26
5	-	3.91	35.03	31.12
6	-	3.91	34.89	30.98
7	-	3.91	34.75	30.84
8	-	3.91	34.60	30.69
9	-	3.91	34.46	30.55
10	-	3.91	34.32	30.41
11	-	3.91	34.18	30.27
12	-	3.91	34.04	30.13
13	-	3.91	33.90	29.99
14	-	3.91	33.76	29.85
15	-	3.91	33.62	29.71
16	-	3.91	33.47	29.56
17	-	3.91	33.33	29.42
18	-	3.91	33.19	29.28
19	-	3.91	33.05	29.14
20	-	3.91	32.91	29.00
21	-	3.91	32.77	28.86
22	-	3.91	32.63	28.72
23	-	3.91	32.49	28.58
24	-	3.91	32.34	28.43
25	-	3.91	32.20	28.29
26	-	3.91	32.06	28.15
27	-	3.91	31.92	28.01
28	-	3.91	31.78	27.87
29	-	3.91	31.64	27.73
30	-	3.91	31.50	27.59

ANNEXURE V — SOIL INVESTIGATION REPORT

Bore logs, laboratory results and foundation recommendations for nine boreholes.

ANNEXURE V — Sheet 1

Barrier-free access has been provided in accordance with the Harmonised Guidelines and Standards for Universal Accessibility. Ramps, lifts, signage and accessible toilets are provided on every floor of the block.

Consultations were held with the head of each clinical department on adjacency, patient flow and equipment layout. Minutes of the consultation meetings are placed at the Annexure to this chapter.

Soil investigation was carried out by rotary drilling at the locations shown on the layout plan. The safe bearing capacity adopted for design is based on the lower of the shear and settlement criteria as reported in the geotechnical report.

Consultations were held with the head of each clinical department on adjacency, patient flow and equipment layout. Minutes of the consultation meetings are placed at the Annexure to this chapter.

Reference: drawing no. PH-948, revision C.

ANNEXURE V — Sheet 2

Departmental Area Statement

Department	Floor	Units	Area per unit (sqm)	Total (sqm)
ICU	First	6	24.65	147.90
Pharmacy	Ground	7	86.74	607.15
Diagnostics	Second	17	50.14	852.45
Diagnostics	First	3	25.32	75.96
Operation Theatre	Second	10	14.70	146.96
IPD Ward	Third	18	88.46	1592.22
Diagnostics	Ground	10	139.79	1397.90
Operation Theatre	Second	3	38.27	114.82
Administration	Ground	14	57.96	811.42
Pharmacy	First	17	136.37	2318.37
Diagnostics	Ground	18	108.98	1961.60
Pharmacy	First	8	75.97	607.78
IPD Ward	Third	16	71.90	1150.42
Diagnostics	Third	3	44.75	134.24
Diagnostics	Second	12	132.38	1588.61
Operation Theatre	First	13	50.15	651.89
Operation Theatre	Ground	11	41.10	452.08
Diagnostics	First	6	12.23	73.39
Pharmacy	Third	11	45.85	504.40
IPD Ward	Ground	2	63.91	127.82
Staff Quarters	First	6	36.15	216.90
Pharmacy	Third	18	119.57	2152.34
ICU	Third	16	139.21	2227.39
ICU	First	13	26.96	350.43

The space programme has been developed from the functional brief agreed with the user department. Departmental areas, circulation and the gross-to-carpet ratio are tabulated in the area statement enclosed at the Annexure.

ANNEXURE V — Sheet 3

Design Computation

Moment = 34.647 × 11.303 = 391.601 kNm   (permissible 721.725) — safe

Axial load = 68.191 × 2.883 = 196.625 kN   (permissible 253.841) — safe

Shear = 7.060 × 7.216 = 50.945 kN   (permissible 86.575) — safe

Moment = 19.422 × 5.971 = 115.971 kNm   (permissible 125.552) — safe

Shear = 83.794 × 8.508 = 712.911 kN   (permissible 986.047) — safe

Moment = 20.460 × 1.676 = 34.295 kNm   (permissible 40.863) — safe

Building services comprise electrical distribution, water supply and sanitation, HVAC for the identified conditioned areas, medical gas pipeline and vertical transportation. Connected load and demand factors are given in the load schedule.

ANNEXURE V — Sheet 4

Consultations were held with the head of each clinical department on adjacency, patient flow and equipment layout. Minutes of the consultation meetings are placed at the Annexure to this chapter.

Quantities have been computed from the approved architectural and structural drawings and cross-checked against the detailed measurement sheets enclosed at the Annexure. Rates are as per the schedule of rates in force with lead and lift adjustments.

The structural design conforms to IS 456 for reinforced concrete and IS 1893 for earthquake resistant design. The site falls in seismic zone V and the importance factor applicable to a hospital building has been adopted.

Consultations were held with the head of each clinical department on adjacency, patient flow and equipment layout. Minutes of the consultation meetings are placed at the Annexure to this chapter.

Reference: drawing no. EL-622, revision A.

ANNEXURE V — Sheet 5

Room Data Sheet

Room	Department	Carpet area (sqm)	Occupancy	Floor finish
Room 593	General Medicine	44.17	27	Kota stone
Room 387	Pathology	63.23	11	Kota stone
Room 619	Surgery	38.76	8	Antistatic vinyl
Room 389	Radiology	63.91	29	Epoxy screed
Room 791	Obstetrics	48.02	25	Antistatic vinyl
Room 155	Pathology	57.44	17	Ceramic tile
Room 703	Casualty	48.17	28	Vitrified tile
Room 863	Casualty	15.97	13	Antistatic vinyl
Room 850	Pathology	32.66	10	Vitrified tile
Room 726	Radiology	74.49	4	Antistatic vinyl
Room 462	Obstetrics	50.88	27	Kota stone
Room 298	Surgery	68.24	4	Ceramic tile
Room 204	Casualty	13.68	3	Vitrified tile
Room 765	Surgery	66.15	15	Vitrified tile
Room 812	Radiology	12.07	9	Antistatic vinyl
Room 842	Radiology	44.04	25	Ceramic tile
Room 489	Obstetrics	19.44	27	Ceramic tile
Room 535	Casualty	29.63	24	Antistatic vinyl
Room 457	Pathology	50.04	25	Vitrified tile
Room 545	General Medicine	57.90	21	Vitrified tile

Consultations were held with the head of each clinical department on adjacency, patient flow and equipment layout. Minutes of the consultation meetings are placed at the Annexure to this chapter.

ANNEXURE V — Sheet 6

Design Computation

Axial load = 33.588 × 10.106 = 339.427 kN   (permissible 371.043) — safe

Moment = 48.270 × 10.507 = 507.180 kNm   (permissible 889.347) — safe

Axial load = 25.470 × 8.763 = 223.200 kN   (permissible 412.148) — safe

Axial load = 73.085 × 4.246 = 310.292 kN   (permissible 376.071) — revise section

Axial load = 48.984 × 9.347 = 457.864 kN   (permissible 650.392) — revise section

Deflection = 79.005 × 6.413 = 506.690 mm   (permissible 582.862) — safe

Shear = 6.267 × 3.439 = 21.554 kN   (permissible 27.619) — safe

Axial load = 55.438 × 10.645 = 590.113 kN   (permissible 1057.707) — safe

Soil investigation was carried out by rotary drilling at the locations shown on the layout plan. The safe bearing capacity adopted for design is based on the lower of the shear and settlement criteria as reported in the geotechnical report.

ANNEXURE V — Sheet 7

The structural design conforms to IS 456 for reinforced concrete and IS 1893 for earthquake resistant design. The site falls in seismic zone V and the importance factor applicable to a hospital building has been adopted.

Soil investigation was carried out by rotary drilling at the locations shown on the layout plan. The safe bearing capacity adopted for design is based on the lower of the shear and settlement criteria as reported in the geotechnical report.

Fire safety provisions follow Part 4 of the National Building Code. Means of egress, travel distances, staircase widths and the fire fighting installations have been designed for the occupancy classification of the block.

Quantities have been computed from the approved architectural and structural drawings and cross-checked against the detailed measurement sheets enclosed at the Annexure. Rates are as per the schedule of rates in force with lead and lift adjustments.

Reference: drawing no. AR-350, revision A.

ANNEXURE V — Sheet 8

Room Data Sheet

Room	Department	Carpet area (sqm)	Occupancy	Floor finish
Room 158	Surgery	24.68	13	Antistatic vinyl
Room 444	Surgery	82.59	29	Kota stone
Room 461	Pathology	46.28	11	Kota stone
Room 157	Paediatrics	82.86	20	Epoxy screed
Room 229	Pathology	13.33	21	Epoxy screed
Room 420	Pathology	61.40	25	Epoxy screed
Room 321	Radiology	65.81	23	Kota stone
Room 544	Paediatrics	87.21	3	Vitrified tile
Room 186	Pathology	14.85	17	Kota stone
Room 821	Paediatrics	69.67	2	Ceramic tile
Room 290	Obstetrics	51.38	9	Ceramic tile
Room 293	Radiology	16.64	28	Antistatic vinyl
Room 177	Surgery	29.42	26	Kota stone
Room 319	Casualty	22.84	4	Epoxy screed
Room 650	Pathology	74.94	10	Vitrified tile
Room 735	Paediatrics	86.63	1	Vitrified tile
Room 123	Casualty	88.91	13	Epoxy screed

Quantities have been computed from the approved architectural and structural drawings and cross-checked against the detailed measurement sheets enclosed at the Annexure. Rates are as per the schedule of rates in force with lead and lift adjustments.

ANNEXURE V — Sheet 9

Design Computation

Axial load = 41.197 × 4.345 = 178.998 kN   (permissible 250.884) — revise section

Moment = 41.811 × 10.577 = 442.246 kNm   (permissible 738.769) — safe

Bearing pressure = 60.837 × 9.445 = 574.609 kPa   (permissible 933.450) — revise section

Bearing pressure = 76.847 × 9.822 = 754.792 kPa   (permissible 1182.414) — safe

Bearing pressure = 73.277 × 3.884 = 284.619 kPa   (permissible 370.771) — safe

Axial load = 71.971 × 9.644 = 694.087 kN   (permissible 1201.891) — safe

Shear = 82.059 × 11.546 = 947.431 kN   (permissible 1009.769) — safe

The structural design conforms to IS 456 for reinforced concrete and IS 1893 for earthquake resistant design. The site falls in seismic zone V and the importance factor applicable to a hospital building has been adopted.

ANNEXURE V — Sheet 10

Building services comprise electrical distribution, water supply and sanitation, HVAC for the identified conditioned areas, medical gas pipeline and vertical transportation. Connected load and demand factors are given in the load schedule.

Consultations were held with the head of each clinical department on adjacency, patient flow and equipment layout. Minutes of the consultation meetings are placed at the Annexure to this chapter.

Building services comprise electrical distribution, water supply and sanitation, HVAC for the identified conditioned areas, medical gas pipeline and vertical transportation. Connected load and demand factors are given in the load schedule.

Fire safety provisions follow Part 4 of the National Building Code. Means of egress, travel distances, staircase widths and the fire fighting installations have been designed for the occupancy classification of the block.

Reference: drawing no. ST-329, revision B.

ANNEXURE V — Sheet 11

Detailed Measurement Sheet

Item	L (m)	B (m)	D (m)	Nos	Quantity
Item 120	18.98	5.21	2.77	18	26.246
Item 550	38.85	1.14	1.10	8	234.898
Item 760	34.30	1.27	0.25	24	200.717
Item 841	5.59	4.82	1.70	5	152.138
Item 471	18.81	5.45	0.32	2	316.277
Item 765	37.75	1.00	2.73	5	389.873
Item 736	33.80	0.34	0.61	22	15.829
Item 770	35.38	3.31	1.45	24	337.354
Item 770	16.34	2.92	1.20	12	313.339
Item 857	26.95	5.01	0.61	12	16.539
Item 495	20.58	2.68	0.35	13	280.031
Item 718	35.27	1.61	0.31	18	372.393
Item 448	18.42	5.61	1.42	20	370.628
Item 731	9.74	2.47	1.45	21	138.284
Item 442	18.41	1.16	1.26	2	125.587
Item 370	10.64	0.89	1.52	5	48.891
Item 271	4.99	0.33	2.23	12	364.832
Item 532	27.28	5.22	0.78	19	328.542
Item 309	4.94	4.01	1.41	22	8.911
Item 192	12.70	5.74	0.20	19	345.243
Item 146	37.17	1.87	1.80	15	277.268
Item 683	16.38	0.65	0.33	1	145.499
Item 578	8.12	0.92	1.01	12	200.919
Item 716	32.19	1.10	1.09	21	130.579
Item 180	3.62	3.76	1.93	3	391.555

Building services comprise electrical distribution, water supply and sanitation, HVAC for the identified conditioned areas, medical gas pipeline and vertical transportation. Connected load and demand factors are given in the load schedule.

ANNEXURE V — Sheet 12

Design Computation

Moment = 69.905 × 8.332 = 582.441 kNm   (permissible 837.891) — safe

Shear = 10.347 × 6.998 = 72.412 kN   (permissible 99.361) — safe

Moment = 56.866 × 1.180 = 67.075 kNm   (permissible 105.970) — safe

Bearing pressure = 34.298 × 11.382 = 390.361 kPa   (permissible 427.316) — safe

Deflection = 52.873 × 11.218 = 593.123 mm   (permissible 968.574) — safe

Bearing pressure = 68.941 × 5.549 = 382.585 kPa   (permissible 697.204) — safe

Soil investigation was carried out by rotary drilling at the locations shown on the layout plan. The safe bearing capacity adopted for design is based on the lower of the shear and settlement criteria as reported in the geotechnical report.

ANNEXURE V — Sheet 13

Consultations were held with the head of each clinical department on adjacency, patient flow and equipment layout. Minutes of the consultation meetings are placed at the Annexure to this chapter.

Soil investigation was carried out by rotary drilling at the locations shown on the layout plan. The safe bearing capacity adopted for design is based on the lower of the shear and settlement criteria as reported in the geotechnical report.

Fire safety provisions follow Part 4 of the National Building Code. Means of egress, travel distances, staircase widths and the fire fighting installations have been designed for the occupancy classification of the block.

Quantities have been computed from the approved architectural and structural drawings and cross-checked against the detailed measurement sheets enclosed at the Annexure. Rates are as per the schedule of rates in force with lead and lift adjustments.

Reference: drawing no. ST-937, revision B.

ANNEXURE V — Sheet 14

Bore Log Summary

Depth (m)	Strata	N value	Sample
17.6	Gravelly sand	30	SPT
23.3	Weathered rock	12	UDS
3.9	Coarse sand	13	DS
8.5	Medium sand	43	UDS
21.4	Silty clay	39	DS
21.7	Silty clay	33	UDS
3.3	Coarse sand	53	DS
17.2	Coarse sand	40	UDS
28.0	Coarse sand	25	UDS
28.7	Clayey silt	52	DS
17.7	Silty clay	8	SPT
17.5	Medium sand	11	DS
11.7	Gravelly sand	19	UDS
15.0	Clayey silt	14	SPT
16.8	Weathered rock	10	SPT
12.9	Medium sand	42	DS
12.8	Silty clay	56	SPT
26.6	Medium sand	33	UDS
27.1	Silty clay	47	SPT
10.5	Coarse sand	34	SPT
3.7	Medium sand	44	DS
20.5	Coarse sand	8	SPT
16.3	Clayey silt	54	DS

Soil investigation was carried out by rotary drilling at the locations shown on the layout plan. The safe bearing capacity adopted for design is based on the lower of the shear and settlement criteria as reported in the geotechnical report.

ANNEXURE V — Sheet 15

Design Computation

Shear = 76.292 × 7.056 = 538.323 kN (permissible 708.221) — safe

Shear = 12.249 × 8.544 = 104.656 kN (permissible 154.863) — safe

Moment = 83.470 × 1.147 = 95.738 kNm (permissible 156.347) — safe

Axial load = 73.215 × 5.863 = 429.269 kN (permissible 638.072) — safe

Bearing pressure = 7.647 × 9.233 = 70.598 kPa (permissible 92.287) — safe

The structural design conforms to IS 456 for reinforced concrete and IS 1893 for earthquake resistant design. The site falls in seismic zone V and the importance factor applicable to a hospital building has been adopted.

ANNEXURE V — Sheet 16

Barrier-free access has been provided in accordance with the Harmonised Guidelines and Standards for Universal Accessibility. Ramps, lifts, signage and accessible toilets are provided on every floor of the block.

Barrier-free access has been provided in accordance with the Harmonised Guidelines and Standards for Universal Accessibility. Ramps, lifts, signage and accessible toilets are provided on every floor of the block.

Quantities have been computed from the approved architectural and structural drawings and cross-checked against the detailed measurement sheets enclosed at the Annexure. Rates are as per the schedule of rates in force with lead and lift adjustments.

Fire safety provisions follow Part 4 of the National Building Code. Means of egress, travel distances, staircase widths and the fire fighting installations have been designed for the occupancy classification of the block.

Reference: drawing no. AR-673, revision B.

ANNEXURE V — Sheet 17

Room Data Sheet

Room	Department	Carpet area (sqm)	Occupancy	Floor finish
Room 630	Radiology	48.46	27	Kota stone
Room 872	Radiology	69.68	4	Antistatic vinyl
Room 701	Pathology	23.13	11	Antistatic vinyl
Room 722	Pathology	9.32	27	Antistatic vinyl
Room 864	Surgery	34.44	29	Kota stone
Room 872	Paediatrics	17.76	19	Antistatic vinyl
Room 559	Radiology	61.88	5	Epoxy screed
Room 204	Obstetrics	33.27	29	Ceramic tile
Room 126	Surgery	61.06	1	Kota stone
Room 744	Radiology	15.30	28	Kota stone
Room 644	Pathology	27.89	3	Kota stone
Room 437	General Medicine	58.96	5	Epoxy screed
Room 584	General Medicine	81.67	12	Antistatic vinyl
Room 777	Paediatrics	70.30	7	Kota stone
Room 861	General Medicine	69.18	8	Epoxy screed
Room 457	Surgery	31.35	25	Antistatic vinyl
Room 551	General Medicine	40.57	5	Ceramic tile
Room 133	Radiology	43.40	28	Antistatic vinyl
Room 163	Obstetrics	10.52	29	Ceramic tile
Room 539	Pathology	57.69	21	Epoxy screed
Room 720	Paediatrics	81.31	23	Antistatic vinyl
Room 124	Pathology	15.03	8	Vitrified tile
Room 340	Paediatrics	68.01	6	Ceramic tile
Room 411	General Medicine	38.99	23	Antistatic vinyl
Room 337	Pathology	83.44	2	Vitrified tile
Room 339	Paediatrics	72.96	1	Antistatic vinyl

Consultations were held with the head of each clinical department on adjacency, patient flow and equipment layout. Minutes of the consultation meetings are placed at the Annexure to this chapter.

ANNEXURE V — Sheet 18

Design Computation

Shear = 44.811 × 3.141 = 140.729 kN   (permissible 232.732) — safe

Axial load = 47.369 × 11.777 = 557.850 kN   (permissible 1059.082) — safe

Shear = 57.081 × 8.760 = 500.051 kN   (permissible 549.894) — safe

Shear = 64.322 × 10.371 = 667.093 kN   (permissible 817.995) — safe

Axial load = 23.645 × 3.102 = 73.349 kN   (permissible 119.506) — safe

Bearing pressure = 35.419 × 3.176 = 112.498 kPa   (permissible 151.023) — safe

Fire safety provisions follow Part 4 of the National Building Code. Means of egress, travel distances, staircase widths and the fire fighting installations have been designed for the occupancy classification of the block.

ANNEXURE V — Sheet 19

Quantities have been computed from the approved architectural and structural drawings and cross-checked against the detailed measurement sheets enclosed at the Annexure. Rates are as per the schedule of rates in force with lead and lift adjustments.

Quantities have been computed from the approved architectural and structural drawings and cross-checked against the detailed measurement sheets enclosed at the Annexure. Rates are as per the schedule of rates in force with lead and lift adjustments.

Quantities have been computed from the approved architectural and structural drawings and cross-checked against the detailed measurement sheets enclosed at the Annexure. Rates are as per the schedule of rates in force with lead and lift adjustments.

Soil investigation was carried out by rotary drilling at the locations shown on the layout plan. The safe bearing capacity adopted for design is based on the lower of the shear and settlement criteria as reported in the geotechnical report.

Reference: drawing no. ST-451, revision C.

ANNEXURE V — Sheet 20

Bore Log Summary

Depth (m)	Strata	N value	Sample
6.6	Clayey silt	43	UDS
18.3	Medium sand	12	DS
1.4	Clayey silt	28	DS
3.2	Weathered rock	41	DS
13.5	Silty clay	31	UDS
29.5	Coarse sand	43	SPT
8.6	Medium sand	28	SPT
4.3	Medium sand	45	DS
17.5	Clayey silt	45	SPT
2.6	Silty clay	37	DS
28.7	Gravelly sand	55	DS
14.4	Clayey silt	46	SPT
5.1	Medium sand	48	UDS
22.2	Weathered rock	27	UDS
6.2	Clayey silt	53	SPT
13.2	Weathered rock	25	SPT
24.6	Gravelly sand	33	SPT
27.4	Clayey silt	51	UDS
14.9	Gravelly sand	37	DS
19.4	Coarse sand	44	UDS

The space programme has been developed from the functional brief agreed with the user department. Departmental areas, circulation and the gross-to-carpet ratio are tabulated in the area statement enclosed at the Annexure.

ANNEXURE V — Sheet 21

Design Computation

Axial load = 71.792 × 11.495 = 825.256 kN   (permissible 1348.175) — safe

Bearing pressure = 49.095 × 6.868 = 337.184 kPa   (permissible 639.174) — safe

Bearing pressure = 32.611 × 11.120 = 362.647 kPa   (permissible 531.428) — safe

Bearing pressure = 89.607 × 11.007 = 986.271 kPa   (permissible 1430.620) — revise section

Axial load = 81.368 × 7.344 = 597.610 kN   (permissible 815.846) — safe

Deflection = 56.445 × 11.999 = 677.270 mm   (permissible 818.799) — safe

Moment = 63.216 × 6.078 = 384.245 kNm   (permissible 613.081) — safe

Bearing pressure = 55.977 × 6.088 = 340.806 kPa   (permissible 480.902) — safe

Building services comprise electrical distribution, water supply and sanitation, HVAC for the identified conditioned areas, medical gas pipeline and vertical transportation. Connected load and demand factors are given in the load schedule.

ANNEXURE V — Sheet 22

Consultations were held with the head of each clinical department on adjacency, patient flow and equipment layout. Minutes of the consultation meetings are placed at the Annexure to this chapter.

Fire safety provisions follow Part 4 of the National Building Code. Means of egress, travel distances, staircase widths and the fire fighting installations have been designed for the occupancy classification of the block.

The structural design conforms to IS 456 for reinforced concrete and IS 1893 for earthquake resistant design. The site falls in seismic zone V and the importance factor applicable to a hospital building has been adopted.

Fire safety provisions follow Part 4 of the National Building Code. Means of egress, travel distances, staircase widths and the fire fighting installations have been designed for the occupancy classification of the block.

Reference: drawing no. AR-212, revision B.

ANNEXURE V — Sheet 23

Detailed Measurement Sheet

Item	L (m)	B (m)	D (m)	Nos	Quantity
Item 101	21.91	3.36	1.01	6	22.558
Item 638	34.49	1.27	2.84	5	121.660
Item 247	32.19	1.04	0.25	15	102.773
Item 168	34.38	5.45	2.97	4	54.521
Item 100	11.81	1.35	2.94	3	371.421
Item 797	3.26	1.85	2.44	3	344.135
Item 188	22.14	4.99	0.51	11	262.395
Item 560	2.53	4.13	2.61	22	1.642
Item 647	39.49	2.07	2.89	8	299.485
Item 317	33.54	2.97	1.18	15	71.332
Item 432	9.84	2.99	2.25	12	202.431
Item 865	5.74	2.21	2.10	12	179.056
Item 119	2.16	1.70	1.73	10	91.331
Item 186	9.74	0.43	0.96	21	211.918
Item 631	8.39	0.70	1.66	6	36.454
Item 112	37.65	5.70	1.41	12	222.971
Item 272	5.76	4.16	1.11	17	274.822
Item 319	29.86	3.92	2.68	4	190.493

The structural design conforms to IS 456 for reinforced concrete and IS 1893 for earthquake resistant design. The site falls in seismic zone V and the importance factor applicable to a hospital building has been adopted.

ANNEXURE V — Sheet 24

Design Computation

Deflection = 64.922 × 2.713 = 176.110 mm   (permissible 320.862) — safe

Bearing pressure = 28.504 × 4.293 = 122.368 kPa   (permissible 207.599) — safe

Shear = 81.653 × 8.275 = 675.668 kN   (permissible 1000.358) — safe

Axial load = 43.886 × 5.194 = 227.964 kN   (permissible 271.194) — safe

Bearing pressure = 21.592 × 9.503 = 205.194 kPa   (permissible 297.166) — safe

Moment = 13.846 × 11.367 = 157.385 kNm   (permissible 286.230) — safe

Moment = 45.295 × 11.919 = 539.896 kNm   (permissible 934.090) — safe

Moment = 13.292 × 7.593 = 100.919 kNm   (permissible 132.471) — safe

The structural design conforms to IS 456 for reinforced concrete and IS 1893 for earthquake resistant design. The site falls in seismic zone V and the importance factor applicable to a hospital building has been adopted.

ANNEXURE V — Sheet 25

Consultations were held with the head of each clinical department on adjacency, patient flow and equipment layout. Minutes of the consultation meetings are placed at the Annexure to this chapter.

Consultations were held with the head of each clinical department on adjacency, patient flow and equipment layout. Minutes of the consultation meetings are placed at the Annexure to this chapter.

Soil investigation was carried out by rotary drilling at the locations shown on the layout plan. The safe bearing capacity adopted for design is based on the lower of the shear and settlement criteria as reported in the geotechnical report.

Soil investigation was carried out by rotary drilling at the locations shown on the layout plan. The safe bearing capacity adopted for design is based on the lower of the shear and settlement criteria as reported in the geotechnical report.

Reference: drawing no. EL-208, revision A.

ANNEXURE V — Sheet 26

Bore Log Summary

Depth (m)	Strata	N value	Sample
1.9	Medium sand	30	UDS
29.0	Gravelly sand	59	DS
24.5	Coarse sand	59	DS
20.7	Coarse sand	22	SPT
14.2	Clayey silt	14	DS
13.5	Gravelly sand	42	DS
25.8	Coarse sand	27	SPT
10.4	Silty clay	33	SPT
8.4	Coarse sand	8	SPT
10.2	Clayey silt	20	DS
5.9	Medium sand	15	SPT
27.6	Coarse sand	17	SPT
8.8	Gravelly sand	44	UDS
20.8	Weathered rock	60	DS
11.2	Medium sand	48	SPT
19.9	Gravelly sand	14	UDS
11.3	Gravelly sand	7	DS
5.5	Gravelly sand	14	DS
18.4	Gravelly sand	36	DS
28.1	Clayey silt	58	UDS
11.7	Coarse sand	37	SPT
11.6	Gravelly sand	27	SPT

Quantities have been computed from the approved architectural and structural drawings and cross-checked against the detailed measurement sheets enclosed at the Annexure. Rates are as per the schedule of rates in force with lead and lift adjustments.

ANNEXURE V — Sheet 27

Design Computation

Axial load = 56.374 × 3.836 = 216.257 kN   (permissible 308.499) — safe

Shear = 9.793 × 4.418 = 43.261 kN   (permissible 79.803) — safe

Moment = 37.159 × 4.691 = 174.306 kNm   (permissible 297.060) — revise section

Axial load = 61.514 × 5.979 = 367.778 kN   (permissible 471.496) — safe

Bearing pressure = 22.561 × 9.657 = 217.876 kPa   (permissible 358.515) — safe

Deflection = 80.008 × 1.964 = 157.121 mm   (permissible 259.303) — safe

Bearing pressure = 50.663 × 4.049 = 205.128 kPa   (permissible 257.976) — safe

Consultations were held with the head of each clinical department on adjacency, patient flow and equipment layout. Minutes of the consultation meetings are placed at the Annexure to this chapter.

ANNEXURE V — Sheet 28

Soil investigation was carried out by rotary drilling at the locations shown on the layout plan. The safe bearing capacity adopted for design is based on the lower of the shear and settlement criteria as reported in the geotechnical report.

Fire safety provisions follow Part 4 of the National Building Code. Means of egress, travel distances, staircase widths and the fire fighting installations have been designed for the occupancy classification of the block.

Consultations were held with the head of each clinical department on adjacency, patient flow and equipment layout. Minutes of the consultation meetings are placed at the Annexure to this chapter.

Fire safety provisions follow Part 4 of the National Building Code. Means of egress, travel distances, staircase widths and the fire fighting installations have been designed for the occupancy classification of the block.

Reference: drawing no. ST-632, revision C.

ANNEXURE V — Sheet 29

Electrical Load Schedule

Panel	Description	Connected load (kW)	Demand factor	Demand (kVA)
DB-15	Pumps	114.54	0.58	73.90
DB-40	HVAC	86.13	0.57	54.32
DB-37	HVAC	58.14	0.77	50.05
DB-30	Pumps	107.83	0.75	90.22
DB-32	Sterilisation	46.40	0.44	22.61
DB-29	Pumps	94.90	0.80	84.77
DB-17	Medical equipment	42.16	0.87	40.58
DB-17	Pumps	101.82	0.81	92.11
DB-24	Lighting	177.81	0.92	181.55
DB-05	Power outlets	113.23	0.59	73.83
DB-40	HVAC	49.55	0.73	39.92
DB-34	Medical equipment	149.06	0.91	150.28
DB-30	Pumps	102.37	0.86	97.41
DB-16	Sterilisation	5.89	0.40	2.62
DB-21	HVAC	157.33	0.89	155.46
DB-11	Lighting	40.01	0.68	30.32
DB-15	Medical equipment	58.60	0.42	27.27
DB-08	Sterilisation	8.84	0.61	6.01
DB-14	Medical equipment	26.94	0.54	16.05
DB-36	Sterilisation	3.79	0.56	2.34
DB-01	Power outlets	151.70	0.95	159.89
DB-16	Lighting	120.77	0.82	110.67
DB-39	Lifts	48.80	0.73	39.49

Building services comprise electrical distribution, water supply and sanitation, HVAC for the identified conditioned areas, medical gas pipeline and vertical transportation. Connected load and demand factors are given in the load schedule.

ANNEXURE VI — SERVICES DRAWINGS

Electrical, plumbing, HVAC and medical gas layouts. Drawings SE-100 to SE-146.

ANNEXURE VI — Sheet 1

The space programme has been developed from the functional brief agreed with the user department. Departmental areas, circulation and the gross-to-carpet ratio are tabulated in the area statement enclosed at the Annexure.

Soil investigation was carried out by rotary drilling at the locations shown on the layout plan. The safe bearing capacity adopted for design is based on the lower of the shear and settlement criteria as reported in the geotechnical report.

Fire safety provisions follow Part 4 of the National Building Code. Means of egress, travel distances, staircase widths and the fire fighting installations have been designed for the occupancy classification of the block.

Building services comprise electrical distribution, water supply and sanitation, HVAC for the identified conditioned areas, medical gas pipeline and vertical transportation. Connected load and demand factors are given in the load schedule.

Reference: drawing no. ST-324, revision A.

ANNEXURE VI — Sheet 2

Departmental Area Statement

Department	Floor	Units	Area per unit (sqm)	Total (sqm)
Operation Theatre	First	9	129.31	1163.76
Diagnostics	Third	5	106.95	534.75
IPD Ward	Second	10	74.14	741.42
Pharmacy	Second	3	132.10	396.30
ICU	Third	11	98.45	1082.92
Staff Quarters	Ground	7	24.14	168.95
IPD Ward	Ground	18	20.27	364.85
Pharmacy	Third	5	57.75	288.76
Operation Theatre	Ground	6	103.23	619.40
IPD Ward	Second	5	19.39	96.97
Operation Theatre	Second	13	40.25	523.24
IPD Ward	Ground	12	19.22	230.66
Staff Quarters	Second	1	81.54	81.54
Diagnostics	First	9	120.17	1081.50
ICU	Third	1	56.27	56.27
Diagnostics	Third	2	13.11	26.21
OPD	First	11	48.55	534.01
ICU	First	1	131.48	131.48
Staff Quarters	First	12	72.60	871.23
Diagnostics	Second	14	139.02	1946.33
Operation Theatre	Second	8	137.78	1102.23

Building services comprise electrical distribution, water supply and sanitation, HVAC for the identified conditioned areas, medical gas pipeline and vertical transportation. Connected load and demand factors are given in the load schedule.

ANNEXURE VI — Sheet 3

Design Computation

Axial load = 77.200 × 2.255 = 174.094 kN   (permissible 315.396) — safe

Moment = 1.552 × 6.258 = 9.712 kNm   (permissible 12.445) — revise section

Moment = 50.181 × 7.732 = 388.009 kNm   (permissible 677.689) — safe

Axial load = 17.043 × 0.851 = 14.503 kN   (permissible 21.509) — safe

Deflection = 19.717 × 8.167 = 161.028 mm   (permissible 289.442) — safe

Deflection = 47.680 × 6.630 = 316.127 mm   (permissible 493.965) — safe

Bearing pressure = 82.253 × 11.083 = 911.631 kPa   (permissible 1454.319) — safe

The structural design conforms to IS 456 for reinforced concrete and IS 1893 for earthquake resistant design. The site falls in seismic zone V and the importance factor applicable to a hospital building has been adopted.

ANNEXURE VI — Sheet 4

Building services comprise electrical distribution, water supply and sanitation, HVAC for the identified conditioned areas, medical gas pipeline and vertical transportation. Connected load and demand factors are given in the load schedule.

Building services comprise electrical distribution, water supply and sanitation, HVAC for the identified conditioned areas, medical gas pipeline and vertical transportation. Connected load and demand factors are given in the load schedule.

Building services comprise electrical distribution, water supply and sanitation, HVAC for the identified conditioned areas, medical gas pipeline and vertical transportation. Connected load and demand factors are given in the load schedule.

Soil investigation was carried out by rotary drilling at the locations shown on the layout plan. The safe bearing capacity adopted for design is based on the lower of the shear and settlement criteria as reported in the geotechnical report.

Reference: drawing no. PH-980, revision A.

ANNEXURE VI — Sheet 5

Electrical Load Schedule

Panel	Description	Connected load (kW)	Demand factor	Demand (kVA)
DB-04	Sterilisation	51.92	0.82	47.02
DB-23	Medical equipment	65.82	0.80	58.15
DB-02	Pumps	9.96	0.50	5.51
DB-34	HVAC	43.12	0.58	27.89
DB-03	Lifts	113.47	0.42	52.97
DB-32	Pumps	132.31	0.93	136.42
DB-11	Medical equipment	172.79	0.68	131.41
DB-04	Lighting	174.44	0.75	144.79
DB-04	HVAC	161.43	0.72	128.68
DB-30	Power outlets	33.23	0.57	20.88
DB-31	Pumps	92.34	0.65	67.14
DB-27	Lighting	157.15	0.94	164.75
DB-15	Medical equipment	119.97	0.82	108.96
DB-14	Lighting	132.89	0.92	135.66
DB-28	Pumps	142.23	0.89	141.28
DB-19	Lifts	48.96	0.71	38.88
DB-07	Lifts	84.17	0.64	59.45
DB-04	Lifts	131.03	0.61	89.48
DB-33	Power outlets	119.61	0.68	89.91
DB-39	Pumps	106.23	0.82	96.37
DB-26	Lighting	95.23	0.87	91.54
DB-03	Pumps	83.49	0.76	70.48
DB-25	Lifts	133.45	0.94	139.51
DB-29	Medical equipment	115.14	0.91	116.23
DB-33	Pumps	130.98	0.89	129.12
DB-23	Lighting	45.03	0.86	42.91

The structural design conforms to IS 456 for reinforced concrete and IS 1893 for earthquake resistant design. The site falls in seismic zone V and the importance factor applicable to a hospital building has been adopted.

ANNEXURE VI — Sheet 6

Design Computation

Axial load = 42.416 × 3.683 = 156.236 kN (permissible 241.139) — safe

Bearing pressure = 79.727 × 5.545 = 442.061 kPa (permissible 699.412) — safe

Moment = 21.935 × 0.906 = 19.866 kNm (permissible 22.889) — safe

Axial load = 76.670 × 2.653 = 203.400 kN (permissible 247.724) — safe

Moment = 31.020 × 10.735 = 333.015 kNm (permissible 589.597) — safe

Consultations were held with the head of each clinical department on adjacency, patient flow and equipment layout. Minutes of the consultation meetings are placed at the Annexure to this chapter.

ANNEXURE VI — Sheet 7

Quantities have been computed from the approved architectural and structural drawings and cross-checked against the detailed measurement sheets enclosed at the Annexure. Rates are as per the schedule of rates in force with lead and lift adjustments.

Building services comprise electrical distribution, water supply and sanitation, HVAC for the identified conditioned areas, medical gas pipeline and vertical transportation. Connected load and demand factors are given in the load schedule.

Building services comprise electrical distribution, water supply and sanitation, HVAC for the identified conditioned areas, medical gas pipeline and vertical transportation. Connected load and demand factors are given in the load schedule.

Soil investigation was carried out by rotary drilling at the locations shown on the layout plan. The safe bearing capacity adopted for design is based on the lower of the shear and settlement criteria as reported in the geotechnical report.

Reference: drawing no. ST-501, revision A.

ANNEXURE VI — Sheet 8

Bore Log Summary

Depth (m)	Strata	N value	Sample
4.2	Silty clay	35	DS
8.5	Clayey silt	7	UDS
26.0	Medium sand	7	SPT
5.1	Silty clay	14	DS
2.7	Weathered rock	54	UDS
13.1	Medium sand	6	SPT
27.8	Weathered rock	8	SPT
13.8	Weathered rock	55	DS
4.4	Coarse sand	50	UDS
12.5	Medium sand	32	SPT
9.8	Coarse sand	24	SPT
25.7	Clayey silt	10	SPT
18.4	Medium sand	30	SPT
14.9	Silty clay	4	DS
9.8	Medium sand	51	UDS
1.7	Weathered rock	5	SPT
18.8	Coarse sand	19	SPT
11.3	Coarse sand	20	SPT
10.5	Gravelly sand	8	UDS
28.4	Clayey silt	55	UDS

Barrier-free access has been provided in accordance with the Harmonised Guidelines and Standards for Universal Accessibility. Ramps, lifts, signage and accessible toilets are provided on every floor of the block.

ANNEXURE VI — Sheet 9

Design Computation

Moment = 27.903 × 9.791 = 273.195 kNm (permissible 510.122) — safe

Bearing pressure = 15.901 × 0.446 = 7.089 kPa (permissible 13.247) — safe

Axial load = 46.585 × 6.367 = 296.596 kN (permissible 410.434) — safe

Axial load = 68.507 × 9.997 = 684.842 kN (permissible 863.016) — safe

Moment = 45.493 × 2.747 = 124.955 kNm (permissible 133.332) — safe

Moment = 83.960 × 9.517 = 799.090 kNm (permissible 931.281) — safe

Moment = 76.888 × 4.030 = 309.845 kNm (permissible 403.136) — safe

Bearing pressure = 14.977 × 5.781 = 86.579 kPa (permissible 147.103) — safe

Quantities have been computed from the approved architectural and structural drawings and cross-checked against the detailed measurement sheets enclosed at the Annexure. Rates are as per the schedule of rates in force with lead and lift adjustments.

ANNEXURE VI — Sheet 10

The space programme has been developed from the functional brief agreed with the user department. Departmental areas, circulation and the gross-to-carpet ratio are tabulated in the area statement enclosed at the Annexure.

Building services comprise electrical distribution, water supply and sanitation, HVAC for the identified conditioned areas, medical gas pipeline and vertical transportation. Connected load and demand factors are given in the load schedule.

Quantities have been computed from the approved architectural and structural drawings and cross-checked against the detailed measurement sheets enclosed at the Annexure. Rates are as per the schedule of rates in force with lead and lift adjustments.

Soil investigation was carried out by rotary drilling at the locations shown on the layout plan. The safe bearing capacity adopted for design is based on the lower of the shear and settlement criteria as reported in the geotechnical report.

Reference: drawing no. EL-851, revision A.

ANNEXURE VI — Sheet 11

Departmental Area Statement

Department	Floor	Units	Area per unit (sqm)	Total (sqm)
Pharmacy	Ground	9	17.79	160.14
Operation Theatre	Third	7	63.68	445.74
OPD	Ground	10	88.20	882.02
IPD Ward	Ground	7	61.43	430.00
Staff Quarters	First	15	53.07	796.03
Diagnostics	First	3	41.29	123.87
Operation Theatre	Second	2	111.05	222.09
ICU	First	1	107.32	107.32
Staff Quarters	First	4	34.39	137.57
Pharmacy	Second	12	79.77	957.26
Operation Theatre	First	8	94.38	755.06
Operation Theatre	First	4	23.74	94.96
Operation Theatre	Third	4	112.13	448.54
Operation Theatre	Second	6	126.40	758.41
Staff Quarters	Second	9	88.58	797.24
Administration	Third	1	24.37	24.37
Operation Theatre	Third	8	139.38	1115.03
Administration	First	15	23.04	345.58
Diagnostics	Third	10	131.22	1312.17
Operation Theatre	Second	5	68.38	341.91
Administration	First	6	103.76	622.59
IPD Ward	First	15	108.27	1624.10
Operation Theatre	Second	2	53.30	106.59
OPD	Third	15	59.02	885.32
Operation Theatre	Second	2	128.50	257.00
Diagnostics	Ground	7	75.23	526.62

The space programme has been developed from the functional brief agreed with the user department. Departmental areas, circulation and the gross-to-carpet ratio are tabulated in the area statement enclosed at the Annexure.

ANNEXURE VI — Sheet 12

Design Computation

Axial load = 15.652 × 8.867 = 138.780 kN   (permissible 167.873) — safe

Moment = 35.419 × 9.079 = 321.583 kNm   (permissible 381.470) — safe

Deflection = 63.175 × 6.123 = 386.843 mm   (permissible 603.884) — safe

Bearing pressure = 87.213 × 2.598 = 226.548 kPa   (permissible 390.178) — safe

Shear = 42.086 × 8.580 = 361.089 kN   (permissible 546.629) — safe

Deflection = 27.382 × 2.257 = 61.791 mm   (permissible 108.772) — safe

Quantities have been computed from the approved architectural and structural drawings and cross-checked against the detailed measurement sheets enclosed at the Annexure. Rates are as per the schedule of rates in force with lead and lift adjustments.

ANNEXURE VI — Sheet 13

Soil investigation was carried out by rotary drilling at the locations shown on the layout plan. The safe bearing capacity adopted for design is based on the lower of the shear and settlement criteria as reported in the geotechnical report.

The structural design conforms to IS 456 for reinforced concrete and IS 1893 for earthquake resistant design. The site falls in seismic zone V and the importance factor applicable to a hospital building has been adopted.

Fire safety provisions follow Part 4 of the National Building Code. Means of egress, travel distances, staircase widths and the fire fighting installations have been designed for the occupancy classification of the block.

Consultations were held with the head of each clinical department on adjacency, patient flow and equipment layout. Minutes of the consultation meetings are placed at the Annexure to this chapter.

Reference: drawing no. EL-841, revision C.

ANNEXURE VI — Sheet 14

Detailed Measurement Sheet

Item	L (m)	B (m)	D (m)	Nos	Quantity
Item 750	12.28	1.30	2.85	14	320.332
Item 135	24.20	3.18	0.39	7	319.039
Item 647	9.53	0.30	2.84	8	140.056
Item 615	35.61	2.34	1.48	18	374.751
Item 432	12.05	3.16	2.47	6	219.793
Item 800	10.55	0.80	0.80	22	232.186
Item 206	16.78	4.61	2.17	17	357.738
Item 853	25.86	3.77	1.59	21	84.624
Item 295	27.46	1.23	1.56	3	359.164
Item 776	13.25	2.19	0.62	16	379.241
Item 168	22.84	0.38	1.16	17	121.473
Item 773	30.78	3.30	2.01	12	129.153
Item 501	34.91	3.50	0.90	2	19.539
Item 689	18.14	3.59	0.42	5	248.601
Item 496	29.17	0.87	0.36	22	300.871
Item 424	20.41	2.99	0.81	1	95.461
Item 148	11.62	4.47	1.87	17	91.326
Item 696	2.92	1.30	1.32	17	27.492
Item 795	31.52	0.48	0.83	22	236.748
Item 309	34.57	5.58	2.33	10	139.884
Item 234	32.23	2.44	2.54	20	223.262
Item 588	18.91	3.43	2.41	18	281.092

Quantities have been computed from the approved architectural and structural drawings and cross-checked against the detailed measurement sheets enclosed at the Annexure. Rates are as per the schedule of rates in force with lead and lift adjustments.

ANNEXURE VI — Sheet 15

Design Computation

Moment = 47.337 × 7.082 = 335.237 kNm (permissible 606.858) — safe

Deflection = 14.467 × 10.648 = 154.044 mm (permissible 176.948) — safe

Axial load = 40.499 × 4.079 = 165.198 kN (permissible 224.585) — safe

Deflection = 50.975 × 5.498 = 280.250 mm (permissible 488.496) — safe

Shear = 75.480 × 4.797 = 362.056 kN (permissible 480.753) — safe

Moment = 40.580 × 3.865 = 156.844 kNm (permissible 198.230) — safe

The structural design conforms to IS 456 for reinforced concrete and IS 1893 for earthquake resistant design. The site falls in seismic zone V and the importance factor applicable to a hospital building has been adopted.

ANNEXURE VI — Sheet 16

The structural design conforms to IS 456 for reinforced concrete and IS 1893 for earthquake resistant design. The site falls in seismic zone V and the importance factor applicable to a hospital building has been adopted.

Barrier-free access has been provided in accordance with the Harmonised Guidelines and Standards for Universal Accessibility. Ramps, lifts, signage and accessible toilets are provided on every floor of the block.

The space programme has been developed from the functional brief agreed with the user department. Departmental areas, circulation and the gross-to-carpet ratio are tabulated in the area statement enclosed at the Annexure.

Building services comprise electrical distribution, water supply and sanitation, HVAC for the identified conditioned areas, medical gas pipeline and vertical transportation. Connected load and demand factors are given in the load schedule.

Reference: drawing no. AR-526, revision A.

ANNEXURE VI — Sheet 17

Departmental Area Statement

Department	Floor	Units	Area per unit (sqm)	Total (sqm)
OPD	First	6	57.26	343.55
Pharmacy	Second	11	108.51	1193.65
Operation Theatre	First	16	51.51	824.08
Operation Theatre	Third	16	75.59	1209.44
Pharmacy	Third	8	19.85	158.79
Pharmacy	Second	7	133.21	932.44
ICU	Ground	1	59.77	59.77
Administration	First	8	26.59	212.73
ICU	Third	1	123.97	123.97
Staff Quarters	Third	16	47.67	762.66
OPD	Ground	4	43.19	172.75
Operation Theatre	Second	18	102.37	1842.71
Diagnostics	First	12	93.95	1127.45
Pharmacy	Second	3	99.56	298.68
Pharmacy	Second	17	114.62	1948.59
Operation Theatre	Second	12	106.83	1281.91
OPD	First	4	74.08	296.32
Operation Theatre	Third	1	86.54	86.54
Pharmacy	First	1	134.99	134.99
OPD	Second	13	99.62	1295.00
Diagnostics	Third	18	108.87	1959.66
Diagnostics	Ground	18	102.79	1850.21
Administration	First	8	103.49	827.89
ICU	First	5	125.11	625.54
IPD Ward	Ground	4	45.43	181.72
Staff Quarters	Ground	3	36.64	109.91

Building services comprise electrical distribution, water supply and sanitation, HVAC for the identified conditioned areas, medical gas pipeline and vertical transportation. Connected load and demand factors are given in the load schedule.

ANNEXURE VI — Sheet 18

Design Computation

Bearing pressure = 70.200 × 9.227 = 647.730 kPa   (permissible 747.544) — safe

Axial load = 47.629 × 2.745 = 130.753 kN   (permissible 169.175) — safe

Deflection = 40.254 × 3.335 = 134.263 mm   (permissible 217.119) — safe

Shear = 82.790 × 1.628 = 134.816 kN   (permissible 213.912) — safe

Bearing pressure = 44.321 × 9.552 = 423.364 kPa   (permissible 626.647) — safe

The space programme has been developed from the functional brief agreed with the user department. Departmental areas, circulation and the gross-to-carpet ratio are tabulated in the area statement enclosed at the Annexure.

ANNEXURE VI — Sheet 19

Fire safety provisions follow Part 4 of the National Building Code. Means of egress, travel distances, staircase widths and the fire fighting installations have been designed for the occupancy classification of the block.

The structural design conforms to IS 456 for reinforced concrete and IS 1893 for earthquake resistant design. The site falls in seismic zone V and the importance factor applicable to a hospital building has been adopted.

Barrier-free access has been provided in accordance with the Harmonised Guidelines and Standards for Universal Accessibility. Ramps, lifts, signage and accessible toilets are provided on every floor of the block.

Soil investigation was carried out by rotary drilling at the locations shown on the layout plan. The safe bearing capacity adopted for design is based on the lower of the shear and settlement criteria as reported in the geotechnical report.

Reference: drawing no. ST-741, revision A.

ANNEXURE VI — Sheet 20

Departmental Area Statement

Department	Floor	Units	Area per unit (sqm)	Total (sqm)
Operation Theatre	First	11	118.85	1307.31
Staff Quarters	Ground	18	109.11	1964.01
Pharmacy	Third	18	17.02	306.40
Staff Quarters	Second	18	54.73	985.20
Staff Quarters	First	2	98.00	196.01
Staff Quarters	Second	16	71.92	1150.72
Diagnostics	First	18	116.42	2095.53
Administration	Third	5	97.64	488.18
Operation Theatre	First	6	49.36	296.18
Staff Quarters	Second	16	114.93	1838.86
Administration	Third	12	122.45	1469.34
Staff Quarters	First	16	97.91	1566.57
Pharmacy	Second	3	89.53	268.58
Operation Theatre	Third	11	60.21	662.36
Administration	Second	12	28.92	347.01
Administration	First	1	94.39	94.39
Staff Quarters	Third	15	69.48	1042.16
OPD	First	16	98.88	1582.09
Pharmacy	Second	7	104.41	730.84
Staff Quarters	Third	17	87.56	1488.49
Pharmacy	First	11	67.73	745.00
Staff Quarters	First	14	126.23	1767.28

Fire safety provisions follow Part 4 of the National Building Code. Means of egress, travel distances, staircase widths and the fire fighting installations have been designed for the occupancy classification of the block.

ANNEXURE VI — Sheet 21

Design Computation

Axial load = 59.452 × 1.867 = 110.981 kN   (permissible 174.145) — safe

Deflection = 60.982 × 7.892 = 481.251 mm   (permissible 714.532) — safe

Axial load = 72.242 × 8.939 = 645.784 kN   (permissible 737.984) — safe

Shear = 1.392 × 11.327 = 15.768 kN   (permissible 28.643) — safe

Moment = 68.418 × 6.756 = 462.201 kNm   (permissible 687.624) — revise section

Moment = 31.772 × 2.880 = 91.491 kNm   (permissible 160.655) — safe

Building services comprise electrical distribution, water supply and sanitation, HVAC for the identified conditioned areas, medical gas pipeline and vertical transportation. Connected load and demand factors are given in the load schedule.

ANNEXURE VI — Sheet 22

The structural design conforms to IS 456 for reinforced concrete and IS 1893 for earthquake resistant design. The site falls in seismic zone V and the importance factor applicable to a hospital building has been adopted.

Fire safety provisions follow Part 4 of the National Building Code. Means of egress, travel distances, staircase widths and the fire fighting installations have been designed for the occupancy classification of the block.

Building services comprise electrical distribution, water supply and sanitation, HVAC for the identified conditioned areas, medical gas pipeline and vertical transportation. Connected load and demand factors are given in the load schedule.

Quantities have been computed from the approved architectural and structural drawings and cross-checked against the detailed measurement sheets enclosed at the Annexure. Rates are as per the schedule of rates in force with lead and lift adjustments.

Reference: drawing no. PH-394, revision C.

ANNEXURE VI — Sheet 23

Electrical Load Schedule

Panel	Description	Connected load (kW)	Demand factor	Demand (kVA)
DB-38	Lighting	16.97	0.60	11.35
DB-20	Pumps	141.34	0.44	68.72
DB-40	Pumps	151.30	0.66	110.43
DB-11	Lifts	171.58	0.56	107.70
DB-02	Medical equipment	155.08	0.53	91.41
DB-11	Sterilisation	129.98	0.53	76.84
DB-26	Power outlets	98.47	0.77	84.15
DB-08	Power outlets	131.65	0.88	128.01
DB-02	HVAC	101.18	0.90	100.89
DB-31	Pumps	3.27	0.93	3.38
DB-06	Lighting	81.15	0.86	77.44
DB-19	Pumps	172.92	0.79	151.20
DB-20	Lifts	73.38	0.87	70.83
DB-02	HVAC	62.38	0.46	31.79
DB-40	Pumps	101.58	0.64	72.49
DB-33	Lighting	11.93	0.87	11.55
DB-12	Medical equipment	58.18	0.43	27.72
DB-39	Power outlets	174.85	0.72	140.00
DB-35	HVAC	95.05	0.85	90.17
DB-15	Medical equipment	52.24	0.76	44.13
DB-30	Sterilisation	3.81	0.92	3.89
DB-10	Lifts	125.48	0.57	79.45
DB-33	Lifts	47.35	0.85	44.92
DB-17	HVAC	162.98	0.58	104.60

Soil investigation was carried out by rotary drilling at the locations shown on the layout plan. The safe bearing capacity adopted for design is based on the lower of the shear and settlement criteria as reported in the geotechnical report.

ANNEXURE VI — Sheet 24

Design Computation

Bearing pressure = 65.639 × 2.589 = 169.962 kPa   (permissible 221.237) — safe

Deflection = 5.542 × 2.821 = 15.634 mm   (permissible 22.925) — safe

Bearing pressure = 2.352 × 1.330 = 3.128 kPa   (permissible 4.110) — safe

Bearing pressure = 82.209 × 8.867 = 728.922 kPa   (permissible 1094.586) — safe

Bearing pressure = 63.701 × 7.146 = 455.180 kPa   (permissible 698.118) — safe

Moment = 40.353 × 2.672 = 107.817 kNm   (permissible 195.385) — safe

Soil investigation was carried out by rotary drilling at the locations shown on the layout plan. The safe bearing capacity adopted for design is based on the lower of the shear and settlement criteria as reported in the geotechnical report.

ANNEXURE VII — CONSULTATION AND APPROVAL RECORDS

Minutes of stakeholder consultations and copies of approvals obtained.

ANNEXURE VII — Sheet 1

Building services comprise electrical distribution, water supply and sanitation, HVAC for the identified conditioned areas, medical gas pipeline and vertical transportation. Connected load and demand factors are given in the load schedule.

The structural design conforms to IS 456 for reinforced concrete and IS 1893 for earthquake resistant design. The site falls in seismic zone V and the importance factor applicable to a hospital building has been adopted.

The space programme has been developed from the functional brief agreed with the user department. Departmental areas, circulation and the gross-to-carpet ratio are tabulated in the area statement enclosed at the Annexure.

Quantities have been computed from the approved architectural and structural drawings and cross-checked against the detailed measurement sheets enclosed at the Annexure. Rates are as per the schedule of rates in force with lead and lift adjustments.

Reference: drawing no. ST-811, revision B.

ANNEXURE VII — Sheet 2

Room Data Sheet

Room	Department	Carpet area (sqm)	Occupancy	Floor finish
Room 422	Paediatrics	19.69	3	Kota stone
Room 683	Radiology	25.78	13	Antistatic vinyl
Room 659	Surgery	78.69	11	Vitrified tile
Room 622	Casualty	23.65	4	Kota stone
Room 877	Casualty	85.68	9	Epoxy screed
Room 770	Casualty	19.55	10	Vitrified tile
Room 473	Pathology	47.19	21	Kota stone
Room 476	Obstetrics	39.31	26	Kota stone
Room 808	Casualty	70.93	29	Antistatic vinyl
Room 353	Pathology	71.74	10	Ceramic tile
Room 616	Pathology	34.96	30	Ceramic tile
Room 610	Surgery	25.62	29	Antistatic vinyl
Room 119	Pathology	32.31	21	Ceramic tile
Room 461	General Medicine	75.40	17	Ceramic tile
Room 184	Radiology	63.67	14	Ceramic tile
Room 560	Casualty	61.13	3	Kota stone

The structural design conforms to IS 456 for reinforced concrete and IS 1893 for earthquake resistant design. The site falls in seismic zone V and the importance factor applicable to a hospital building has been adopted.

ANNEXURE VII — Sheet 3

Design Computation

Axial load = 3.074 × 3.097 = 9.520 kN   (permissible 15.191) — safe

Axial load = 79.433 × 9.014 = 716.015 kN   (permissible 753.651) — safe

Moment = 75.181 × 11.973 = 900.108 kNm   (permissible 1211.170) — safe

Axial load = 42.230 × 6.306 = 266.286 kN   (permissible 476.212) — safe

Moment = 35.771 × 4.009 = 143.393 kNm   (permissible 178.794) — safe

Deflection = 76.041 × 7.103 = 540.154 mm   (permissible 820.309) — revise section

Shear = 81.510 × 2.497 = 203.504 kN   (permissible 362.805) — safe

Building services comprise electrical distribution, water supply and sanitation, HVAC for the identified conditioned areas, medical gas pipeline and vertical transportation. Connected load and demand factors are given in the load schedule.

ANNEXURE VII — Sheet 4

Building services comprise electrical distribution, water supply and sanitation, HVAC for the identified conditioned areas, medical gas pipeline and vertical transportation. Connected load and demand factors are given in the load schedule.

Barrier-free access has been provided in accordance with the Harmonised Guidelines and Standards for Universal Accessibility. Ramps, lifts, signage and accessible toilets are provided on every floor of the block.

Soil investigation was carried out by rotary drilling at the locations shown on the layout plan. The safe bearing capacity adopted for design is based on the lower of the shear and settlement criteria as reported in the geotechnical report.

The structural design conforms to IS 456 for reinforced concrete and IS 1893 for earthquake resistant design. The site falls in seismic zone V and the importance factor applicable to a hospital building has been adopted.

Reference: drawing no. AR-821, revision A.

ANNEXURE VII — Sheet 5

Bore Log Summary

Depth (m)	Strata	N value	Sample
10.3	Silty clay	43	UDS
18.5	Gravelly sand	60	SPT
29.9	Medium sand	58	UDS
11.2	Gravelly sand	17	UDS
10.4	Clayey silt	49	DS
15.2	Silty clay	35	DS
23.2	Coarse sand	40	UDS
13.5	Clayey silt	32	UDS
26.9	Clayey silt	35	UDS
28.4	Weathered rock	16	UDS
0.9	Clayey silt	39	SPT
15.1	Silty clay	18	UDS
1.3	Coarse sand	11	UDS
6.8	Gravelly sand	4	DS
5.4	Silty clay	58	DS
30.0	Silty clay	22	DS
22.0	Clayey silt	57	DS
3.7	Medium sand	51	SPT
9.7	Coarse sand	42	DS

Fire safety provisions follow Part 4 of the National Building Code. Means of egress, travel distances, staircase widths and the fire fighting installations have been designed for the occupancy classification of the block.

ANNEXURE VII — Sheet 6

Design Computation

Moment = 6.890 × 11.422 = 78.701 kNm   (permissible 129.918) — safe

Axial load = 37.173 × 9.620 = 357.607 kN   (permissible 573.164) — safe

Bearing pressure = 68.663 × 8.509 = 584.273 kPa   (permissible 711.369) — safe

Bearing pressure = 42.976 × 10.667 = 458.433 kPa   (permissible 553.110) — safe

Deflection = 76.127 × 11.916 = 907.123 mm   (permissible 1214.384) — safe

The space programme has been developed from the functional brief agreed with the user department. Departmental areas, circulation and the gross-to-carpet ratio are tabulated in the area statement enclosed at the Annexure.

ANNEXURE VII — Sheet 7

Barrier-free access has been provided in accordance with the Harmonised Guidelines and Standards for Universal Accessibility. Ramps, lifts, signage and accessible toilets are provided on every floor of the block.

Consultations were held with the head of each clinical department on adjacency, patient flow and equipment layout. Minutes of the consultation meetings are placed at the Annexure to this chapter.

The structural design conforms to IS 456 for reinforced concrete and IS 1893 for earthquake resistant design. The site falls in seismic zone V and the importance factor applicable to a hospital building has been adopted.

Consultations were held with the head of each clinical department on adjacency, patient flow and equipment layout. Minutes of the consultation meetings are placed at the Annexure to this chapter.

Reference: drawing no. PH-367, revision B.

ANNEXURE VII — Sheet 8

Departmental Area Statement

Department	Floor	Units	Area per unit (sqm)	Total (sqm)
IPD Ward	First	16	109.11	1745.77
Pharmacy	Ground	10	118.97	1189.73
Staff Quarters	Second	1	45.89	45.89
OPD	Ground	12	82.28	987.36
Administration	Second	12	61.85	742.21
IPD Ward	Second	18	79.53	1431.53
OPD	Second	5	138.13	690.64
Diagnostics	Second	5	70.48	352.38
Staff Quarters	First	3	115.95	347.84
Administration	Ground	4	120.78	483.11
Pharmacy	Second	18	16.74	301.24
Diagnostics	Second	15	61.26	918.85
Pharmacy	Ground	15	41.71	625.61
Pharmacy	First	16	122.08	1953.29
Operation Theatre	Third	8	116.39	931.09
Administration	First	4	68.36	273.44
Administration	First	14	97.32	1362.46
Staff Quarters	Third	17	18.84	320.31
ICU	Second	3	19.02	57.07
Pharmacy	Third	11	109.64	1206.04
OPD	Second	11	12.64	139.00
Administration	First	3	116.27	348.80
ICU	Second	14	52.40	733.56

The space programme has been developed from the functional brief agreed with the user department. Departmental areas, circulation and the gross-to-carpet ratio are tabulated in the area statement enclosed at the Annexure.

ANNEXURE VII — Sheet 9

Design Computation

Axial load = 73.660 × 3.259 = 240.080 kN   (permissible 301.212) — safe

Bearing pressure = 8.380 × 7.795 = 65.319 kPa   (permissible 95.125) — safe

Bearing pressure = 36.590 × 9.424 = 344.814 kPa   (permissible 410.330) — safe

Axial load = 6.325 × 4.445 = 28.118 kN   (permissible 37.634) — safe

Moment = 48.045 × 3.609 = 173.371 kNm   (permissible 216.690) — safe

Quantities have been computed from the approved architectural and structural drawings and cross-checked against the detailed measurement sheets enclosed at the Annexure. Rates are as per the schedule of rates in force with lead and lift adjustments.

ANNEXURE VII — Sheet 10

Fire safety provisions follow Part 4 of the National Building Code. Means of egress, travel distances, staircase widths and the fire fighting installations have been designed for the occupancy classification of the block.

Soil investigation was carried out by rotary drilling at the locations shown on the layout plan. The safe bearing capacity adopted for design is based on the lower of the shear and settlement criteria as reported in the geotechnical report.

Fire safety provisions follow Part 4 of the National Building Code. Means of egress, travel distances, staircase widths and the fire fighting installations have been designed for the occupancy classification of the block.

Quantities have been computed from the approved architectural and structural drawings and cross-checked against the detailed measurement sheets enclosed at the Annexure. Rates are as per the schedule of rates in force with lead and lift adjustments.

Reference: drawing no. ST-535, revision C.

ANNEXURE VII — Sheet 11

Room Data Sheet

Room	Department	Carpet area (sqm)	Occupancy	Floor finish
Room 324	Radiology	29.85	7	Antistatic vinyl
Room 863	General Medicine	32.07	27	Ceramic tile
Room 693	Paediatrics	87.88	6	Vitrified tile
Room 747	Pathology	26.87	22	Epoxy screed
Room 813	Casualty	21.76	12	Kota stone
Room 858	Pathology	68.30	3	Epoxy screed
Room 274	Surgery	85.94	15	Ceramic tile
Room 171	Surgery	89.38	24	Ceramic tile
Room 772	Obstetrics	74.31	16	Ceramic tile
Room 162	Obstetrics	63.32	4	Antistatic vinyl
Room 808	Surgery	66.67	29	Antistatic vinyl
Room 297	Pathology	56.44	13	Epoxy screed
Room 729	General Medicine	46.22	1	Antistatic vinyl
Room 842	Obstetrics	9.65	3	Kota stone
Room 726	Obstetrics	27.93	20	Epoxy screed
Room 707	Surgery	24.38	22	Vitrified tile
Room 757	General Medicine	83.12	3	Antistatic vinyl
Room 306	General Medicine	40.11	8	Epoxy screed
Room 868	Paediatrics	41.00	10	Ceramic tile
Room 648	Pathology	77.13	13	Kota stone
Room 545	Obstetrics	43.70	15	Epoxy screed
Room 166	Paediatrics	24.77	11	Vitrified tile
Room 656	General Medicine	10.65	22	Kota stone

The structural design conforms to IS 456 for reinforced concrete and IS 1893 for earthquake resistant design. The site falls in seismic zone V and the importance factor applicable to a hospital building has been adopted.

ANNEXURE VII — Sheet 12

Design Computation

Bearing pressure = 60.800 × 11.491 = 698.641 kPa   (permissible 1081.254) — safe

Deflection = 6.787 × 10.955 = 74.351 mm   (permissible 117.511) — safe

Deflection = 20.240 × 0.958 = 19.393 mm   (permissible 26.697) — safe

Shear = 31.446 × 1.918 = 60.313 kN   (permissible 70.912) — safe

Bearing pressure = 41.812 × 6.884 = 287.829 kPa   (permissible 318.074) — safe

Axial load = 23.973 × 6.893 = 165.235 kN   (permissible 288.279) — safe

Shear = 23.266 × 10.054 = 233.911 kN   (permissible 439.952) — revise section

Moment = 23.933 × 5.468 = 130.871 kNm   (permissible 172.867) — revise section

The space programme has been developed from the functional brief agreed with the user department. Departmental areas, circulation and the gross-to-carpet ratio are tabulated in the area statement enclosed at the Annexure.

ANNEXURE VII — Sheet 13

Quantities have been computed from the approved architectural and structural drawings and cross-checked against the detailed measurement sheets enclosed at the Annexure. Rates are as per the schedule of rates in force with lead and lift adjustments.

Consultations were held with the head of each clinical department on adjacency, patient flow and equipment layout. Minutes of the consultation meetings are placed at the Annexure to this chapter.

The space programme has been developed from the functional brief agreed with the user department. Departmental areas, circulation and the gross-to-carpet ratio are tabulated in the area statement enclosed at the Annexure.

Quantities have been computed from the approved architectural and structural drawings and cross-checked against the detailed measurement sheets enclosed at the Annexure. Rates are as per the schedule of rates in force with lead and lift adjustments.

Reference: drawing no. EL-138, revision B.